# Operation manual of the inSolar solution

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# 1 The PV plant - An overview

A photovoltaic plant is an installation which produces energy using the sun as a source. The PV plant is interconnected to the Electrical Power System (EPS, which is also called utility grid), so that the produced energy is forwarded to the customer loads. The generic architecture of a PV plant is presented in Figure 1.

The PV plant production facilities consist of:

- A number of PV panels, depending on the PV plant nominal power
- A number of PV panel groups, each consisting of a number of PV panels. The grouping of PV panels into PV panel groups is decided at installation time and each panel group is supposed to represent a string, i.e. a serial connection of the panels to the corresponding inverter and is mandatorily considered to include only one PV panel type. Panel groups are equipped with a temperature sensor and a solar irradiance sensor (pyranometer, placed at the inclination of the PV panels), together with the associated transducers. Most often, readings from temperature sensors and pyranometers would have to be correlated to multiple panels, since, for cost reasons, fewer sensors than panel groups are deployed. Additionally, the DC current of each string (PV panel group) may be monitored through a string monitor.
- A number of PV arrays, consisting of a number of PV panel groups and one inverter, converting the DC output of the PV panels to AC power. Which PV panel groups (strings) belong to which PV array is decided at installation time. All modern inverters provide monitoring of a number of operational parameters and events.
- A number of PV array groups, each consisting of a number of PV arrays. Which PV arrays belong to which PV array groups is decided upon by the PV plant designer.
- A number of step-up transformers, connected in parallel, which raise the 220 VAC output of the inverters to the voltage of the EPS. The number of transformers depends on the PV plant nominal power and which PV array groups are connected to which transformer is decided by the PV plant designer. The input to/output/from the transformers are monitored through 2 multimeters, connected on the primary (LV side) and secondary (HV side) windings. A protection device connected on the secondary winding, automatically disconnects the transformer from the EPS whenever certain conditions are met. This protection device may also act as the HV multimeter. Each step-up transformer may provide a set of devices monitoring its proper operation (a DGPT relay, a Buchholz relay and a thermostat, depending on the transformer type). Each step-up transformer and associated multimeters/protection devices are collectively denoted as the Transformer Connection Point (TCP) in the ensuing and multiple TCPs may exist in a PV plant.
- The Point of Common Coupling (PCC), which is the point of interconnection with the EPS as referred to within relevant bibliography (see Reference 5). This is the point where all requirements mandated on the PV plant by the EPS operator apply, so it is considered that only one PCC can exist per PV plant. The quality of the power delivered to the grid is monitored through a multimeter and a protection device undertakes the task of automatically disconnecting the PV plant under certain conditions. The protection device may also act as an additional multimeter.
- The Load Connection Point (LCP), which is the point of interconnection of PV plant loads, which are PV plant auxiliary equipment, consuming part of the produced

power during the morning and power from the EPS during the night, since, of course, the PV plant does not operate when the sun is down. Loads are connected in parallel to the PCC via a step-down transformer. The step-down transformer may provide a set of signals so that its operation and status can be monitored, in analogy to the TCPs. The LCP is monitored through a multimeter. The loads can be fed with electrical power either by the EPS/PV production or by other power sources, depending on the position of a transfer switch. The PCC protection device is usually placed in such a way that the power supply to the loads is not interrupted when the PV plant is disconnected from the grid. Only one LCP is supposed to exist per PV plant.

- A number of electrical panels, which are part of the PV plant electrical installation and may contain a subset of the following component groups, whose state can be monitored through contacts:
    - Circuit breaker groups
    - Surge arrester groups
    - Fuse groups
    - Switch groups

  Electrical panels may also contain a number of multimeters and protection devices, which cannot be assigned to the PCC, LCP or any TCP.
- A number of weather stations, monitoring the environmental conditions at the PV plant, which may be used to quantify the PV plant production performance. These weather stations may include:
    - An ambient temperature sensor and transducer
    - A horizontal solar irradiance sensor (pyranometer) and transducer
    - A wind speed sensor
    - A wind direction sensor
    - A relative humidity sensor
    - A barometric pressure sensor
    - A water precipitation (rain gauge) sensor
- A number of reference cells, which are in fact contain an actual PV cell, which is short-circuited. The short-circuit current is directly analogous to the irradiance, as perceived by the PV cell. The reference cell also contains a temperature sensor, soldered at the back of the PV cell, which provides a measurement of the cell temperature as well as a correction factor for the irradiance measurement. The reference cell measurements are mostly used for PV panel performance modeling and prediction.
- A number of shelters, each of which is assumed to have multiple rooms. Access to the doors of some rooms is controlled and the status of each room is monitored through a set of sensors (fire, flood, occupancy and temperature together with the associated transducers).
- A PV plant site perimeter control system, which consists of a number of alarm and actuator zones. The former indicate security breaches through infrared barriers or motion detectors and the respective actuator zones activate sirens or perimeter lights to avert intruders.
- A multi-camera video surveillance system, which may or may not interact with the perimeter control system

 

- A security system, which may or may not interact with the perimeter control system

Depending on the nominal power of the PV plant and the country, the PV plant may be connected to either low voltage (220 VAC) or high voltage. In the former case, the generic architecture presented in Figure 1 is different, since all transformers are absent and the loads are connected directly in parallel to the inverters.

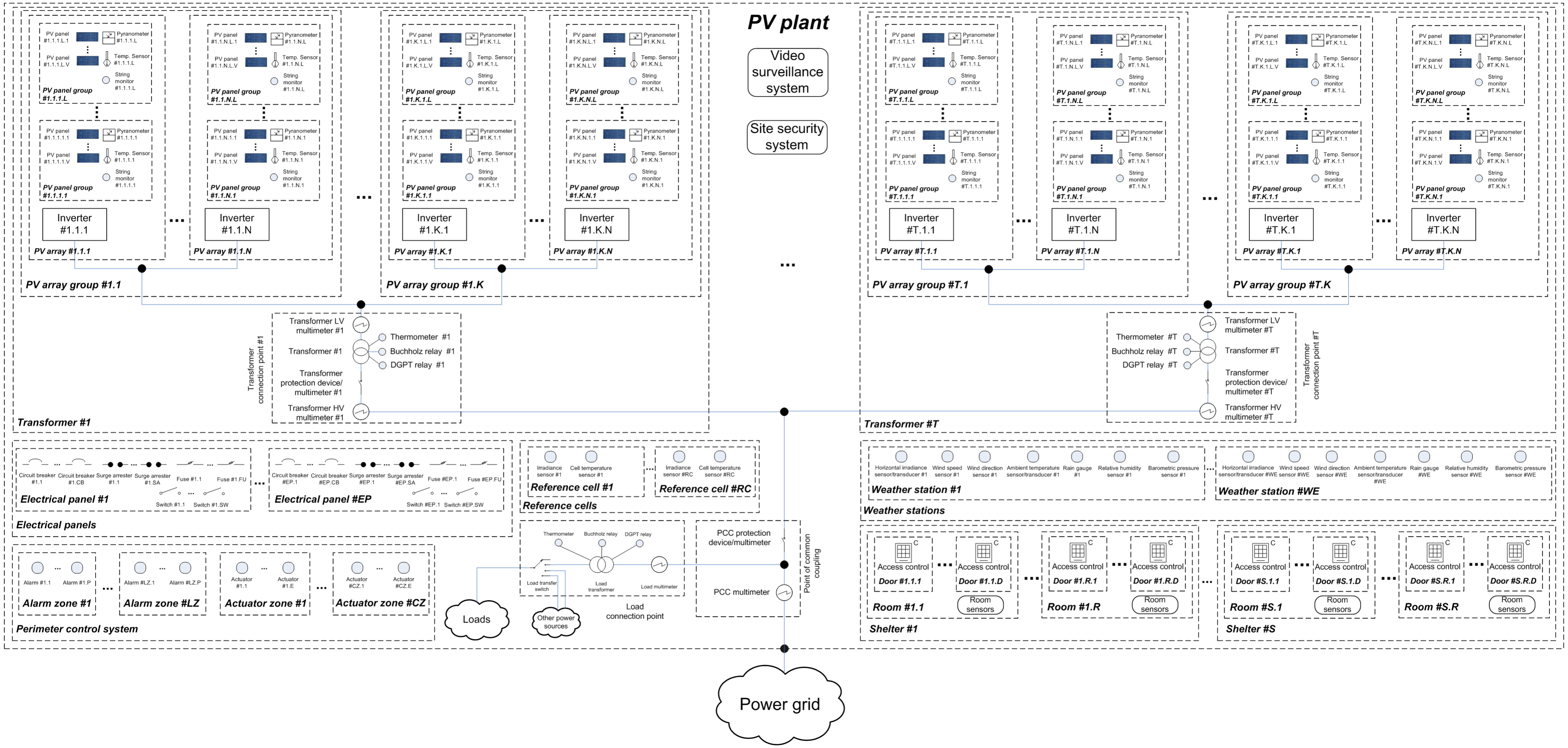

**Figure 1. PV plant components and interconnections**

# 2 The inSolar solution

The inSolar solution aims at providing centralized monitoring & control of a number of remote PV plants (as per Figure 1) to a number of solar park owners, maintenance contractors and security companies. The interaction of the users with the solution relies on the public internet and the GSM/3G network. The overall abstract architecture of the solution is presented in the figure below.

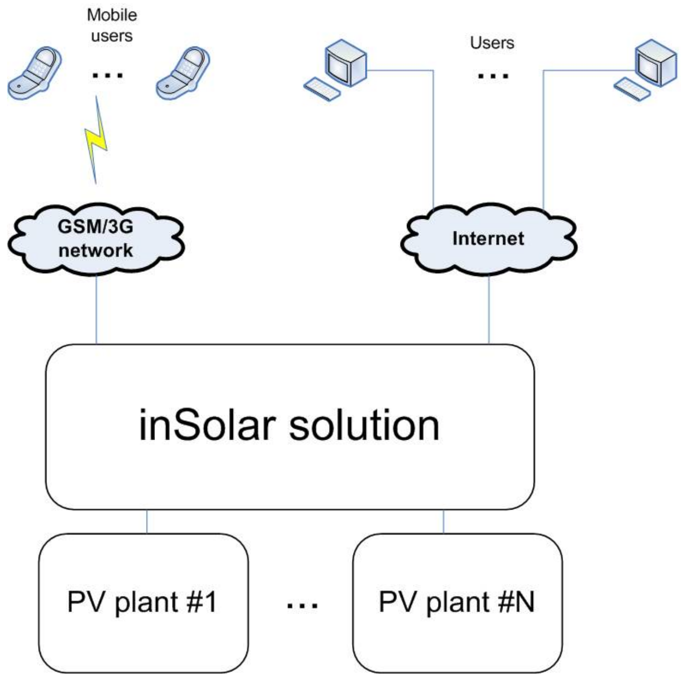


**Figure 2. The overall abstract architecture of the inSolar solution**

The inSolar solution has been designed to provide 6 different services to its users:

1. The monitoring service, operating over the public internet (Web). The user is capable, through an Internet browser, to get information (parameters, events and status) regarding the PV plants of interest to him/her.
2. The control service, also operating over the public internet (Web). The user is capable, through an Internet browser, to control certain aspects of the PV plant operation.
3. The notification service, operating both over the Web and the mobile network and delivering notifications to the mobile phones or e-mails of the users who have subscribed to the service whenever certain events related to the PV plants of interest to them occur
4. The report generation & delivery service, operating over the Web and delivering reports to the e-mails of users who have subscribed to the service in a periodical manner. These reports include extensive information on the plants regarding operational, maintenance and financial aspects.

5. The graphing service, creating comprehensive graphs of parameters and events to be presented through the Web interface of the PV plant monitoring service or to be included in reports, created by the report generation & delivery service
6. The export service, allowing the user to store locally in his/her computer data retained within the inSolar solution

These services are provided to the end user for all PV plants related to the enterprise he/she is working for. This set of plants is collectively referred to as “group of PV plants” in the ensuing.

# 3 Information modeling supported by the inSolar solution

The inSolar solution models the PV plant in the hierarchical manner presented below which is a tree representation of the PV plant architecture depicted in Figure 1.

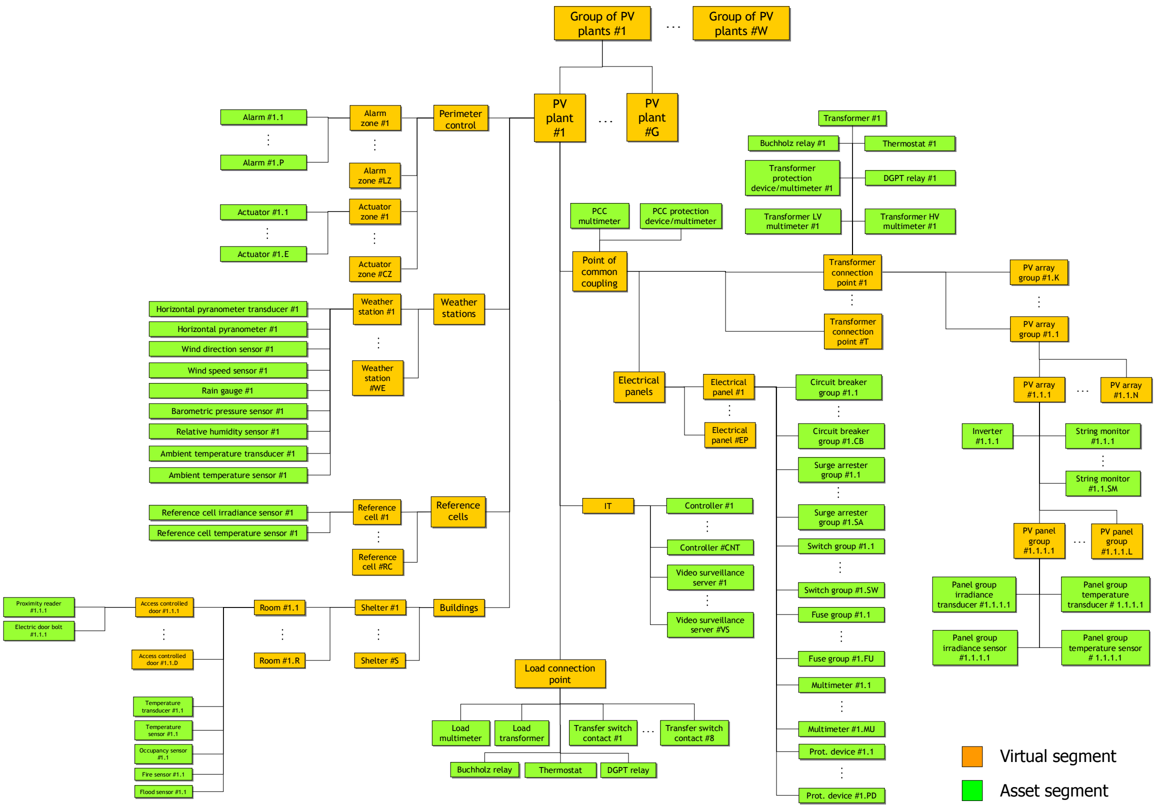


**Figure 3. Hierarchical model supported by the inSolar solution**

The nodes of the tree can be broadly categorized as:

- Nodes representing virtual entities, i.e. entities that do not contain actual equipment installed at the plant. Such entities are for example the PV plant, the PCC, the PV array etc. These nodes are denoted as virtual segments in the ensuing.
- Nodes representing equipment entities, i.e. entities that contain a specific piece of equipment installed at the plant. Such entities are for example the numerous sensors/transducers, the multimeters, the protection devices etc. These nodes are denoted as asset segments in the ensuing.

Both virtual and asset segments are used in the ensuing as placeholders for the data that are handled by the inSolar solution. It has to be noted that certain parts of the tree may be absent in certain cases. For example, a low-power PV plant does not incorporate transformers. As another example, a certain PV plant may not employ perimeter or access control. In such cases, the simplifications that should be applied are evident.

# 4 PV plant monitoring basics

In order to successfully present the functionality provided by the inSolar solution, the IEC standard 61724 (Reference 1) was used as the basis for this section and the remainder of the document. Reference 1 outlines the parameters that will have to be measured in order to rate the production performance of a PV plant.

## *4.1 The IEC61724 approach*

Reference 1 provides extensive information on the physical quantities which have to be measured from site-deployed equipment and sensors as well as the data acquisition method to accomplish such measurements and the post-processing necessary to end up with meaningful results that must then be displayed and recorded. The methodology is depicted in the figure below.

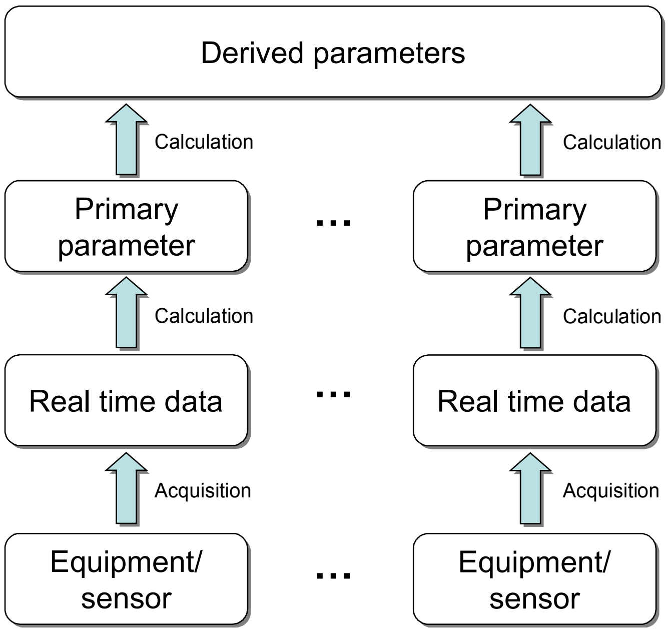


**Figure 4. The IEC 61724 approach**

Compared to Figure 1, IEC 61724 is a subset, since it does not consider the step-up transformers but rather models a PV plant directly connected to the power grid at low voltage. All parameters can, however, be modified accordingly to match the general architecture of Figure 1, which applies to higher power PV plants and this is presented in the sections that follow.
Based on customer requirements, it is necessary (as shown in the figure below) to extend the IEC61724 approach depicted in Figure 4 in order to include state-sets and state change events, which implement the notification of the users on the occurrence of specific phenomena, maximum/minimum parameter values and settings, which orchestrate the operation of the inSolar solution. This extension applies for the remainder of this document.

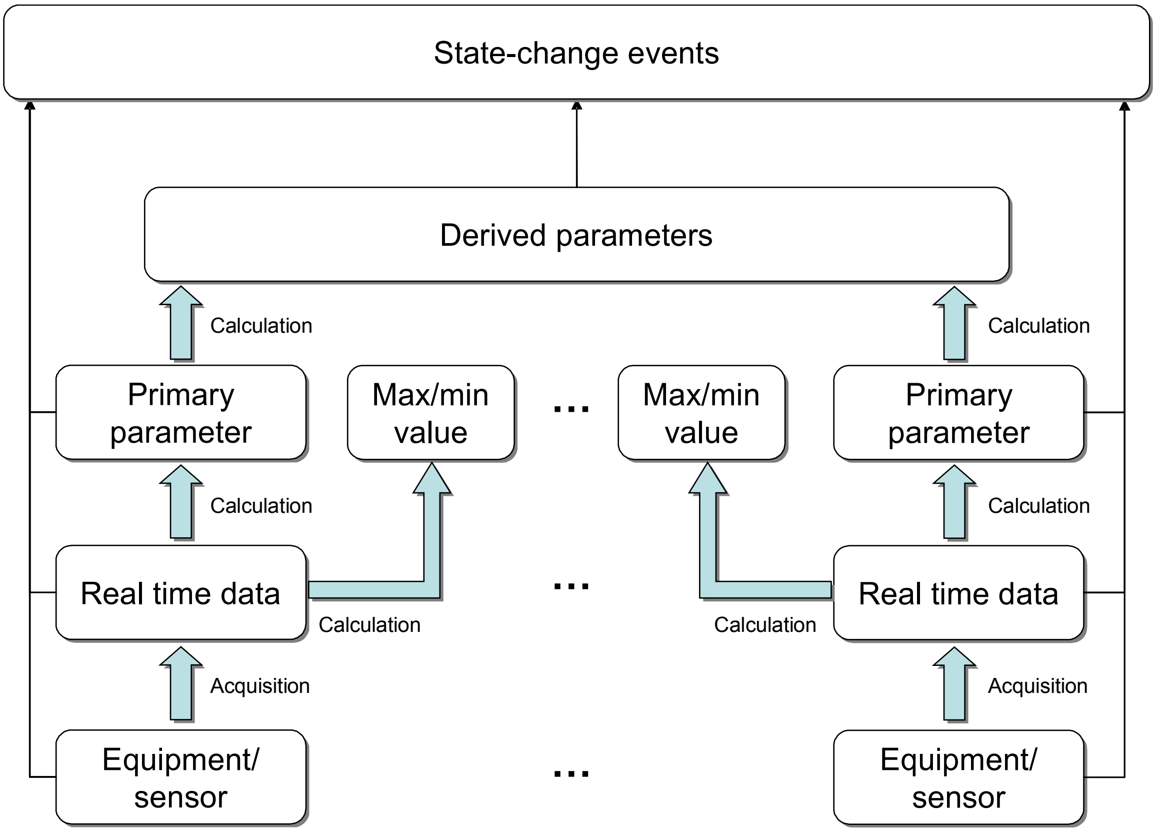


**Figure 5. The extended IEC61724 approach**

## *4.2 Data acquisition*

Reference 1 states that real-time data should be acquired from the field equipment and sensors every 1 minute or less, depending on the physical quantity being measured, so requirements on the sampling rate are explicitly recorded per measurement type in the ensuing. Reference 1 also specifies requirements on the acquisition accuracy per measured parameter, so whenever this document is quoted, the related accuracies are implied. It has to be noted that both the sampling rate and accuracy of the measurements depend on the equipment deployed at the PV plant.

 

## 4.3 Primary parameter calculation

Once the real-time data have been acquired, certain processing steps have to be applied on them. These steps aim at the reduction of the number of data values and produce the so-called primary parameters to be used in subsequent calculations.

First of all, a proper processing window has to be determined. Reference 1 suggests that primary parameters should be produced on an hourly basis but other intervals are also possible. Reference 1 encodes this interval using the “recording period” notion and mandates that this period must always be an integer multiple or sub-multiple of an hour. Only one recording period is supported per PV plant. The remainder of the present document takes the recording period into consideration for all calculations.

Afterwards, certain real-time data values have to be eliminated from subsequent calculations. These include:

- **Missing data:** The user must take into consideration that sensors and equipment deployed in the field can fail and assuming that failures can be detected, real-time data produced during such periods are discarded.
- **Out-of-scale data:** A measurement range is determined and values outside this range are discarded, since, most probably, they do not properly represent the physical quantities that are measured.
- **Out-of-scale derivative of the data values:** The absolute value of the derivative of 2 successive real-time data values is calculated and if it is larger than a pre-determined value, both values are discarded.

Finally, the completeness of each real-time data set has to be determined. It is understandable that the processing steps described above may reduce the number of valid real-time data values. In order to proceed, this number has to be compared to a pre-determined threshold. If it is smaller, the value of the respective primary parameter for the specific time window is also declared invalid, since it cannot be accurately calculated. If it is larger, the respective primary parameter value for the specific time window os derived using a mean operation on the valid real-time data values and stored for further use. Reference 2 proposes that at least 75% of the real-time data values must be valid in order to declare the respective primary parameter value as valid. The entire approach is presented in the 2 figures below. The first one depicts an exemplary data set, assuming a real-time data sampling rate of 1 value per minute and a recording period of 1 hour, so 4 hours are recorded (240 real-time data samples). A sensor malfunction has eliminated real-time data from sample (minute) 125 to sample (minute) 155. The second one presents the same real-time data set after the elimination of values and derivative values that lie outside their pre-determined ranges. A primary parameter value cannot be calculated for recording period 3 (samples 121-180), because the completeness criterion is not satisfied.

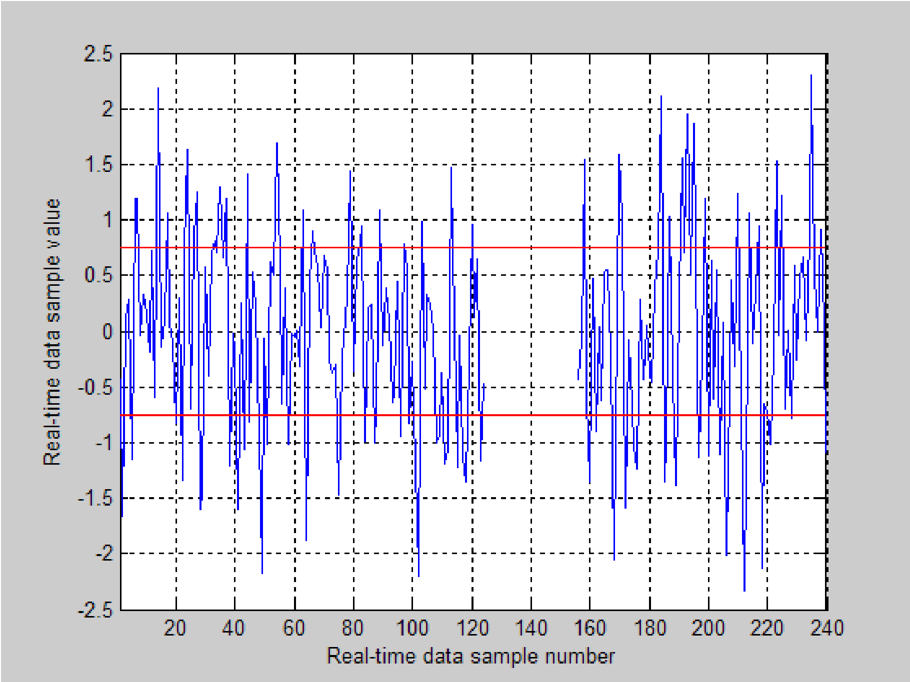


**Figure 6. Real-time data (sensor malfunction from sample 125 to sample 155)**

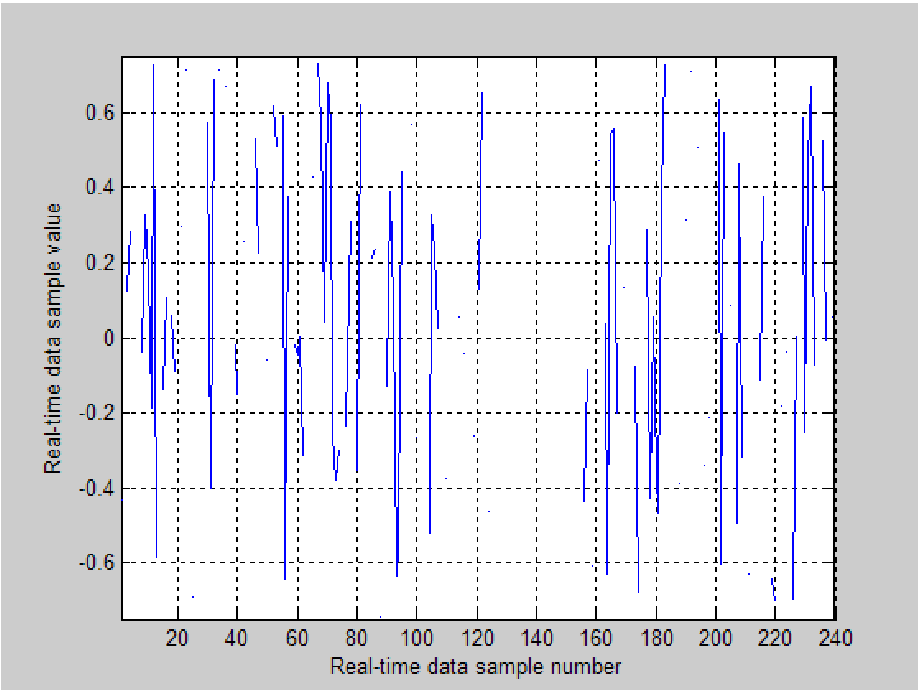


**Figure 7. Real-time data with value and derivative value invalidation**

## 4.4 Maximum/minimum value calculation

Using much the same approach as the one described in the section above, the inSolar solution calculates the maximum/minimum value of each set of real-time data spanning the recording period. The same real time data sample elimination procedure, described in section 4.3 is followed and the same completeness threshold applies.

## 4.5 Derived parameter calculation

Derived parameters are calculated using a specific mathematic formula which operates on a specific set of values or vectors of values of primary (and/or other derived) parameters covering a set of specific time spans (always in multiples of the recording period), which are called “reporting periods” within Reference 1, which also defines the necessary derived parameters, together with the formulas to be used for their calculation.

It is understandable that since primary (or other derived) parameter values may be missing (since they are invalid), it is necessary to define the set of values of primary (and/or other derived) parameters that can be considered in subsequent calculations. Since derived parameters have to be calculated over a time period (reporting period) covering multiple recording periods, only the recording period intervals during which all primary (or other derived) parameters participating in the calculations with valid values are used. If the number of recording period intervals containing valid values for all primary (or other derived) parameters that participate in the calculation of the derived one is less than 50% of the number of recording periods that the reporting period spans, the derived parameter is declared invalid for the specific reporting period and it is not calculated.

The approach is presented in the figure below, assuming a derived parameter in the calculation of which 2 different primary and 1 derived parameter are involved. The recording period equals an hour and an arbitrary reporting period of 5 hours is used for pictorial purposes, so ideally, all 5 samples of each parameter shall have to be used. Since valid samples coexist for all input parameters only during hours 2, 4 and 5, the derived parameter can be calculated using 3 samples out of 5. The calculation shall have to be carried out, since 60% of the values are present.

The arithmetic of all calculations carried out within the inSolar solution adheres to IEEE 754 (see Reference 9).

| | Hour 1 | Hour 2 | Hour 3 | Hour 4 | Hour 5 |
|---|---|---|---|---|---|
| **Primary parameter 1** | ✗ | ✓ | ✓ | ✓ | ✓ |
| **Primary parameter 2** | ✓ | ✓ | ✗ | ✓ | ✓ |
| **Derived parameter 1** | ✓ | ✓ | ✓ | ✓ | ✓ |

| | Hour 2 | Hour 4 | Hour 5 |
|---|---|---|---|
| **Primary parameter 1** | ✓ | ✓ | ✓ |
| **Primary parameter 2** | ✓ | ✓ | ✓ |
| **Derived parameter 1** | ✓ | ✓ | ✓ |

**Figure 8. Exemplary derived parameter calculation**

## *4.6 Reporting periods for derived parameters*

According to Reference 1, the inSolar solution must be capable of calculating derived parameters for a variety of time spans, which extend into the past and are called reporting periods. It is understandable that since the values of all primary parameters are stored within the inSolar solution for every recording period, it is possible to post-process that data to produce derived parameters for any reporting period. However, baseline requirements mandate that the inSolar solution is capable of calculating and storing derived parameter values for:

1. Each recording period of the current day (recording period values)
2. Past days (daily values)
3. Past months (monthly values)
4. Past years (yearly values)

The inSolar solution, depending on the derived parameter, may support all abovementioned reporting periods or a subset of those. In any case, the supported reporting periods are clearly indicated in the sections that follow.

As far as the derived parameter calculation for each recording period of the current day is concerned, 2 different modes of operation are supported:

- Calculation using primary parameter values of the last recording period (instantaneous value calculation)
- Calculation using primary parameter values from 00:01 hours until the last recording period (day-cumulative value calculation)

In the remainder of this document, the use of the second mode is clearly indicated whenever applicable.

If the derived parameter calculation is carried out for past days, recording period values of the related primary parameters from 00:01 hours to 00:00 hours of the following day are used. Monthly and yearly values are calculated on a calendar basis, using the respective recording period values of the primary parameters. As far as energies are concerned, their

values have to be calculated for each recording period in a cumulative manner since the beginning of the operation of a PV plant. Such cases are clearly designated in the ensuing.
The inSolar solution stores and presents to the users each derived parameter value with its timestamp and engineering unit. The timestamp of a recording period derived parameter value is set to the beginning of the first minute of the recording period following the one for which the recording period derived parameter value is calculated. The timestamp of a daily derived parameter value is set to the beginning of the first minute of the day following the one for which the derived parameter daily value is calculated. The timestamp of a monthly derived parameter value is set to the beginning of the first minute of the first day of the month following the one for which the monthly derived parameter value is calculated. The timestamp of a yearly derived parameter value is set to the beginning of the first minute of the first day of the year following the one for which the yearly derived parameter value is calculated. All timestamps refer to the timezone of the specific PV plant.

## *4.7 Reporting periods for primary parameters and max/min values*

Since, according to the paragraph above, all derived parameters are calculated for a set of reporting periods (each recording period of the current day, past days, months and years) it makes sense to calculate daily, monthly and yearly values for primary parameters as well so as for the user to be capable to directly compare primary and derived parameter values. In that case, the daily, monthly and yearly estimates of primary parameters can also be considered as derived parameters. There is no such function defined within Reference 1, however the reasonable choice for the inSolar solution is to use, depending on the parameter:

- A time average (sum of available recording period primary parameter values over the number of available recording period primary parameter values), spanning over past days, months and years
- A sum, across past days, months and years
- A maximum/minimum operation (for max/min values)

It has to be noted that as far as the calculation of daily, monthly and yearly primary parameter estimates is concerned, the completeness threshold of section 4.5 applies, i.e. if the available data are less than 75% complete, a value is not produced.
The inSolar solution stores and presents to the users each primary/derived parameter and max/min value sample for each reporting period with their timestamps and engineering units. The same timestamp rule used for derived parameters applies (see section 4.6).
The inSolar solution, depending on the primary parameter, may support all abovementioned reporting periods or a subset of those. In any case, the supported reporting periods is clearly indicated in the sections that follow.

## *4.8 Raw real-time data*

Apart from primary, derived and maximum/minimum parameters, the inSolar solution is capable of storing and presenting to the user selected sets of real-time data values with their acquisition timestamps and engineering units. The validation procedure of section 4.5 does not apply for such data and specific reporting periods have no meaning.

## 4.9 Real-time monitored data

In certain cases, it is necessary to present to the user for monitoring purposes the real-time data of Figure 5 as they are recorded by the inSolar solution with an acquisition timestamp and engineering unit but without any of the processing of section 4.3. In that case, the parameter is denoted as real-time monitored and its values are not stored within the inSolar solution.

## 4.10 Status, state-sets, states, state-change events and the concept of severity

Apart from the parameters presented in the sections above, the inSolar solution is required to record and present to the user the current status of the entities presented in section 3, regardless of whether they are virtual or asset segments. Though IEC 61724 does not define this concept, it is considered to be a powerful tool in terms of the user getting an immediate grasp on whether any PV plant entity or even the inSolar solution itself operate properly or not. Status information actually encodes the severity of the condition of each entity independently, which is a number on a scale from 0 (everything normal) to 100 (complete disaster). The status update mechanism shall become apparent in the ensuing.

Apart from status information, the inSolar solution also supports an arbitrary number of state-sets per entity. Each state set encodes a set of states, which are mutually exclusive, i.e. the entity can be found in only one of them. Each state of each state set also has a preset severity level associated with it.

Changes among states are triggered by state-change events, which are either directly generated by sensors and equipment connected to the inSolar solution (primary events), or by the inSolar solution itself (derived events).

The inSolar solution provides all means necessary for the user to acknowledge that he/she has been notified on any specific state-change event and has taken appropriate actions. This acknowledgement is a simple form of bookkeeping of user actions for further investigation. Each state-change event is stored and presented to the user per state set with the following data:

- The entity generating the event (virtual or asset segment)
- The state the event led to in a descriptive form
- The event occurrence date/time, in the timezone of the PV plant
- The event reception date/time, in the timezone of the PV plant
- The event acknowledgment status

Since all entities are organized in a hierarchical structure (see Figure 3), the current status of a certain entity cannot depend solely on the state (or states) it is into, but the statuses of all underlying entities within the hierarchy have to be used as well. This is a basic concept within the inSolar solution, since it enables the efficient notification of the user if something calls for his/her attention. Moreover, this hierarchical status propagation, as it will be denoted in the ensuing, can help the user realize the effect that an entity being in a certain state of a certain state set has on the operation of the PV plant as a whole.

The algorithm is very simple. The status of each entity of the plant hierarchy is always be equal to the maximum severity of the states of all underlying entities at any given point in time. In order to further elaborate on this hierarchical status propagation, let us refer back to Figure 3 and imagine that a certain inverter breaks down. An event encoding this state shall be generated and presented to the user. Furthermore, the status of the inverter shall change since it is in error. Should such an event occur, it is also expected to affect the group of inverters the specific inverter belongs to, the PV plant as a whole and consequently, the group of PV plants the specific PV plant belongs to. The statuses of all involved entities are presented to the user in an easily understandable form (i.e. color encoding, based on a set of severity thresholds) so that his/her attention is drawn not only to a specific event but also to the effect it has on the operation of the whole PV plant. The approach is further clarified through the following figure, presenting an exemplary hierarchy with the related statuses and state-sets for each node. The related severities are also depicted, so that the hierarchical status propagation is apparent.

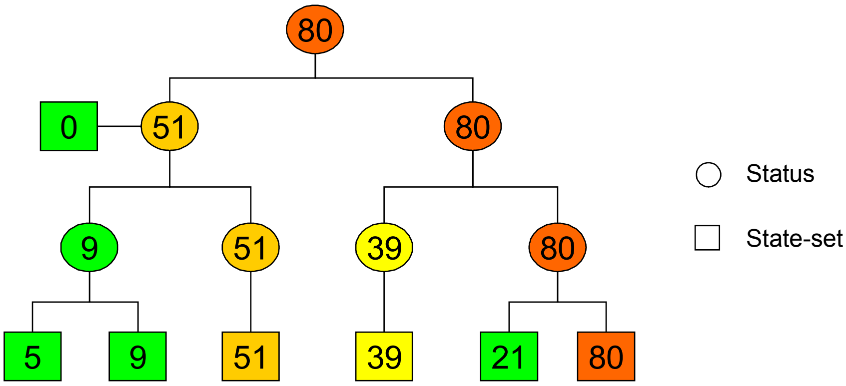


**Figure 9. The hierarchical status propagation concept**

2 hierarchical status propagation trees are supported, one propagating the status of the acknowledged and one propagating the status of the unacknowledged events.

## *4.11 Settings*

Settings are parameters set by the users of the inSolar solution in order to configure its operation. The values of the settings are considered constant from the time they are set until they are reset. The inSolar solution stores each setting value with the timestamp in the timezone of the PV plant it refers to and engineering unit. It also stores and presents the status of each setting separately (pending, failed and successful).

## *4.12 The treatment of energy and power measurements*

A number of parameters that are necessary according to Reference 1 relate to power and energy, the second being the integral across time of the first. If the continuous power is not available but instead this physical quantity is sampled, Reference 1 proposes that the energy is calculated using a summation, i.e. assuming that each sample of the power corresponds to the average power for the sampling interval. This may or may not be true, depending on the variation of the power as a function of time and errors (larger or smaller) are induced in the calculation.

 

Depending on the sensors and other equipment which provide the power or energy measurements, either the former or the latter shall be available for the inSolar solution. If the energy is directly provided (usually in a cumulative way, i.e. the measurement equipment does the integration of power over time), the energy for the recording period can be easily calculated as the derivative of the cumulative value measured at the end of the recording period from the value read out at the end of the previous recording period. This approach implies that the energy is sampled once per recording period and the time-averaging procedure of section 4.3 does not apply. This mode of operation is by far more beneficiary in terms of accuracy of the results.

However, there exist metering equipment that provide only power and not energy measurements or the energy measurements are inaccurate. In that case and if energy is necessary as a primary parameter, the procedure of section 8 of Reference 1 is employed to yield the recording period energy estimates. This approach is the equivalent of numerical integration of the power for the recording period. As already mentioned, the sampling rate employed for the process is expected to affect the measurement (the faster is usually the better, though this is not always possible).

In any case, the recording period values of the power and cumulative energy are stored within the inSolar solution as primary parameters, whenever available. In the case one is missing, it is calculated from the other as a derived parameter.

## *4.13 Conventions regarding the monitoring of the electrical power flow*

In order for the inSolar solution to provide proper monitoring of the electrical power flow throughout the PV plant certain conventions have to be introduced. Electrical parameter monitoring equipment, such as multimeters, protection devices and the inverters themselves follow certain rules regarding the signs of the active and reactive power and the power factor measurements they provide (the apparent power is always positive). There exist 2 sets of such rules, one followed by IEC-compliant devices (European standards body) and one by IEEE-compliant devices (US standards body). Their differences are presented in the figure below.

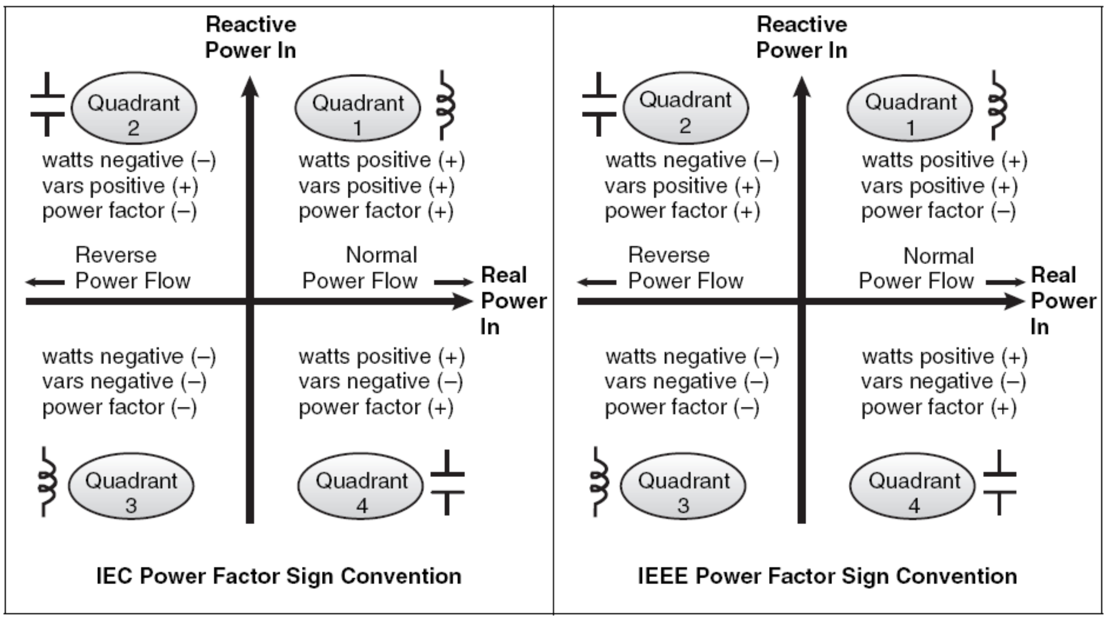


**Figure 10. The IEC and IEEE power factor sign conventions**

The inSolar solution adheres to the IEC power factor sign convention (left hand side of the figure above), provided that field-deployed equipment also support it. The definition of the power flow as normal or reverse depends on the entity being monitored. Referring back to Figure 1, these entities may be:

- The PV arrays/array groups
- The TCPs
- The PCC
- The LCP

For monitored entities along the production hierarchy (PV arrays/array groups, TCPs and the PCC), let us define as “normal” the power flow from the PV plant to the EPS. This means that during the day, when the PV plant produces electrical energy, the power and the power factor for the abovementioned monitored points are recorded and presented to the user as positive numbers, while during the night, these 2 numbers are negative. The sign of reactive power can be positive or negative irrespective of whether the PV plant is producing or not. The energy (either active or reactive) provided by the PV plant to the EPS, which is equal to the integral of the respective power over time is denoted as “export energy”, whilst the energy provided by the EPS to the PV plant is denoted as “import energy”. As far as the LCP is concerned, the power flow towards the loads is denoted as “normal” and the energy consumed by the loads (either active or reactive) as “import” energy.

The concerned reader has to be informed that the orientation of the coils used by metering equipment to sense the direction of the current in the power lines affects the definition of the power flow as “normal” or “reverse” and unfortunately, the installers do not always follow the rules. The inSolar solution provides the same user experience, regardless of the orientation of the current sensing coils.

# 5 The inSolar solution service architecture

PV plants are almost always placed at hard to reach areas, away from terrestrial communication infrastructures. This is why data communication to/from the field has to be carried out using mostly either the GSM/3G network or satellite means. Both share the common characteristic of scarce bandwidth and/or occasional service unavailability. Data reduction and local data storage in the field is thus a mandatory requirement. Additionally, due to the vast variety of equipment deployed in the plant, it is expected that the inSolar solution has to interface them using an equivalent variety of physical connection methods and protocols. Finally, there are certain requirements that mandate the existence of logic in the field, which operates irrespective of whether a connection to the outside world is operational or not.

The 3 abovementioned facts intonate the need for a stand-alone infrastructure, installed at the field which undertakes the responsibilities of:

- Interfacing with all equipment installed at the PV plant (multimeters, protection devices, inverters, sensors, actuators etc) to acquire data as per section 4.2, irrespective of the connection method and/or protocol employed by the external device
- Reducing the volume of data to something representative, yet less bandwidth hungry. This function is essentially the primary and maximum/minimum parameter calculation (sections 4.3 & 4.4)
- Conveying primary state-change events that are generated directly by field equipment upon their occurrence
- Generating primary state-change events whenever certain conditions on real-time data are met (section 4.2, a typical example is the temperature thresholding for modules, see section 6.2.9.2) and conveying them upon their occurrence
- Providing access to the real-time data of section 4.9 to enable real-time monitoring of parameters
- Storing all necessary parameters and events in a cyclical manner in the case of failure of the communication with the outside world
- Implementing all necessary control loops that have to operate locally and in an independent manner

This infrastructure is essentially a set of programmable networked controllers employing all necessary interfacing and storage functionality deployed at each PV plant separately.

Continuing with this approach on the service architecture, a central infrastructure is necessary in order for the users to monitor and control a number of PV plants in a centralized manner. This means that data from a large number of PV plants have to be concentrated, stored and managed by a single entity, which is referred to as the service back-end. The responsibilities of this back-end are:

- To collect, store and manage information (parameters, raw real-time data, state-change events, status, event acknowledgements and general/system data) from a variety of PV plants up to the available level of detail
- To relay real-time monitored parameters to the user (such parameters are not stored and further processed)
- To implement the hierarchical status propagation tree, presented in section 4.10
- To store and manage information regarding the users accessing the service

- To generate and send notifications towards users
- To generate and send reports towards users
- To generate exports of data for the user
- To allow the users to configure the inSolar solution via the application of settings either to the service back-end itself or the field-deployed controllers
- To calculate the derived parameters and state-sets presented in the present document. This is not mandatorily a function that belongs to the service back-end, since the field-deployed controller could carry out this task. However, this is not the optimal approach for a variety of reasons:
    - The necessary communication bandwidth must be increased to cater for a bigger volume of data from the PV plant, since it is considered mandatory to store primary parameters and state-sets as well for a variety of reasons:
        - User access to the primary parameters
        - Direct comparison of primary and derived parameters
        - Further derived parameter & state-set calculations (involving primary parameters & state-sets) to be carried out by the service back-end and not the field-deployed controller
    - The implementation of additional derived parameter calculations shouldn't imply an update in the firmware of field-deployed controllers

The interface among the service back-end and the cloud of field-deployed controllers (and consequently the PV plant equipment) is now clear and it has to allow for the transmission of:

- PV plant to service back-end direction:
    - Recording period primary parameters and raw real-time data, conveyed using a schedule aligned with the recording period
    - Primary state-change events, conveyed either in a scheduled manner or upon their occurrence
    - Real-time parameters, conveyed as they are collected from the field upon user request
- Service back-end to PV plant direction:
    - Settings, conveyed upon change

As far as the field-deployed controllers are concerned, inAccess normally uses their RSC product family. However, it is also understandable that the service back-end is supported to provide for interoperation with a variety of third-party controllers and PLCs. For this reason, an independent module, denoted as the Communications Adapter Module (CAM), has been implemented, which undertakes the task of acquiring data from the remote PV plant and feeding them to the service back-end as well as conveying data from the service back-end to the remote PV plant.

The inSolar solution is, of course complemented by a proper front-end, capable of operation with a standard Web browser (Internet explorer, Mozzila Firefox or Google Chrome), which provides users with access to the services and the data managed by the service back-end. The architecture is presented in the figure below.

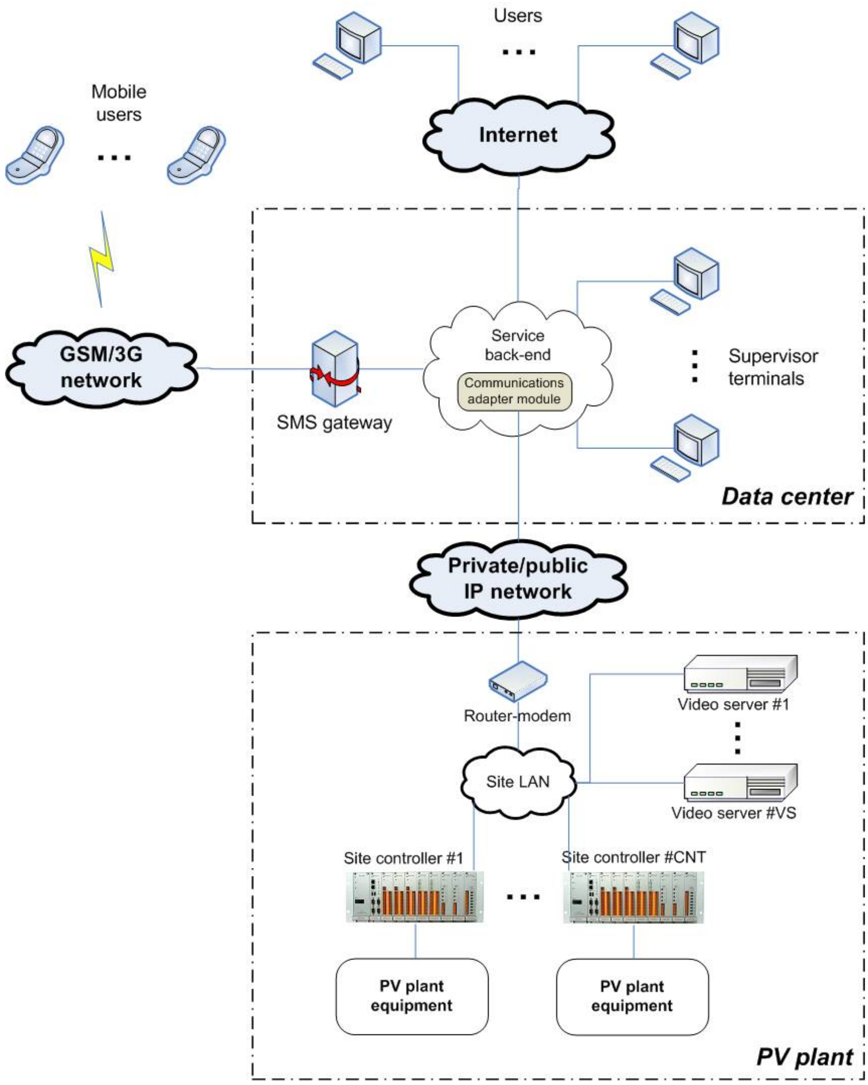


**Figure 11. The inSolar solution service architecture**

It has to be noted that the video servers presented in the figure above are part of the video surveillance system of Figure 1, which is outside the scope of the present document. However, since they may be networked equipment using the same IP connection between the service back-end and the PV plant, they are also monitored by the inSolar solution.

All real-time data, primary parameters, max/min values, derived parameters, state-change events and settings refer to the same timebase, irrespective of the PV plant they are related with. Moreover, the limits of a recording period, a day, a month and a year coincide, at least within 10 seconds, provided that field-deployed equipment remain in synchrony with the central office infrastructure. All timestamps reported to the user for a specific PV plant always refer to the timezone of that PV plant.

# 6 The monitoring service

The main duty of the inSolar solution is to provide, in a timely and concise manner, information to its user on the operation of a group of PV plants. The monitoring service functionality can be broadly categorized into 6 categories:

- Monitoring of the weather conditions at the PV plant
- Monitoring of the production of the PV plant
- Monitoring of the PV plant electrical installation (PCC, LCP, TCPs and electrical panels)
- Monitoring of the PV plant building status
- Monitoring of the PV plant perimeter status
- Monitoring of the PV plant equipment operational and communication status

The following sections present in a structured manner, the monitored parameters, state-sets as well as settings and establish relations among them in order to aid the users in the understanding of the operation of the inSolar solution.

## *6.1 Monitoring of the weather conditions at the PV plant*

Several parameters, representative of the weather conditions at the PV plant are recorded by the inSolar solution:

- The total irradiance, at the horizontal plane
- The ambient air temperature
- The wind speed
- The wind direction
- The rain precipitation
- The barometric pressure
- The relative humidity

### 6.1.1 Total irradiance, at the horizontal plane

The total irradiance (at the horizontal plane), denoted as *G* in the ensuing, is measured in $W \bullet m^{-2}$ as per section 4.1 of Reference 1. According to Reference 1, the irradiance is measured through an irradiance sensor (pyranometer) placed horizontally. This sensor is part of the weather stations presented in Figure 1, so the total irradiance is recorded for each weather station separately. If only one weather station exists, this parameter is recorded only for the PV plant entity.

Daily, monthly and yearly total irradiance (in the horizontal plane) parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$G_{T,\tau} = \frac{\sum_{t=T-\tau}^{T} G_t}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $G_t$ is a vector of the values of the recording period total irradiance primary parameter spanning the reporting period. If the irradiance value provided by the pyranometer is below a certain threshold which is configurable per pyranometer, the value of the primary parameter is set to zero by the inSolar solution.
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

### 6.1.2 Ambient air temperature

The ambient air temperature, denoted as $T_a$ in the ensuing, is measured in degrees Celsius (°C) as per section 4.2 of Reference 1. According to Reference 1, the ambient air temperature is measured through the use of a temperature sensor/trnasducer. This sensor is part of the weather stations presented in Figure 1, so the ambient air temperature is recorded for each weather station separately. The respective recording period primary parameter is calculated as per section 4.3 of the present document. If only one weather station exists, this parameter is recorded only for the PV plant entity.

Daily, monthly and yearly ambient air temperature parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$T_{a,T,\tau} = \frac{\sum_{t=T-\tau}^{T} T_{a,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $T_{a,t}$ is a vector of the values of the recording period ambient air temperature primary parameter spanning the reporting period
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

### 6.1.3 Wind speed

The wind speed, denoted as $S_w$ in the ensuing, is measured in $m \cdot s^{-1}$ as per section 4.3 of Reference 1. According to Reference 1, the wind speed is measured through the use of a wind speed sensor. This sensor is part of the weather stations presented in Figure 1, so the wind speed is recorded for each weather station separately. The respective recording period primary parameter is calculated as per section 4.3 of the present document. If only one weather station exists, this parameter is recorded only for the PV plant entity.

Daily, monthly and yearly wind speed parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$S_{w,T,\tau} = \frac{\sum_{t=T-\tau}^{T} S_{w,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $S_{w,t}$ is a vector of the values of the recording period wind speed primary parameter spanning the reporting period
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

### 6.1.4 Wind direction

The wind direction, denoted as $D_w$ in the ensuing, is measured in azimuth degrees (degrees from the north). The wind direction is measured through the use of a wind direction sensor. This sensor is part of the weather stations presented in Figure 1, so the wind direction is recorded for each weather station separately. The respective recording period primary parameter is calculated as per section 4.3 of the present document. If only one weather station exists, this parameter is recorded only for the PV plant entity.
Daily, monthly and yearly wind direction parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$D_{w,T,\tau} = \frac{\sum_{t=T-\tau}^{T} D_{w,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $D_{w,t}$ is a vector of the values of the recording period wind direction primary parameter spanning the reporting period
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

 

### 6.1.5 Rain precipitation

The rain precipitation, denoted as $P_r$ in the ensuing, is measured in mm/h. The rain precipitation is measured through the use of a rain gauge sensor. This sensor is part of the weather stations presented in Figure 1, so the rain precipitation is recorded for each weather station separately. The respective recording period primary parameter is calculated as per section 4.3 of the present document. If only one weather station exists, this parameter is recorded only for the PV plant entity.
Daily, monthly and yearly rain precipitation parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$P_{r,T,\tau} = \frac{\sum_{t=T-\tau}^{T} P_{r,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $P_{r,t}$ is a vector of the values of the recording period rain precipitation primary parameter spanning the reporting period
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

### 6.1.6 Barometric pressure

The barometric pressure, denoted as $P$ in the ensuing, is measured in kPa. The barometric pressure is measured through the use of a barometric pressure sensor. This sensor is part of the weather stations presented in Figure 1, so the barometric pressure is recorded for each weather station separately. The respective recording period primary parameter is calculated as per section 4.3 of the present document. If only one weather station exists, this parameter is recorded only for the PV plant entity.
Daily, monthly and yearly barometric pressure parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$P_{T,\tau} = \frac{\sum_{t=T-\tau}^{T} P_t}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out

- $P_t$ is a vector of the values of the recording period barometric pressure primary parameter spanning the reporting period
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

### 6.1.7 Relative humidity

The relative humidity, denoted as *RH* in the ensuing, is measured in %. The relative humidity is measured through the use of a relative humidity sensor. This sensor is part of the weather stations presented in Figure 1, so the relative humidity is recorded for each weather station separately. The respective recording period primary parameter is calculated as per section 4.3 of the present document. If only one weather station exists, this parameter is recorded only for the PV plant entity.

Daily, monthly and yearly relative humidity parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$RH_{T,\tau} = \frac{\sum_{t=T-\tau}^{T} RH_t}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $RH_t$ is a vector of the values of the recording period relative humidity primary parameter spanning the reporting period
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

### 6.1.8 Average spatial values of environmental parameters

If multiple weather stations exist at a PV plant, the spatial averages of the environmental parameters are calculated by the inSolar solution and recorded for the PV plant entity. These average spatial parameters are calculated using the formula below:

$$ENV_T = \frac{\sum_{i=1}^{WE} ENV_i}{\sum_{i=1}^{WE} V_i}$$

where:

- $T$ denotes the recording period during which the calculation is carried out

- *WE* is the number of weather stations in the PV plant
- $ENV_i$ is a vector of the values of the respective recording period environmental primary parameter (total irradiance, ambient air temperature etc.) from all weather stations of the PV plant for the recording period during which the calculation is carried out
- $V_i$ is a data validity vector of length equal to the number of weather stations in the PV plant, containing ones for the parameters containing valid values and zeros otherwise

Daily, monthly and yearly spatial average parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$ENV_{T,\tau} = \frac{\sum_{t=T-\tau}^{T} ENV_t}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $ENV_t$ is a vector of the values of the respective recording period spatial average of the environmental parameter (total irradiance, ambient air temperature etc.) spanning the reporting period
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

### 6.1.9 Primary state-sets related to weather conditions

The current state of these state-sets is updated directly depending on the output of the weather station sensors. The real-time data used for the calculation of the primary parameters related to weather conditions presented in the section above is thresholded against limits treated as settings by the inSolar solution and the respective states are updated correspondingly. More specifically, there exist 2 state-sets for each weather station, encoding:

- The ambient temperature status with the following 3 states (high/low threshold settings are supported):
  - Temperature normal
  - Temperature low
  - Temperature high
- The wind speed status with the following 2 states (a high threshold setting is supported):
  - Wind-speed normal
  - Wind-speed high

If only one weather station exists, these state-sets are supported only for the PV plant entity.

## 6.2 Monitoring of the PV plant production

A number of parameters and state sets related to the PV plant production are monitored by the inSolar solution in order to satisfy the requirements of IEC 61724 and other customer requirements.

### 6.2.1 Monitoring of environmental parameters related to the PV plant production

There are certain environmental parameters which are necessary for the calculation of a multitude of performance metrics of a variety of PV plant components and other purposes. These are the total irradiance, in the plane of the array and the temperature of the PV panel modules.
2 additional parameters that are monitored are the panel group tracker tilt/azimuth angles, though they are not used in any additional calculations. Finally, the solar energy derived parameter is calculated by the inSolar solution.

#### 6.2.1.1 Total irradiance, in the plane of the array

The total irradiance, denoted as $G_I$ in the ensuing, is measured in W•m$^{-2}$ as per section 4.1 of Reference 1. It has to be noted that within Reference 1, the total irradiance is defined only for PV panel groups, but this definition has been extended to PV arrays, PV array groups and the PV plant as a whole in the ensuing.
According to Reference 1, the irradiance is measured through an irradiance sensor (pyranometer) placed at the inclination of the PV panels. Since pyranometers are very costly devices, it is understandable that the PV plant owners ideally want to use just one of them. In small PV plants, this is feasible but if the area occupied by the PV panels is large, multiple pyranometers (pyranometer #T.K.N.L of Figure 1) shall have to be installed in order to properly capture and treat phenomena such as partial shading of a portion of the PV plant by a cloud. In the former case, the single recording period irradiance reading has to be used for all calculations; in the latter, the multiple recording period readings have to be used so that the accuracy of the calculations of the performance metrics is maximized.
Referring back to Figure 1, the recording period total irradiance for each PV panel group may be monitored in 2 different ways:

1. If the PV panel group has a pyranometer associated with it, the recording period primary parameter value of the total irradiance this pyranometer measures is used. In that case, the recording period total irradiance is considered a primary parameter, calculated according to section 4.3 of the present document. This parameter is also real-time monitored as per 4.9.
2. If a PV panel group has no pyranometer associated with it, the average of recording period total irradiance primary parameters from selected pyranometers is used. These selected pyranometers may be those that belong to neighboring PV panel groups, the site pyranometer or a combination of these sources. In that case, the recording period total irradiance is considered a derived parameter.

Which pyranometers belong to which PV panel group and which recording period total irradiance readings are averaged for each PV panel group that does not incorporate a pyranometer is determined at installation time.

The recording period total irradiance for each PV array is equal to the average of the recording period total irradiance parameter values of all PV panel groups belonging to the specific PV array, irrespective of whether they are primary or derived. The recording period total irradiance for each PV array group is equal to the average of the recording period total irradiance parameter values of all PV arrays belonging to the specific PV array group.
The recording period total irradiance for the PV plant may be monitored in 2 different ways:

1. If the PV plant has a single pyranometer associated with it (the site pyranometer), the recording period primary parameter value of the total irradiance this pyranometer measures is used. In that case, the recording period total irradiance is considered a primary parameter, calculated according to section 4.3 of the present document. This parameter is also real-time monitored as per 4.9.
2. If the PV plant has multiple pyranometers associated with it, the spatial average of recording period total irradiance primary parameters from all pyranometers is used. In that case, the recording period total irradiance is considered a derived parameter.

Whenever spatial averaging across pyranometers is employed to calculate the respective derived parameter, non-available data do not take part in the procedure and if no input data are available for a specific recording period, the respective derived parameter value is also unavailable for that specific recording period.
Daily, monthly and yearly total irradiance parameter values, irrespective of the entity they refer to, are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$G_{I,T,\tau} = \frac{\sum_{t=T-\tau}^{T} G_{I,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $G_{I,t}$ is a vector of the values of the recording period total irradiance primary/derived parameter spanning the reporting period, depending on whether the daily/monthly/yearly derived parameter is calculated for a specific PV array, PV array group or the PV plant as a whole. If the primary irradiance value provided by the pyranometer is below a certain threshold which is configurable per pyranometer, this value is set to zero by the inSolar solution.
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

### *6.2.1.2 Module temperature*

The module temperature, denoted as $T_m$ in the ensuing, is measured in degrees Celsius (°C) as per section 4.4 of Reference 1. It has to be noted that within Reference 1, the module

temperature is defined only for PV panel groups, but this definition has been extended to PV arrays, PV array groups and the PV plant as a whole in the ensuing.
According to Reference 1, the temperatures of the modules of the PV panels are measured through the use of a temperature sensor placed on the back of the PV panels (temperature sensor #T.K.N.L of Figure 1). Generally, the number of temperature sensors is expected to be smaller than the number of PV panel groups, so the values from existing temperature sensors have to be used to fill in values for the PV panel groups which do not incorporate one in a way that the accuracy of the calculations of the performance metrics is maximized.
Referring back to Figure 1, the recording period module temperature for each PV panel group may be monitored in 2 different ways:

1. If the PV panel group has a temperature sensor associated with it, the recording period primary parameter value of the temperature this sensor measures is used. In that case, the recording period module temperature is considered a primary parameter, calculated according to section 4.3 of the present document. This parameter is also real-time monitored as per 4.9.
2. If a PV panel group has no temperature sensor associated with it, the average of recording period module temperature primary parameters from selected temperature sensors is used. These selected sensors are those that belong to neighboring PV panel groups. In that case, the recording period module temperature is considered a derived parameter.

Which temperature sensors belong to which PV panel group and which recording period temperature readings are averaged for each PV panel group that does not incorporate a temperature sensor is determined at installation time.
The recording period module temperature for each PV array is equal to the average of the recording period module temperature parameter values of all PV panel groups belonging to the specific PV array, irrespective of whether they are primary or derived. The recording period module temperature for each PV array group is equal to the average of the recording period module temperature parameter values of all PV arrays belonging to the specific PV array group. The recording period module temperature for the PV plant is equal to the average of the recording period module temperature parameter values of all PV array groups belonging to the plant.
Whenever spatial averaging across temperature sensors is employed to calculate the respective recording period derived parameter, non-available data do not take part in the procedure and if no input data are available for a specific recording period, the respective derived parameter value is also unavailable for that specific recording period.
Daily, monthly and yearly module temperature parameter values, irrespective of the entity they refer to, are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$T_{m,T,\tau} = \frac{\sum_{t=\mathrm{T}-\tau}^{T} T_{m,t}}{\sum_{t=\mathrm{T}-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $T_{m,t}$ is a vector of the values of the recording period module temperature primary/derived parameter spanning the reporting period, depending on whether the daily/monthly/yearly derived parameter is calculated for a specific PV array, PV array group or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

### 6.2.1.3 Parameters provided by reference cells

2 environmental parameters related to the PV plant production are provided by reference cells (possibly multiple) installed at the PV plant. These are the reference cell irradiance and the reference cell temperature.

#### 6.2.1.3.1 Reference cell irradiance

The reference cell irradiance, denoted as $G_{rc}$ in the ensuing, is measured in W•m$^{-2}$ as per section 4.1 of Reference 1. Irradiance readings are recorded for each reference cell separately. If only one reference cell exists, this parameter is recorded only for the PV plant entity.
Daily, monthly and yearly reference cell irradiance parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$G_{RC,T,\tau} = \frac{\sum_{t=T-\tau}^{T} G_{RC,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $G_{RC,t}$ is a vector of the values of the recording period reference cell irradiance primary parameter spanning the reporting period.
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

#### 6.2.1.3.2 Reference cell temperature

The reference cell temperature, denoted as $T_{rc}$ in the ensuing, is measured in degrees Celsius (ºC) as per section 4.4 of Reference 1. Cell temperature readings are recorded for each reference cell separately. If only one reference cell exists, this parameter is recorded only for the PV plant entity.
Daily, monthly and yearly reference cell temperature parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$T_{RC,T,\tau} = \frac{\sum_{t=T-\tau}^{T} T_{RC,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)

- *T* denotes the recording period during which the calculation is carried out
- $T_{RC,t}$ is a vector of the values of the recording period reference cell temperature primary parameter spanning the reporting period.
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

### 6.2.1.3.3 Average spatial values of parameters provided by reference cells

If multiple reference cells exist at a PV plant, the spatial averages of the parameters are calculated by the inSolar solution and recorded at the PV plant entity. These average spatial parameters are calculated using the formula below:

$$RC_T = \frac{\sum_{i=1}^{RC} RC_i}{\sum_{i=1}^{RC} V_i}$$

where:

- *T* denotes the recording period during which the calculation is carried out
- *RC* is the number of reference cells in the PV plant
- $RC_i$ is a vector of the values of the respective recording period primary parameter provided by the reference cell (cell irradiance, cell temperature etc.) from all reference cells of the PV plant for the recording period during which the calculation is carried out
- $V_i$ is a data validity vector of length equal to the number of reference cells in the PV plant, containing ones for the parameters containing valid values and zeros otherwise

Daily, monthly and yearly spatial average parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$RC_{T,\tau} = \frac{\sum_{t=T-\tau}^{T} RC_t}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- *T* denotes the recording period during which the calculation is carried out
- $RC_t$ is a vector of the values of the respective recording period spatial average of the parameter provided by the reference cell (cell irradiance, cell temperature etc.) spanning the reporting period

- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

### 6.2.1.4 Panel group tracker tilt/azimuth angle

Tracker angles are optional for systems with tracking PV panel groups. For single axis trackers, the tracker tilt angle is used to describe the position of the panel group about its tracking axis. For example, for a horizontal single axis tracker this parameter would give the angle from horizontal, east is negative and west is positive. These angles are measured directly from the tracker and the related recording period primary parameters are calculated according to section 4.3 of the present document. The units for panel group tracker tilt and azimuth angles, denoted as $\varphi_T$ and $\varphi_A$ in the ensuing, are degrees (°). The tracker tilt/azimuth angles are monitored only at the PV panel group entity.
Daily, monthly and yearly tracker tilt and azimuth angle parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$\varphi_{i,T,\tau} = \frac{\sum_{t=T-\tau}^{T} \varphi_{i,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $\varphi_{i,t}$ is a vector of the values of the recording period tracker tilt/azimuth angle primary parameter spanning the reporting period ($i$ = $T$ or $A$ respectively)
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

These parameters are also real-time monitored as per 4.9.

### 6.2.1.5 Solar energy

The solar energy is calculated according to the formula of section 8.1 of Reference 1 replicated below for PV arrays, PV array groups and the PV plant as a whole, taking into consideration the completeness threshold of section 4.5:

$$H_{I,T,\tau} = \frac{\tau_r \times \sum_{t=T-\tau}^{T} G_{I,t}}{1000} \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $G_{I,t}$ is a vector of the values of the proper recording period primary/derived parameter as per section 6.2.1.1, which spans the reporting period, depending on whether the solar energy is calculated for a specific PV array, PV array group or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The recording period solar energy values are calculated only in a day-cumulative manner. The solar energy units are $kWh \bullet m^{-2} \bullet d^{-1}$ if the solar energy is calculated for the present day or any past days, $kWh \bullet m^{-2} \bullet mo^{-1}$ if the solar energy is calculated for a month and $kWh \bullet m^{-2} \bullet y^{-1}$ if the solar energy is calculated for a year. The divisor of 1000 is inserted to convert $W \bullet m^{-2}$ (as provided by the pyranometers) into $kW \bullet m^{-2}$.

### 6.2.2 Monitoring of the flow of active electrical power

One of the most important functions of the inSolar solution is the monitoring of the flow of electrical power from the output of the PV panels towards the utility grid and the loads. The present section refers only to the active power component, since it is the one characterizing the production chain and generating income.

#### *6.2.2.1 Parameters related to active electrical power monitored through the inverter*

Certain parameters related to electrical power are monitored through the inverter and this poses a specific problem, when dealing with a wide range of inverters, independent of the manufacturer. During the night, these components shut down, since they are powered by the sun, so communication with them is interrupted and there is no way of acquiring the measurements provided by them. Due to this phenomenon, the recording period power values have to be corrected by replacing unavailable values with zero.

##### ***6.2.2.1.1 PV panel group output power***

PV panels produce DC power whenever the irradiance is above a certain level. This power is not constant but rather analogous to the irradiance levels. The PV panel group output power can generally be measured in 2 ways:

1. Through the inverter of the PV array the PV panel group belongs to (either per PV panel group, if supported, or in a combined manner)
2. Through the dedicated string monitors depicted in Figure 1 (per PV panel group). The strings communicate with the inSolar solution via the inverter.

The PV panel group power recording period primary parameter, denoted as $P_{A,orig}$ in the ensuing (units: kW), is measured and/or calculated as per section 4.6 of Reference 1, section 4.12 and section 4.3 of the present document.

Whenever the PV panel group power can be measured independently per PV panel group, it is recorded for the PV panel group entity as a recording period primary parameter. In the case the PV panel group output power can be measured only per PV array, it is recorded at each PV array entity as a recording period primary parameter.

Then, the recording period PV panel group output power primary parameter (irrespective of the entity for which it has been recorded) is corrected according to the methodology described above, yielding the recording period derived parameter $P_A$, which is used in subsequent calculations.

In the case the PV panel group output power is recorded at the PV panel group entity, the respective PV array entity recording period parameter os calculated as the sum of the respective corrected parameter values of the PV panel groups belonging to the specific PV array as a derived parameter.

Referring back to Figure 1, if the recording period PV panel group output power is calculated for a specific PV array group, it is equal to the sum of corrected PV panel group output power recording period parameters of PV arrays belonging to this specific PV array group (either primary or derived). If the recording period PV panel group output power is calculated for the whole of the PV plant, it is equal to the sum of the PV panel group output power recording period parameters of all PV array groups belonging to the plant.

 

Whenever a summation is employed to calculate the respective derived parameter, all input data have to present available data and if input data are missing for at most one PV panel group for a specific recording period, the respective derived parameter value is also unavailable for that specific recording period.
Daily, monthly and yearly PV panel group output power parameter values, irrespective of the entity they refer to, are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$P_{A,T,\tau} = \frac{\sum_{t=T-\tau}^{T} P_{A,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $P_{A,t}$ is a vector of the values of the proper recording period corrected PV panel group output power derived parameter spanning the reporting period, depending on whether the daily/monthly/yearly derived parameter is calculated for a specific PV array, PV array group or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

### *6.2.2.1.2 PV array output power*

Inverters convert the DC power that is output by the PV panels to AC power which is, in turn injected in the utility grid either directly or through step-up transformers. A certain amount of the input power is dissipated as heat. Since the power factor of the inverters is close to unity, the whole of the power they produce can be considered as active. In the case of 3-phase inverters, the total active power across phases is monitored for each PV array depicted in Figure 1.
The PV array output power recording period primary parameter, denoted as $P_{AC,orig}$ in the ensuing (units: kW), is measured through the inverter and/or calculated as per sections 4.12 and 4.3 of the present document. It has to be noted that Reference 1 does not define this parameter, since it considers only 1 inverter per PV plant.
Then, the recording period PV array output power primary parameter is corrected according to the methodology described above, yielding the recording period derived parameter $P_{AC}$, which is used in subsequent calculations.
Referring back to Figure 1, if the recording period PV array output power is calculated for a specific PV array group, it is equal to the sum of corrected recording period PV array output power parameters of PV arrays belonging to this specific PV array group. If the recording period PV array output power is calculated for the whole of the PV plant, it is equal to the sum of the recording period PV array output power parameters of all PV array groups belonging to the plant.

Whenever a summation is employed to calculate the respective derived parameter, all input data have to present available data and if input data are missing for at most one PV array for a specific recording period, the respective derived parameter value is also unavailable for that specific recording period.

Daily, monthly and yearly PV array output power parameter values, irrespective of the entity they refer to, are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$P_{AC,T,\tau} = \frac{\sum_{t=T-\tau}^{T} P_{AC,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $P_{AC,t}$ is a vector of the values of the proper recording period corrected PV array output power derived parameter spanning the reporting period, depending on whether the daily/monthly/yearly derived parameter is calculated for a specific PV array, PV array group or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

This parameter is also real-time monitored as per 4.9.

### *6.2.2.2 Parameters related to active electrical power monitored through other devices*

If the parameters related to active electrical power are monitored through dedicated equipment such as multimeters and protection devices, problematic operation during the night is not an issue, so the related correction is not applied.

#### ***6.2.2.2.1 Transformer input active power***

Transformers convert the output voltage of the PV array (which is 220 VAC) to a higher voltage as mandated by the EPS. A certain amount of the power input to the transformers is lost as heat. If the PV plant is directly connected to the utility grid, there exist no transformers. As far as production monitoring is concerned, only the active component of the transformer input power needs to be monitored as a recording period primary parameter for each transformer depicted in Figure 1, though in a total manner and not per phase.

The transformer input active power is measured through the transformer LV multimeter according to sections 6.3.2, 4.12 and 4.3 of the present document. The units for the active power at the transformer input, denoted as $P_{AC,transf,in}$ in the ensuing, are kW. It has to be noted that Reference 1 does not define this parameter, since it does not consider transformers.

Referring back to Figure 1, if the recording period transformer input power is calculated for the PV plant as a whole, it is equal to the sum of recording period transformer input power parameters of all transformers belonging to the PV plant. Non-available data do not take part in the summation procedure and if no input data are available for a specific recording period, the respective derived parameter value is also unavailable for that specific recording period.

Daily, monthly and yearly transformer input power parameter values, irrespective of the entity they refer to, are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$P_{AC,transf,in,T,\tau} = \frac{\sum_{t=T-\tau}^{T} P_{AC,transf,in,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $P_{AC,transf,in,t}$ is a vector of the values of the proper recording period transformer input power primary/derived parameter spanning the reporting period, depending on whether the daily/monthly/yearly derived parameter is calculated for a specific transformer or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

### 6.2.2.2.2 Transformer output active power

The transformer output power is measured through the transformer HV multimeter according to sections 6.3.2, 4.12 and 4.3 of the present document. As far as production monitoring is concerned, only the active component of the transformer output power needs to be monitored as a recording period primary parameter for each transformer depicted in Figure 1, though in a total manner and not per phase. The units for the active power at the transformer output, denoted as $P_{AC,transf,out}$ in the ensuing, are kW. It has to be noted that Reference 1 does not define this parameter, since it does not consider transformers.

Referring back to Figure 1, if the recording period transformer output power is calculated for the PV plant as a whole, it is equal to the sum of recording period transformer output power parameters of all transformers belonging to the PV plant. Non-available data do not take part in the summation procedure and if no input data are available for a specific recording period, the respective derived parameter value is also unavailable for that specific recording period. The transformer output active power can also be recorded as a primary parameter for the PV plant entity (output of all inverters) via the PCC protection device presented in Figure 1 as per section 6.3.1.

Daily, monthly and yearly transformer output power parameter values, irrespective of the entity they refer to, are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$P_{AC,transf,out,T,\tau} = \frac{\sum_{t=T-\tau}^{T} P_{AC,transf,out,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $P_{AC,transf,out,t}$ is a vector of the values of the proper recording period transformer output power primary/derived parameter spanning the reporting period, depending on whether the daily/monthly/yearly derived parameter is calculated for a specific transformer or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

### *6.2.2.2.3 Active power towards the loads*

Auxiliary equipment (lighting, air-conditioning and others) always consume power, so part of the active power produced during the day goes towards them. During the night, active power coming from the utility grid is consumed in order for the loads to operate. It has to be noted that the load transformer, should there exist one, is also considered a load. As far as production monitoring is concerned, only the active component of the active power towards the loads needs to be monitored as a recording period primary parameter at the load connection point (see section 6.3.3) as well as at the PV plant entity depicted in Figure 1, though in a total manner and not per phase.

The active component of the power towards the loads is measured by the load multimeter depicted in Figure 1 and/or calculated as per section 4.6 of Reference 1 as well as sections 4.12, 6.3.3 and 4.3 of the present document. The units for active load power, denoted as $P_L$ in the ensuing, are kW.

Daily, monthly and yearly load power parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$P_{L,T,\tau} = \frac{\sum_{t=T-\tau}^{T} P_{L,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $P_{L,t}$ is a vector of the values of the recording period primary parameter of the active power towards the loads spanning the reporting period

- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

#### 6.2.2.2.4 Active power to/from the utility grid

During the day, the PV plant produces active power which is fed, partly to the PV plant loads and partly to the utility grid. During the night since the production falls to zero, loads consume power from the utility grid, so there is an inflow to the plant. As far as production monitoring is concerned, only the active component of the active power to/from the utility grid needs to be monitored as a recording period primary parameter at the Point of Common Coupling (see section 6.3.1) as well as at the PV plant entity depicted in Figure 1, though in a total manner and not per phase.

The active components of the power to/from the utility grid are measured by the PCC multimeter depicted in Figure 1 and/or calculated as per section 4.6 of Reference 1 as well as sections 4.12, 6.3.1 and 4.3 of the present document. The units for active power to/from the utility network, denoted as $P_{TU}$ and $P_{FU}$ in the ensuing, are kW.

Daily, monthly and yearly power to/from the utility grid parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$P_{TU/FU,T,\tau} = \frac{\sum_{t=T-\tau}^{T} P_{TU/FU,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $P_{TU/FU,t}$ is a vector of the values of the proper recording period primary parameter of the active power to/from the utility network spanning the reporting period
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

### 6.2.3 Monitoring of electrical voltage and current

Electrical voltage and current are monitored by the inSolar solution at various points of the electrical installation presented in Figure 1. Though these parameters are not used in any subsequent calculations according to Reference 1, they are extremely useful for monitoring the proper operation of the PV plant and carry out troubleshooting.

#### *6.2.3.1 Parameters related to electrical voltage/current monitored through the inverter*

Certain parameters related to electrical voltage and current are monitored through the inverter and this poses the problem already described within section 6.2.2.1. Though it makes sense to replace missing power measurements with zero, this approach is not feasible for voltage and current.

The daily, monthly and yearly averages of voltage and current related parameters cannot be calculated since the completeness threshold of section 4.7 is never fulfilled (the period during which an inverter is operational is always smaller than 75% of the 24 hours of any given day).

##### ***6.2.3.1.1 PV panel group output voltage & current***

The PV panel group output voltage and current (DC quantities) are generally measured as per section 4.5 of Reference 1 and section 4.3 of the present document in 2 ways:

1. Through the inverter of the PV array the PV panel group belongs to (either per PV panel group, if supported, or in a combined manner)
2. Through the dedicated string monitors depicted in Figure 1 (per PV panel group)

The units for PV panel group output voltage and current are Volts DC and Amps respectively.

Whenever the PV panel group output voltage and current can be measured independently per PV panel group, they are recorded for the PV panel group entity as recording period primary parameters. In the case the PV panel group output voltage and current can be measured only per PV array, they are recorded for the PV array entity as recording period primary parameters. This voltage and current are denoted as $V_A$ and $I_A$ in the ensuing. Daily, monthly and yearly voltage/current parameter values are not supported according to the rationale presented in the introduction. Both parameters are also real-time monitored as per 4.9.

##### ***6.2.3.1.2 PV array output voltage & current***

The PV array output voltage and current (AC quantities) are measured as per section 4.5 of Reference 1 and section 4.3 of the present document through the inverter and are recorded as recording period primary parameters for the PV array entity. In the case of multi-phase inverters, both the voltage and current are recorded per phase. This voltage and current are denoted as $V_{AC}$ and $I_{AC}$ in the ensuing. The units for PV array group output voltage and current are Volts AC and Amps respectively. Daily, monthly and yearly voltage/current parameter values are not supported according to the rationale presented in the introduction. Both parameters are also real-time monitored as per 4.9.

### *6.2.3.2 Parameters related to voltage/current monitored through other devices*

If the parameters related to voltage and current are monitored through dedicated equipment such as multimeters and protection devices, problematic operation during the night is not an issue, so daily/monthly/yearly values can be calculated.

#### *6.2.3.2.1 Transformer input/output voltage & current*

The transformer input/output voltage and current (AC quantities) are measured as per section 4.5 of Reference 1 and section 4.3 of the present document through the transformer connection point multimeters (LV/HV) depicted in Figure 1 and are recorded as primary parameters for the TCP entity per phase. The units for transformer input/output voltage and current (denoted as $V_{AC,transf,in/out}$ and $I_{AC,transf,in/out}$ in the ensuing) are Volts AC and Amps respectively. For further information regarding the monitoring of the transformer connection point, please consult section 6.3.2.
Daily, monthly and yearly voltage/current parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$V/I_{AC,transf,in/out,T,\tau} = \frac{\sum_{t=T-\tau}^{T} V/I_{AC,transf,in/out,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $V/I_{AC,transf,in/out,t}$ is a vector of the values of the proper recording period primary parameter spanning the reporting period, depending on whether the daily/monthly/yearly derived parameter is calculated for the input or output voltage or current and for which phase the calculation is carried out
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

#### *6.2.3.2.2 Load voltage & current*

The load voltage and current (AC quantities) are measured as per section 4.5 of Reference 1 and section 4.3 of the present document through the load connection point multimeter depicted in Figure 1 and are recorded as primary parameters for the LCP entity per phase. The units for load voltage and current (denoted as $V_{AC,L}$ and $I_{AC,L}$ in the ensuing) are Volts AC and Amps respectively. For further information regarding the monitoring of the load connection point, please consult section 6.3.3.
Daily, monthly and yearly voltage/current parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$V/I_{AC,L,T,\tau} = \frac{\sum_{t=T-\tau}^{T} V/I_{AC,L,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $V/I_{AC,L,t}$ is a vector of the values of the proper recording period primary parameter spanning the reporting period, depending on whether the daily/monthly/yearly derived parameter is calculated for the voltage or current and for which phase the calculation is carried out
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

#### *6.2.3.2.3 Utility grid voltage and current*

The utility grid voltage and current (AC quantities) are measured as per section 4.5 of Reference 1 and section 4.3 of the present document through the PCC multimeter depicted in Figure 1 and are recorded as primary parameters for the PCC entity per phase. The units for utility grid voltage and current (denoted as $V_{AC,U}$ and $I_{AC,U}$ in the ensuing) are Volts AC and Amps respectively. For further information regarding the monitoring of the PCC, please consult section 6.3.1.

Daily, monthly and yearly voltage/current parameter values are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$V/I_{AC,U,T,\tau} = \frac{\sum_{t=T-\tau}^{T} V/I_{AC,U,t}}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $V/I_{AC,U,t}$ is a vector of the values of the proper recording period primary parameter spanning the reporting period, depending on whether the daily/monthly/yearly derived parameter is calculated for the voltage or current and for which phase the calculation is carried out
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

### 6.2.4 Monitoring of parameters related to active electrical energy

All active energies within the present section are measured in kWh.

#### *6.2.4.1 Parameters related to active electrical energy recorded in a cumulative manner*

Certain equipment types, such as multimeters, protection devices and, in some cases inverters, record energy in a cumulative manner since the beginning of their operation. If cumulative values are available, they are mandatorily recorded by the inSolar solution every recording period.

The active energies of this category are assumed to be recorded within the related devices depicted in Figure 1 (multimeters, protection devices and inverters) in a cumulative manner and captured by the inSolar solution as recording period maximum value parameters as per sections 4.4 and 4.12 and include the following:

- PV panel group output energy (recorded for the PV array entity and denoted as $E_{A,tot,measured}$)
- PV array output energy (recorded for the PV array entity and denoted as $E_{AC,tot,measured}$)
- Active energies input to/output from each transformer (recorded for the TCP entity and denoted as $E_{AC,transf,in,tot,measured}$ and $E_{AC,transf,out,tot,measured}$)
- Active energy to the loads (recorded for the LCP entity and denoted as $E_{L,tot,measured}$)
- Active energies related to the utility grid (recorded for the PCC entity):
    - Active energy export (denoted as $E_{TU,tot,measured}$)
    - Active energy import (denoted as $E_{FU,tot,measured}$)

For cumulative energies recorded by the inverters, the problem aready described within section 6.2.2.1 appears. For that reason, the inSolar solution records the last known energy value for each recording period value that is missing.

#### *6.2.4.2 Parameters related to active electrical energy calculated using active electrical power parameters*

Whenever cumulative active energies are not measured by equipment external to the inSolar solution, they are calculated by the latter using the respective active powers and integration over time. In that case, the following formula applies:

$$E_{i,T} = \begin{cases} E_{i,T_{closest\_valid}} + \tau_r \times P_{i,T_{closest\_valid}}, for \quad positive \quad energy \quad cumulation \\ E_{i,T_{closest\_valid}} - \tau_r \times P_{i,T_{closest\_valid}}, for \quad negative \quad energy \quad cumulation \end{cases}$$

where:

- $T$ denotes the recording period during which the calculation is carried out
- $T_{closest\_valid}$ is the last reporting period (in hours) for which a valid cumulative energy estimate was available. If $E_{i,Tclosest\_valid}$ does not exist (i.e. during the initialization of energy cumulation), its value is considered equal to zero.
- $\tau_r$ is the recording period in hours
- $P_{i,Tclosest\_valid}$ is the last valid value in the past of one of the following:
    - A recording period value of the PV panel group output power primary parameter (if available). In that case, $P_i$ must be replaced by $P_A$ and the

corresponding calculated energy is denoted as $E_{A,tot,measured}$. This power is cumulated using positive cumulation.

- A recording period value of the PV array output power primary parameter (if available). In that case, $P_i$ must be replaced by $P_{AC}$ and the corresponding calculated energy is denoted as $E_{AC,tot,measured}$. This power is cumulated using positive cumulation.
- A recording period value of the transformer input power primary parameter (if available). In that case, $P_i$ must be replaced by $P_{AC,transf,in}$ and the corresponding calculated energy is denoted as $E_{AC,transf,in,tot,measured}$. This power is cumulated using positive cumulation.
- A recording period value of the transformer output power primary parameter (if available). In that case, $P_i$ must be replaced by $P_{AC,transf,out}$ and the corresponding calculated energy is denoted as $E_{AC,transf,out,tot,measured}$. This power is cumulated using positive cumulation.
- A recording period value of the power to the loads primary parameter (if available). In that case, $P_i$ must be replaced by $P_L$ and the corresponding calculated energy is denoted as $E_{L,tot,measured}$. This power is cumulated using positive cumulation.
- A recording period value of the power to the utility network primary parameter (if available). In that case, $P_i$ must be replaced by $P_{TU}$ and the corresponding calculated energy is denoted as $E_{TU,tot,measured}$. This power is cumulated using positive cumulation.
- A recording period value of the power to the utility network primary parameter (if available). In that case, $P_i$ must be replaced by $P_{FU}$ and the corresponding calculated energy is denoted as $E_{FU,tot,measured}$. This power is cumulated using negative cumulation.

### *6.2.4.3 Correction of cumulative energies*

Recording energy in a cumulative manner poses certain problems:

- It is quite common to exchange devices or use devices that have not been reset to zero prior to their use. In that case and without any warning, the energy parameter which is measured cumulatively is expected to present large deviations from one moment to the next.
- When there exists a second device installed and monitored by another entity (a typical example is the multimeter installed at the PCC, where the EPS operator mandatorily installs its own energy meter for billing purposes), these 2 are certainly expected to drift apart after a certain period of time. This is a significant problem, which cannot be avoided, so the 2 cumulative energy readings shall have to be aligned from time to time.

In order to resolve the 2 abovementioned issues, the introduction of independent offset settings each related to a measured cumulative energy is necessary, the values of which are used in conjunction with the cumulative energy readings from the external device or the respective cumulative energies calculated by the inSolar solution to yield a corrected derived parameter. To further enhance the reader's understanding on the approach, the following figure depicts an exemplary evolution of the recording period primary parameter values of the energy towards the utility grid (denoted as active energy export) over time for a fictitious PV plant. 2 discontinuities are presented, one because of the replacement of the PCC multimeter and the other due to the alignment of the PCC meter. The related offset setting values are also presented.

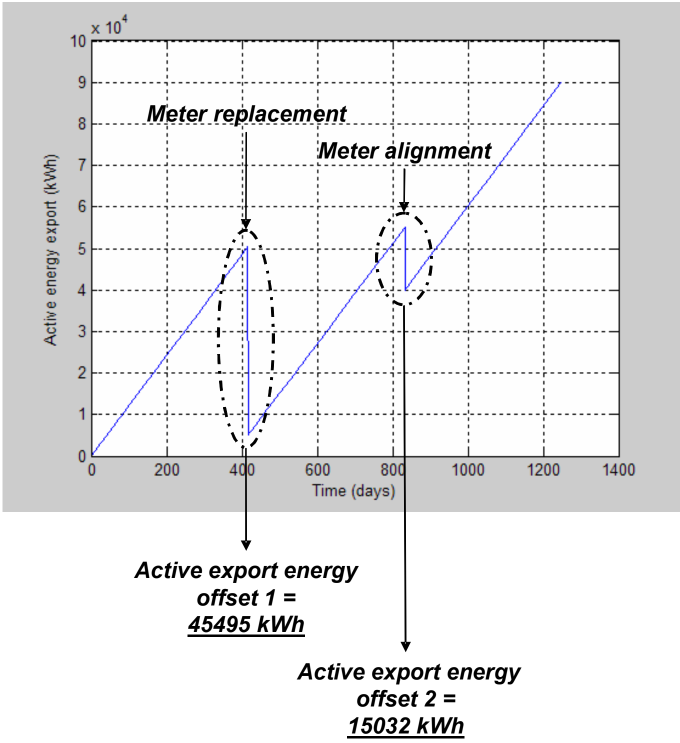


**Figure 12. The active energy export and its discontinuities**

Offset setting value 1 shall have to be added to all recording period primary parameter values of the active energy export after it has been set. After setting offset value 2, both offset value 1 and offset value 2 shall have to be added to the recording period active energy export primary parameter values to yield the corrected derived parameter, presented in the figure below.

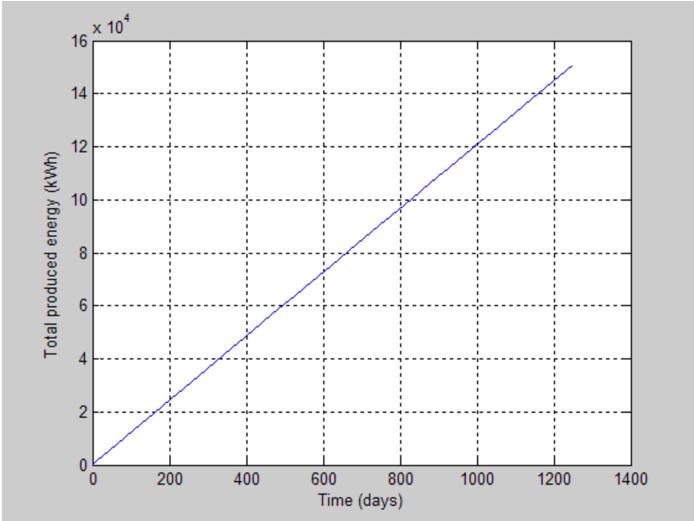


**Figure 13. The corrected derived energy**

Each of these energies needs to have an offset associated with them. Then, the respective corrected derived parameter recording period values can be calculated through the addition of the respective offset values as outlined above, yielding:

- The corrected PV panel group output energy, denoted as $E_{A,tot}$. This parameter is recorded for the PV array entity. The output energy from all PV panel groups belonging to a specific PV array group is calculated for every PV array group through the addition of the respective energies of the PV arrays belonging to the group. Finally, the output energy from all PV panel groups belonging to a specific PV plant is calculated for the PV plant through the addition of the respective energies from all PV array groups.
- The corrected PV array output energy, denoted as $E_{AC,tot}$. This parameter is recorded for the PV array entity. The output energy from all PV arrays belonging to a specific PV array group is calculated for every PV array group through the addition of the respective energies of the PV arrays belonging to the group. Finally, the output energy from all PV arrays belonging to a specific PV plant is calculated for the PV plant through the addition of the respective energies from all PV array groups.
- The corrected energies related to the transformers:
    - Active energy input to the transformer, denoted as $E_{AC,transf,in,tot}$ in the ensuing. This parameter is recorded for the TCP entity. The active energy input to all transformers is also calculated for the PV plant as a whole through the addition of the respective recording period values for all transformers belonging to the specific PV plant.
    - Active energy output from the transformer, denoted as $E_{AC,transf,out,tot}$ in the ensuing. This parameter is recorded for the TCP entity. The active energy output from all transformers is also calculated for the PV plant as a whole through the addition of the respective recording period values for all transformers belonging to the specific PV plant.
- The corrected active energy to the loads, denoted as $E_{L,tot}$ in the ensuing. This parameter is recorded for the PV plant and LCP entities

- The corrected energies related to the utility grid:
    - Active energy to the utility network, denoted as $E_{TU,tot}$ in the ensuing. This parameter is recorded for the PV plant and PCC entities
    - Active energy from the utility network, denoted as $E_{FU,tot}$ in the ensuing. This parameter is recorded for the PV plant and PCC entities

If no valid primary parameter values are available for a certain recording period, the corrected energies are also invalid for the specific recording period. In the case of cumulative energies which are produced through the addition of other cumulative energies (e.g. the transformer input/output active energies), all the recording period values that take part in the calculations have to be valid in order for a valid sum to be produced.

The corrected active energies to/from the utility network are also calculated in a combined manner for a group of PV plants as a derived parameter, summing up the respective corrected active energies for each PV plant separately for every recording period. Only the last available values are used, since energy is recorded in a cumulative manner and non-available data do not take part in the summation procedure. In any case, if no active energy parameters for any PV plant are available for the specific recording period, the respective recording period active energy derived parameters for the PV plant group are also unavailable.

#### 6.2.4.4 Calculating energies for the current day and past days/months/years

Once the corrected cumulative energies have been calculated, the related recording period, daily, monthly and yearly parameter values are derived using the differentiation formula below:

$$E_{i,T,\tau} = \begin{cases} E_{i,T_{last\_valid}} - E_{i,T_{day\_first\_valid}}, & for \ recording \ period \ parameter \ values \\ E_{i,T_{last\_valid}} - E_{i,T_{first\_valid}}, & for \ daily/monthly/yearly \ parameter \ values \end{cases}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $E_{i,Tlast\_valid}$ is the value of the proper recording period corrected cumulative energy parameter closest to the end of the reporting period
- $E_{i,Tday\_first\_valid}$ is the value of the proper recording period corrected cumulative energy parameter closest to the beginning of the day
- $E_{i,Tfirst\_valid}$ is the value of the proper recording period corrected cumulative energy parameter closest to the beginning of the reporting period

Due to the cumulative nature of the energy-related primary parameters, the completeness criterion of section 4.7 had to be modified. More specifically, $E_{i,T,\tau}$ is produced only if the timestamps of the first and last valid recording period values participating in the calculation are apart at least more than 75% of the length of the reporting period.

### 6.2.4.5 Additional energy-related parameters

#### 6.2.4.5.1 Net energy towards the utility grid

The net energy towards the utility grid is calculated only for the PV plant as a whole according to the formula of section 8.2, item (c) of Reference 1 replicated below, taking into consideration the completeness threshold of section 4.5:

$$E_{TUN,T,\tau} = E_{TU,T,\tau} - E_{FU,T,\tau} = \tau_r \times \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t - \tau_r \times \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t$$

The formula has to be modified as follows to account for missing primary parameter values:

$$E_{TUN,T,\tau} = E_{TU,T,\tau} - E_{FU,T,\tau} = \left( \tau_r \times \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t - \tau_r \times \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t \right) \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where $E_{TUN,T,\tau}$ cannot be less than zero and:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The net energy towards the utility grid is measured in kWh and the recording period values are calculated only in a day-cumulative manner.

#### 6.2.4.5.2 Net energy from the utility grid

The net energy from the utility grid is calculated only for the PV plant as a whole according to the formula of section 8.2, item (d) of Reference 1 replicated below, taking into consideration the completeness threshold of section 4.5:

$$E_{FUN,T,\tau} = E_{FU,T,\tau} - E_{TU,T,\tau} = \tau_r \times \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \tau_r \times \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t$$

The formula has to be modified as follows to account for missing primary parameter values:

$$E_{FUN,T,\tau} = E_{FU,T,\tau} - E_{TU,T,\tau} = \left( \tau_r \times \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \tau_r \times \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t \right) \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where $E_{FUN,T,\tau}$ cannot be less than zero and:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The net energy from the utility grid is measured in kWh and the recording period values are calculated only in a day-cumulative manner.

### *6.2.4.5.3 Total system output energy*

The total system output energy is calculated only for the PV plant as a whole according to the formula of section 8.2, item (f) of Reference 1 replicated below, taking into consideration the completeness threshold of section 4.5:

$$E_{use,T,\tau} = E_{L,T,\tau} + E_{TUN,T,\tau} = E_{L,T,\tau} + E_{TU,T,\tau} - E_{FU,T,\tau} =$$
$$\tau_r \times \sum_{t=T-\tau}^{T} P_{L,t} \cdot V_t + \tau_r \times \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t - \tau_r \times \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t$$

The reader has to be reminded that $E_{TUN,T,\tau}$ cannot be less than zero. The formula has to be modified as follows to account for missing primary parameter values:

$$E_{use,T,\tau} = \left( \tau_r \times \sum_{t=T-\tau}^{T} P_{L,t} \cdot V_t + \tau_r \times \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t - \tau_r \times \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t \right) \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out

 

- $P_{L,t}$ is a vector of the values of the recording period primary parameter of the active power towards the loads, spanning the reporting period, as per section 6.2.2.2.3
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The total system output energy is measured in kWh and the recording period values are calculated only in a day-cumulative manner.

### *6.2.4.5.4 Total system input energy*

The total system input energy is calculated only for the PV plant as a whole according to the formula of section 8.2, item (e) of Reference 1 replicated below, taking into consideration the completeness threshold of section 4.5:

$$E_{in,T,\tau} = E_{A,T,\tau} + E_{FUN,T,\tau} = E_{A,T,\tau} + E_{FU,T,\tau} - E_{TU,T,\tau} =$$

$$\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t + \tau_r \times \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \tau_r \times \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t$$

The reader has to be reminded that $E_{FUN,\tau}$ cannot be less than zero. The formula has to be modified as follows to account for missing primary parameter values:

$$E_{in,T,\tau} = \left( \tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t + \tau_r \times \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \tau_r \times \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t \right) \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{A,t}$ is a vector of the values of the recording period derived parameter of the PV panel group output power for the PV plant as a whole, spanning the reporting period, as per section 6.2.2.1.1
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4

- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The total system input energy is measured in kWh and the recording period values are calculated only in a day-cumulative manner.

 

### 6.2.5 Monitoring of the operational parameters of the PV plant

This section includes 2 parameters, namely the nominal output power and the PV panel area, which are necessary for a set of performance metric calculations.

#### *6.2.5.1 Nominal output power*

The nominal output power for a specific PV array, PV array group and the PV plant as a whole (denoted as $P_0$ in the ensuing and measured in kWp) is supposed to be constant for the duration of the life of a PV plant; however experience has shown that this is not the case. Often, inverters are replaced and panels are removed and this alters the nominal output power of the respective PV arrays, PV array groups and finally of the PV plant and the PV plant group. For this reason, the nominal output power of a specific PV panel group is considered a setting and the nominal output powers for each PV array and PV array group are calculated by adding the nominal output powers of the PV panel groups belonging to them. The nominal output power of the PV plant as a whole is calculated by adding the nominal output powers of the PV array groups belonging to it. The nominal output power of a group of PV plants is calculated by adding the nominal output powers of the PV plants belonging to it. Only recording period values of the nominal output power derived parameter are calculated, since it makes no sense to calculate it for past days, months or years. Since all abovementioned calculations do not depend on the availability of data from sensors external to the inSolar solution, such data are always considered available.

#### *6.2.5.2 PV panel area*

The PV panel area for a specific PV panel group, PV array, PV array group or the PV plant as a whole (denoted as $A_a$ in the ensuing and measured in $m^2$) is supposed to be constant for the duration of the life of a PV plant; however experience has shown that this is not the case. Often, PV panels are added or removed and this alters the PV panel area of the respective PV panel groups, PV arrays, PV array groups and finally of the PV plant. For this reason, the area of a specific PV panel group is considered a setting and the total PV panel area for each PV array, PV array group and the PV plant is calculated by adding the areas of the PV panel groups belonging to them. Only recording period values of the PV panel area derived parameter are calculated, since it makes no sense to calculate it for past days, months or years. Since all abovementioned calculations do not depend on the availability of data from sensors external to the inSolar solution, such data are always considered available.

### 6.2.6 Monitoring of the performance metrics

IEC61724 presents a number of derived parameters, related to efficiency, yields, losses and performance ratio which can be used to monitor the performance of a PV plant.

#### *6.2.6.1 Monitoring of efficiencies*

##### *6.2.6.1.1 Mean array efficiency*

The mean array efficiency represents the mean energy conversion efficiency of the PV array, which is useful for comparison with the array efficiency $n_{A0}$ at its nominal output power, $P_0$. The difference in efficiency values represents diode, wiring, and mismatch losses as well as energy wasted during plant operation. The mean array efficiency estimates are calculated according to the formula of section 8.4.3, item (a) of Reference 1 replicated below for PV arrays, PV array groups and the PV plant as a whole, taking into consideration the completeness threshold of section 4.5:

$$n_{Amean,T,\tau} = \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\tau_r \times \sum_{t=T-\tau}^{T} A_{a,t} \cdot G_{I,t} \cdot V_t}$$

where:

- $\tau$ is the reporting period in hours (equal to zero when the parameter is calculated for any recording period of the current day and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $G_{I,t}$ is a vector of the values of the proper recording period primary/derived parameter of the total irradiance, spanning the reporting period, as per section 6.2.1.1, depending on whether the mean array efficiency is calculated for a specific PV array, PV array group or the PV plant as a whole
- $P_{A,t}$ is a vector of the values of the proper recording period corrected derived parameter of the PV panel group output power, spanning the reporting period, as per section 6.2.2.1.1, depending on whether the mean array efficiency is calculated for a specific PV array, PV array group or the PV plant as a whole
- $A_{a,t}$ is a vector of the values of the proper recording period derived parameter of the PV panel area as per section 6.2.5.2 (depending on whether the mean array efficiency is calculated for a specific PV array, PV array group or the PV plant as a whole)
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values AND the recording period value of $G_I$ is above a threshold, treated by the inSolar solution as a user-configurable setting and zeros for recording periods where at least one formula input has an invalid value OR the recording period value of $G_I$ is below the threshold

The completeness is calculated using all valid values, irrespective of whether the total irradiance is below or above the threshold. The mean array efficiency is expressed as a percentage.

### 6.2.6.1.2 Inverter efficiency

Despite the fact that the inverter efficiency is not defined in Reference 1, it is a very useful metric, provided by competitive solutions. Moreover, it is necessary for the calculation of a series of other performance metrics. The necessary formula is presented below and applies for PV arrays, PV array groups and the PV plant as a whole, taking into consideration the completeness threshold of section 4.5:

$$n_{inv,T,\tau} = \frac{E_{AC,T,\tau}}{E_{A,T,\tau}} = \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{AC,t} \cdot V_t}{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}$$

where:

- $\tau$ is the reporting period in hours (equal to zero when the parameter is calculated for any recording period of the current day and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{A,t}$ is a vector of the values of the proper recording period corrected derived parameter of the PV panel group output power, spanning the reporting period, as per section 6.2.2.1.1, depending on whether the mean array efficiency is calculated for a specific PV array, PV array group or the PV plant as a whole
- $P_{AC,t}$ is a vector of the values of the proper recording period corrected derived parameter of the array output power, spanning the reporting period, as per section 6.2.2.1.2, depending on whether the inverter efficiency is calculated for a specific PV array, PV array group or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values AND the recording period value of $G_I$ is above the threshold presented in section 6.2.6.1.1 and zeros for recording periods where at least one formula input has an invalid value OR the recording period value of $G_I$ is below the threshold

The completeness is calculated using all valid values, irrespective of whether the total irradiance is below or above the threshold. The inverter efficiency is expressed as a percentage.

### 6.2.6.1.3 Load efficiency

The load efficiency represents the efficiency with which the energy from the PV arrays is transmitted to the loads. It is calculated only for the PV plant as a whole according to the formula of section 8.2, item (h) of Reference 1 replicated below, taking into consideration the completeness threshold of section 4.5:

$$n_{load,T,\tau} = \frac{E_{use,T,\tau}}{E_{in,T,\tau}} = \frac{E_{L,T,\tau} + E_{TUN,T,\tau}}{E_{A,T,\tau} + E_{FUN,T,\tau}} = \frac{E_{L,T,\tau} + \left(E_{TU,T,\tau} - E_{FU,T,\tau}\right)}{E_{A,T,\tau} + \left(E_{FU,T,\tau} - E_{TU,T,\tau}\right)} =$$

$$\frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{L,t} \cdot V_t + \tau_r \times \left( \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t \right)}{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t + \tau_r \times \left( \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t \right)}$$

where:

- $\tau$ is the reporting period in hours (equal to zero when the parameter is calculated for any recording period of the current day and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{A,t}$ is a vector of the values of the recording period derived parameter of the PV panel group output power for the PV plant as a whole, spanning the reporting period, as per section 6.2.2.1.1
- $P_{L,t}$ is a vector of the values of the recording period primary parameter of the active power towards the loads, spanning the reporting period, as per section 6.2.2.2.3
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The load efficiency is expressed as a percentage. The reader has to be reminded that both $E_{TUN,T,\tau}$ and $E_{FUN,T,\tau}$ cannot be less than zero.

#### *6.2.6.1.4 System efficiency*

The system efficiency is calculated only for the PV plant as a whole according to the formula of section 8.3 of Reference 1 replicated below, taking into consideration the completeness threshold of section 4.5:

$$n_{BOS,T,\tau} = \frac{E_{L,T,\tau} + E_{TUN,T,\tau} - E_{FUN,T,\tau}}{E_{A,T,\tau}} =$$

$$\frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{L,t} \cdot V_t + \tau_r \times \left( \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t \right) - \tau_r \times \left( \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t \right)}{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}$$

where:

- $\tau$ is the reporting period in hours (equal to zero when the parameter is calculated for any recording period of the current day and days, months, years otherwise)

- $\tau_r$ is the recording period in hours
- *T* denotes the recording period during which the calculation is carried out
- $P_{A,t}$ is a vector of the values of the recording period derived parameter of the PV panel group output power for the PV plant as a whole, spanning the reporting period, as per section 6.2.2.1.1
- $P_{L,t}$ is a vector of the values of the recording period primary parameter of the active power towards the loads, spanning the reporting period, as per section 6.2.2.2.3
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values AND the recording period value of $G_I$ is above the threshold presented in section 6.2.2.1 and zeros for recording periods where at least one formula input has an invalid value OR the recording period value of $G_I$ is below the threshold

The completeness is calculated using all valid values, irrespective of whether the total irradiance is below or above the threshold. The reader has to be reminded that both $E_{TUN,T,\tau}$ and $E_{FUN,T,\tau}$ cannot be less than zero. The system efficiency is expressed as a percentage.

### 6.2.6.1.5 Total efficiency

The total efficiency is calculated only for the PV plant as a whole according to the formula of section 8.4.3, item (b) of Reference 1 replicated below, taking into consideration the completeness threshold of section 4.5:

$$n_{tot,T,\tau} = n_{Amean,T,\tau} \times n_{load,T,\tau} = \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\tau_r \times \sum_{t=T-\tau}^{T} A_{a,t} \cdot G_{I,t} \cdot V_t} \times \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{L,t} \cdot V_t + \tau_r \times \left( \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t \right)}{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t + \tau_r \times \left( \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t \right)}$$

where:

- $\tau$ is the reporting period in hours (equal to zero when the parameter is calculated for any recording period of the current day and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- *T* denotes the recording period during which the calculation is carried out
- $P_{A,t}$ is a vector of the values of the recording period derived parameter of the PV panel group output power for the PV plant as a whole, spanning the reporting period, as per section 6.2.2.1.1
- $P_{L,t}$ is a vector of the values of the recording period primary parameter of the active power towards the loads, spanning the reporting period, as per section 6.2.2.2.3

- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $A_{a,t}$ is a vector of the values of the proper recording period derived parameter of the PV panel area for the PV plant as a whole as per section 6.2.5.2
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values AND the recording period value of $G_I$ is above the threshold presented in section 6.2.2.1 and zeros for recording periods where at least one formula input has an invalid value OR the recording period value of $G_I$ is below the threshold

The completeness is calculated using all valid values, irrespective of whether the total irradiance is below or above the threshold. The total efficiency is expressed as a percentage. The reader has to be reminded that both $E_{TUN,T,\tau}$ and $E_{FUN,T,\tau}$ which participate in the calculation of $n_{load}$ cannot be less than zero as per section 6.2.6.1.3.

### *6.2.6.1.6 Solar fraction*

The solar fraction is calculated only for the PV plant as a whole according to the formula of section 8.2, item (g) of Reference 1 replicated below, taking into consideration the completeness threshold of section 4.5:

$$F_{A,T,\tau} = \frac{E_{A,T,\tau}}{E_{in,T,\tau}} = \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t + \tau_r \times \left( \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t \right)}$$

where:

- $\tau$ is the reporting period in hours (equal to zero when the parameter is calculated for any recording period of the current day and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{A,t}$ is a vector of the values of the recording period derived parameter of the PV panel group output power for the PV plant as a whole, spanning the reporting period, as per section 6.2.2.1.1
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The solar fraction is expressed as a percentage. The reader has to be reminded that $E_{FUN,T,\tau}$ which participates in the calculation of $E_{in,T,\tau}$ cannot be less than zero as per section 6.2.4.5.4.

#### *6.2.6.1.7 Transformer efficiency*

Despite the fact that the transformer efficiency is not defined in Reference 1, it is a very useful metric. The necessary formula is presented below and applies for each TCP separately and the PV plant as a whole, taking into consideration the completeness threshold of section 4.5:

$$n_{Transf,T,\tau} = \frac{E_{transf,out,T,\tau}}{E_{transf,in,T,\tau}} = \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{AC,transf,out,t} \cdot V_t}{\tau_r \times \sum_{t=T-\tau}^{T} P_{AC,transf,in,t} \cdot V_t}$$

where:

- $\tau$ is the reporting period in hours (equal to zero when the parameter is calculated for any recording period of the current day and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{AC,transf,out,t}$ is a vector of the values of the proper recording period transformer output power primary/derived parameter spanning the reporting period, as per section 6.2.2.2.1, depending on whether the transformer efficiency is calculated for a specific transformer or the PV plant as a whole
- $P_{AC,transf,in,t}$ is a vector of the values of the proper recording period transformer input power primary/derived parameter spanning the reporting period as per section 6.2.2.2.2, depending on whether the transformer efficiency is calculated for a specific transformer or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values AND the recording period value of $G_I$ is above the threshold presented in section 6.2.2.1 and zeros for recording periods where at least one formula input has an invalid value OR the recording period value of $G_I$ is below the threshold

The transformer efficiency is expressed as a percentage.

### 6.2.6.2 *Monitoring of normalized yields*

Yields represent normalized energies at various points throughout the PV plant. The recording period values for all yields are calculated in a day-cumulative manner.

#### *6.2.6.2.1 Energy yield*

The energy yield is the PV array energy output per kW of installed PV array output power and is calculated according to the formula of section 8.4.1, item (a) of Reference 1 replicated below for PV arrays, PV array groups and the PV plant as a whole, taking into consideration the completeness threshold of section 4.5:

$$Y_{A,T,\tau} = \frac{E_{A,T,\tau}}{P_0} = \tau_r \times \frac{\sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\sum_{t=T-\tau}^{T} P_{0,t} \cdot V_t}$$

This yield represents the number of hours per day that the PV plant, PV array group or PV array would need to operate at its nominal output power $P_0$ to contribute the same energy to the system as was monitored. The formula has to be modified as follows to account for missing primary parameter values:

$$Y_{A,T,\tau} = \frac{E_{A,T,\tau}}{P_0} = \tau_r \times \frac{\sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\sum_{t=T-\tau}^{T} P_{0,t} \cdot V_t} \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{A,t}$ is a vector of the values of the proper recording period corrected derived parameter of the PV panel group output power, spanning the reporting period, as per section 6.2.2.1.1, depending on whether the energy yield is calculated for a specific PV array, PV array group or the PV plant as a whole
- $P_{0,t}$ is a vector of the values of the proper nominal output power derived parameter, spanning the reporting period, as per section 6.2.5.1, depending on whether the energy yield is calculated for a specific PV array, PV array group or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The units for the energy yield are $h \bullet d^{-1}$ if it is calculated for the present day or any past days, $h \bullet mo^{-1}$ if it is calculated for a month and $h \bullet y^{-1}$ if it is calculated for a year.

### 6.2.6.2.2 Reference yield

The reference yield can be calculated by dividing the total in-plane irradiation by the module's reference in-plane irradiance. The reference yield is calculated according to the formula of section 8.4.1, item (c) of Reference 1 replicated below for PV arrays, PV array groups and the PV plant as a whole, taking into consideration the completeness threshold of section 4.5:

$$Y_{r,T,\tau} = \tau_r \times \frac{\sum_{t=T-\tau}^{T} G_{I,t} \cdot V_t}{G_{I,ref}}$$

This yield represents the number of hours per day during which the solar radiation would need to be at reference irradiance levels in order to contribute the same incident energy as was monitored for the PV plant, PV array group or PV array. The formula has to be modified as follows to account for missing primary parameter values:

$$Y_{r,T,\tau} = \tau_r \times \frac{\sum_{t=T-\tau}^{T} G_{I,t} \cdot V_t}{G_{I,ref}} \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to *T* if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- *T* denotes the recording period during which the calculation is carried out
- $G_{I,t}$ is a vector of the values of the proper recording period primary/derived parameter of the total irradiance, spanning the reporting period, as per section 6.2.1.1, depending on whether the reference yield is calculated for a specific PV array, PV array group or the PV plant as a whole
- $G_{I,ref}$ is the module's reference in-plane irradiance which is considered to be constant and identical for the PV array, PV array group and PV plant
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The units for the reference yield are $h{\bullet}d^{-1}$ if it is calculated for the present day or any past days, $h{\bullet}mo^{-1}$ if it is calculated for a month and $h{\bullet}y^{-1}$ if it is calculated for a year.

### 6.2.6.2.3 Final yield

The final PV system yield is the portion of the net energy output of the entire PV plant which was supplied by the array per kW of installed PV array. According to Reference 1, the final yield is calculated only for the PV plant as a whole according to the formula of section

8.4.1, item (b) of Reference 1 replicated below, taking into consideration the completeness threshold of section 4.5:

$$Y_{f,T,\tau} = Y_{A,T,\tau} \times n_{load,T,\tau} =$$

$$\tau_r \times \frac{\sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\sum_{t=T-\tau}^{T} P_{0,t} \cdot V_t} \times \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{L,t} \cdot V_t + \tau_r \times \left( \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t \right)}{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t + \tau_r \times \left( \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t \right)} \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{0,t}$ is a vector of the values of the derived parameter of the nominal output power of the PV plant as a whole, spanning the reporting period, as per section 6.2.5.1
- $P_{A,t}$ is a vector of the values of the recording period derived parameter of the PV panel group output power for the PV plant as a whole, spanning the reporting period, as per section 6.2.2.1.1
- $P_{L,t}$ is a vector of the values of the recording period primary parameter of the active power towards the loads, spanning the reporting period, as per section 6.2.2.2.3
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The reader has to be reminded that both $E_{TUN,T,\tau}$ and $E_{FUN,T,\tau}$ which participate in the calculation of $n_{load,T,\tau}$ cannot be less than zero as per section 6.2.6.1.3.

According to Reference 6, the final yield can also be calculated for specific PV arrays and PV array groups, though this is not mandated by Reference 1. In that case, the following formula is applied, taking into consideration the completeness threshold of section 4.5:

$$Y_{f,T,\tau} = Y_{A,T,\tau} \times n_{inv,T,\tau} = \tau_r \times \frac{\sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\sum_{t=T-\tau}^{T} P_{0,t} \cdot V_t} \times \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{AC,t} \cdot V_t}{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t} \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)

- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{AC,t}$ is a vector of the values of the proper recording period corrected derived parameter of the PV array output power, spanning the reporting period, as per section 6.2.2.1.2, depending on whether the final yield is calculated for a specific PV array or PV array group
- $P_{0,t}$ is a vector of the values of the proper nominal output power derived parameter, spanning the reporting period, as per section 6.2.5.1, depending on whether the energy yield is calculated for a specific PV array or PV array group
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The final yield represents the number of hours per day, month or year that the PV plant, PV array group or PV array would need to operate at its rated output power to equal its monitored contribution. The units for the final yield are $h \bullet d^{-1}$ if the final yield is calculated for the present day or any past days, $h \bullet mo^{-1}$ if the final yield is calculated for a month and $h \bullet y^{-1}$ if the final yield is calculated for a year.

 

### 6.2.6.3 Monitoring of the normalized losses

The normalized losses quantify the energy lost at various points throughout the PV plant. The recording period values for all losses are calculated in a day-cumulative manner.

#### 6.2.6.3.1 Array capture losses

The array capture losses represent the losses due to array operation and indicate the amount of time during which the array would be required to operate at its rated power to provide such losses. They are calculated according to the formula of section 8.4.2, item (a) of Reference 1 replicated below for PV arrays, PV array groups and the PV plant as a whole, taking into consideration the completeness threshold of section 4.5:

$$L_{C,T,\tau} = Y_{r,T,\tau} - Y_{A,T,\tau} = \tau_r \times \left( \frac{\sum_{t=T-\tau}^{T} G_{I,t} \cdot V_t}{G_{I,ref}} - \frac{\sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\sum_{t=T-\tau}^{T} P_{0,t} \cdot V_t} \right) \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t} \tau_r$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to *T* if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- *T* denotes the recording period during which the calculation is carried out
- $G_{I,t}$ is a vector of the values of the proper recording period primary/derived parameter of the total irradiance, spanning the reporting period, as per section 6.2.1.1, depending on whether the array capture loses are calculated for a specific PV array, PV array group or the PV plant as a whole
- $G_{I,ref}$ is the module's reference in-plane irradiance which is considered to be constant and identical for the PV array, PV array group and PV plant
- $P_{A,t}$ is a vector of the values of the proper recording period corrected derived parameter of the PV panel group output power, spanning the reporting period, as per section 6.2.2.1.1, depending on whether the array capture losses are calculated for a specific PV array, PV array group or the PV plant as a whole
- $P_{0,t}$ is a vector of the values of the proper nominal output power derived parameter, spanning the reporting period, as per section 6.2.5.1, depending on whether the array capture losses are calculated for a specific PV array, PV array group or the PV plant as a whole
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The units for the array capture losses are $h \bullet d^{-1}$ if they are calculated for the present day or any past days, $h \bullet mo^{-1}$ if they are calculated for a month and $h \bullet y^{-1}$ if they are calculated for a year.

### 6.2.6.3.2 System losses

The system losses sum up all losses internal to the PV plant because of the operation of the inverters, transformers, wiring and all components other than the PV panel itself. The system losses are not included in Reference 1, so for a definition, please refer to Reference 6.

The system losses are calculated for a specific PV array, a specific PV array group, or for the whole of PV plant and different formulas apply. If the system losses are calculated for the PV plant as a whole, the following formula is used, taking into consideration the completeness threshold of section 4.5:

$$L_{S,T,\tau} = Y_{A,T,\tau} - Y_{f,T,\tau} = Y_{A,T,\tau} \times \left(1 - n_{load,T,t}\right) =$$

$$\tau_r \times \frac{\sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\sum_{t=T-\tau}^{T} P_{0,t} \cdot V_t} \times \left(1 - \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{L,t} \cdot V_t + \tau_r \times \left(\sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t\right)}{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t + \tau_r \times \left(\sum_{t=T-\tau}^{T} P_{FU,t} \cdot V_t - \sum_{t=T-\tau}^{T} P_{TU,t} \cdot V_t\right)}\right) \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{0,t}$ is a vector of the values of the derived parameter of the nominal output power of the PV plant as a whole, spanning the reporting period, as per section 6.2.5.1
- $P_{A,t}$ is a vector of the values of the recording period derived parameter of the PV panel group output power for the PV plant as a whole, spanning the reporting period, as per section 6.2.2.1.1
- $P_{L,t}$ is a vector of the values of the recording period primary parameter of the active power towards the loads, spanning the reporting period, as per section 6.2.2.2.3
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The reader has to be reminded that both $E_{TUN,T,\tau}$ and $E_{FUN,T,\tau}$ which participate in the calculation of $n_{load,T,t}$ cannot be less than zero as per section 6.2.6.1.3.

If the system losses are calculated for a specific PV array or PV array group, the following formula is used:

$$L_{S,T,\tau} = Y_{A,T,\tau} - Y_{f,T,\tau} = \tau_r \times \left( \frac{\sum_{t=T-\tau}^{T} P_{A,t} \cdot V_t}{\sum_{t=T-\tau}^{T} P_{0,t} \cdot V_t} - \frac{\tau_r \times \sum_{t=T-\tau}^{T} P_{AC,t} \cdot V_t}{\tau_r \times \sum_{t=T-\tau}^{T} P_{0,t} \cdot V_t} \right) \times \frac{\tau}{\tau_r \times \sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (equal to the time elapsed from 00:00 hours to $T$ if the parameter is calculated for the current recording period and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $P_{A,t}$ is a vector of the values of the proper recording period corrected derived parameter of the PV panel group output power, spanning the reporting period, as per section 6.2.2.1.1, depending on whether the system losses are calculated for a specific PV array or PV array group
- $P_{AC,t}$ is a vector of the values of the proper recording period corrected derived parameter of the PV array output power, spanning the reporting period, as per section 6.2.2.1.2, depending on whether the system losses are calculated for a specific PV array or PV array group
- $P_{0,t}$ is a vector of the values of the proper nominal output power derived parameter, spanning the reporting period, as per section 6.2.5.1, depending on whether the system losses are calculated for a specific PV array or PV array group
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

The units for system losses are $h \bullet d^{-1}$ if they are calculated for the present day or any past days, $h \bullet mo^{-1}$ if they are calculated for a month and $h \bullet y^{-1}$ if they are calculated for a year.

### *6.2.6.4 Monitoring of the performance ratio*

The traditional availability and reliability factors for electrical generation units stated within Reference 3 do not apply to PV plants due to the fact that they do not operate at night, so the performance ratio (denoted as $R_P$ in Reference 1) has been introduced as a proper substitute, as per Reference 4.

The maintenance performance ratio is calculated per PV array, PV array group and the PV plant as a whole only for values of total irradiance which are above the threshold of section 6.2.6.1.1. This threshold is the same for all performance ratio calculations. If the maintenance performance ratio is calculated for the PV plant as a whole, the following formula is used as per section 8.4.2, item I of Reference 1, taking into consideration the completeness threshold of section 4.5:

$$R_{P,T,\tau} = \frac{Y_{f,T,\tau}}{Y_{r,T,\tau}} = \frac{\tau_r \times \dfrac{\sum_{t=T-\tau}^{T} P_{A,t}\cdot V_t}{\sum_{t=T-\tau}^{T} P_{0,t}\cdot V_t} \times \dfrac{\tau_r \times \sum_{t=T-\tau}^{T} P_{L,t}\cdot V_t + \tau_r \times \left(\sum_{t=T-\tau}^{T} P_{TU,t}\cdot V_t - \sum_{t=T-\tau}^{T} P_{FU,t}\cdot V_t\right)}{\tau_r \times \sum_{t=T-\tau}^{T} P_{A,t}\cdot V_t + \tau_r \times \left(\sum_{t=T-\tau}^{T} P_{FU,t}\cdot V_t - \sum_{t=T-\tau}^{T} P_{TU,t}\cdot V_t\right)}}{\tau_r \times \dfrac{\sum_{t=T-\tau}^{T} G_{I,t}\cdot V_t}{G_{I,ref}}}$$

where:

- $\tau$ is the reporting period in hours (equal to zero when the parameter is calculated for any recording period of the current day and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $G_{I,t}$ is a vector of the values of the recording period derived parameter of the total irradiance for the PV plant as a whole, spanning the reporting period, as per section 6.2.1.1
- $G_{I,ref}$ is the module's reference in-plane irradiance which is considered to be constant for the PV plant as a whole
- $P_{A,t}$ is a vector of the values of the recording period derived parameter of the PV panel group output power for the PV plant as a whole, spanning the reporting period, as per section 6.2.2.1.1
- $P_{L,t}$ is a vector of the values of the recording period primary parameter of the active power towards the loads, spanning the reporting period, as per section 6.2.2.2.3
- $P_{TU,t}$ is a vector of the values of the recording period primary parameter of the active power towards the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{FU,t}$ is a vector of the values of the recording period primary parameter of the active power from the utility grid, spanning the reporting period, as per section 6.2.2.2.4
- $P_{0,t}$ is a vector of the values of the derived parameter of the nominal output power of the PV plant as a whole, spanning the reporting period, as per section 6.2.5.1
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values AND the recording period value of $G_I$ is above

the threshold presented in section 6.2.2.1 and zeros for recording periods where at least one formula input has an invalid value OR the recording period value of $G_I$ is below the threshold

The performance ratio is also calculated for a group of PV plants as the average of the performance ratios of the PV plants belonging to it.

The reader has to be reminded that both $E_{TUN,\tau}$ and $E_{FUN,\tau}$ which participate in the calculation of $Y_{f,\tau}$ cannot be less than zero as per section 6.2.6.1.3.

If the performance ratio is calculated for a specific PV array or PV array group, the following formula is applied, taking into consideration the completeness threshold of section 4.5:

$$R_{P,T,\tau} = \frac{Y_{f,T,\tau}}{Y_{r,T,\tau}} = \frac{\tau_r \times \dfrac{\sum_{t=T-\tau}^{T} P_{AC,t} \cdot V_t}{\sum_{t=T-\tau}^{T} P_{0,t} \cdot V_t}}{\tau_r \times \dfrac{\sum_{t=T-\tau}^{T} G_{I,t} \cdot V_t}{G_{I,ref}}}$$

where:

- $\tau$ is the reporting period in hours (equal to zero when the parameter is calculated for any recording period of the current day and days, months, years otherwise)
- $\tau_r$ is the recording period in hours
- $T$ denotes the recording period during which the calculation is carried out
- $G_{I,t}$ is a vector of the values of the proper recording period primary/derived parameter of the total irradiance, spanning the reporting period, as per section 6.2.1.1, depending on whether the performance ratio is calculated for a specific PV array or PV array group
- $G_{I,ref}$ is the module's reference in-plane irradiance which is considered to be constant and identical for PV arrays and PV array groups
- $P_{AC,t}$ is a vector of the values of the proper recording period corrected derived parameter of the PV array output power, spanning the reporting period, as per section 6.2.2.1.2, depending on whether the performance ratio is calculated for a specific PV array or PV array group
- $P_{0,t}$ is a vector of the values of the proper nominal output power derived parameter, spanning the reporting period, as per section 6.2.5.1, depending on whether the performance ratio is calculated for a specific PV array or PV array group
- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values AND the recording period value of $G_I$ is above the threshold presented in section 6.2.2.1 and zeros for recording periods where at least one formula input has an invalid value OR the recording period value of $G_I$ is below the threshold

The completeness is calculated using all valid values, irrespective of whether the total irradiance is below or above the threshold. All abovementioned performance ratios are expressed as a percentage.

### 6.2.7 Monitoring of additional parameters from the inverters

Apart from the electrical power/energy and voltages/currents, the inverters provide a variety of parameters related to their operation. These parameters, though not mandated in Reference 1, are recorded by the inSolar solution as primary parameters. The exact set of parameters presented within the present paragraph depend on the type of inverter used. Daily, monthly and yearly values for these parameters are not supported following the rationale already presented in section 6.2.3.1.

#### *6.2.7.1 Internal temperature*

The internal temperature, denoted as $T_{inv,int}$ in the ensuing, is measured in °C. The respective recording period primary parameter is calculated as per section 4.3 of the present document for each inverter. This parameter is also real-time monitored as per 4.9.

#### *6.2.7.2 Cooler temperature*

The cooler temperature, denoted as $T_{inv,cool}$ in the ensuing, is measured in °C. The respective recording period primary parameter is calculated as per section 4.3 of the present document for each inverter. This parameter is also real-time monitored as per 4.9.

#### *6.2.7.3 Availability*

The availability, denoted as $AV_{inv}$ in the ensuing, is measured in %. The respective recording period primary parameter is calculated as per section 4.3 of the present document for each inverter.

#### *6.2.7.4 Total hours of operation*

The total hours of operation are recorded by the inverter in a cumulative manner since the beginning of its operation and they are recorded by the inSolar solution every recording period as maximum value parameters and as per section 4.4 for each inverter. The total hours of operation count the number of hours that the inverter was operational, regardless of whether it was feeding in power to the grid or not.

#### *6.2.7.5 Total hours of feed-in operation*

The total hours of feed-in operation are recorded by the inverter in a cumulative manner since the beginning of its operation and they are recorded by the inSolar solution every recording period as maximum value parameters and as per section 4.4 for each inverter. The total hours of operation count the number of hours that the inverter was operational and was producing power.

#### *6.2.7.6 Array reactive output power*

The array reactive output power, denoted as $Q_{AC}$ in the ensuing, is measured in kVAr. The respective recording period primary parameter is calculated as per section 4.3 of the present document for each PV array.

#### *6.2.7.7 Array apparent output power*

The array apparent output power, denoted as $S_{AC}$ in the ensuing, is measured in kVA. The respective recording period primary parameter is calculated as per section 4.3 of the present document for each PV array.

#### *6.2.7.8 Total power factor*

The total power factor at the output of the inverter, denoted as $PF_{inv}$ in the ensuing is dimensionless. The respective recording period primary parameter is calculated as per section 4.3 of the present document for each PV array.

#### *6.2.7.9 Panel group open-circuit voltage*

The panel group open-circuit voltage, denoted as $V_{OC}$ in the ensuing is measured in Volts DC. The respective recording period primary parameter is calculated as per section 4.3 of the present document for each PV array.

#### *6.2.7.10 Panel group setpoint voltage*

The panel group setpoint voltage, denoted as $V_{setpoint}$ in the ensuing is measured in Volts DC. The respective recording period primary parameter is calculated as per section 4.3 of the present document for each PV array.

#### *6.2.7.11 Array output frequency*

The array output frequency, denoted as $F_{AC}$ in the ensuing, is measured in Hz. The respective recording period primary parameter is calculated as per section 4.3 of the present document for each PV array. This parameter is also real-time monitored as per 4.9.

#### *6.2.7.12 Insulation resistance*

The insulation resistance, denoted as $R_{ins}$ in the ensuing, is measured in Ohms. This parameter is only real-time monitored according to section 4.9 for each PV array.

#### *6.2.7.13 Leakage current*

The leakage current, denoted as $I_{leak}$ in the ensuing, is measured in Amps. This parameter is only real-time monitored according to section 4.9 for each PV array.

### 6.2.8 Monitoring of parameters related to financial/environmental issues

This set of parameters is directly related to the cumulative energies to/from the utility network and include the revenue and a number of parameters quantifying the environmental impact of a single or a group of PV plants.

#### *6.2.8.1 Total revenue*

The total revenue (revenue since the beginning of the operation of the PV plant expressed in Euros) is calculated for the PV plant as a whole as well as for a group of PV plants. The total revenue is recorded only for a reporting period equal to the reporting period. If the total revenue is calculated for a specific PV plant, the following formula is used:

$$R_T = R_{T_{closest\_valid}} + \left(E_{TU,tot,T} - E_{TU,tot,T_{closest\_valid}}\right) \times C_{TU,T_{closest\_valid}} - \left(E_{FU,tot,T} - E_{FU,tot,T_{closest\_valid}}\right) \times C_{FU,T_{closest\_valid}}$$

where:

- $T$ denotes the recording period during which the calculation is carried out
- $T_{closest\_valid}$ is the last reporting period (in hours) for which a valid total revenue estimate was available. If $R_{Tclosest\_valid}$ does not exist (i.e. during the initialization of revenue cumulation), its value is considered equal to zero and $R_T$ is equal to $E_{TU,tot,T} \times C_{TU,T} - E_{FU,tot,T} \times C_{FU,T}$
- $E_{TU,tot}$ is the cumulative energy towards the utility network (power grid) measured by the PCC multimeter as per section 6.2.4.3
- $E_{FU,tot}$ is the cumulative energy from the utility network (power grid) towards the PV plant measured by the PCC multimeter as per section 6.2.4.3
- $C_{TU}$ is the setting of the price of the energy sold to the utility network, based on the agreement of the PV plant owner with the distribution company
- $C_{FU}$ is the setting of the price of the energy bought from the utility network, based on the agreement of the PV plant owner with the distribution company

The total revenue is also calculated in a combined manner for a group of PV plants as a derived parameter, summing up the respective total revenues for each PV plant separately for every recording period. Non-available data do not take part in the summation procedure. In any case, if no total revenue parameters for any PV plant are available for the specific recording period, the recording period total revenue derived parameter for the PV plant group is also unavailable.

#### *6.2.8.2 Calculating revenue for the current day and past days/months/years*

Once the total revenue has been calculated, the related recording period, daily, monthly and yearly parameter values are derived using the differentiation formula below:

$$R_{T,\tau} = \begin{cases} R_{T_{last\_valid}} - R_{T_{day\_first\_valid}}, & \textit{for recording period parameter values} \\ R_{T_{last\_valid}} - R_{T_{first\_valid}}, & \textit{for daily/monthly/yearly parameter values} \end{cases}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $R_{Tlast\_valid}$ is the value of the recording period total revenue parameter closest to the end of the reporting period
- $R_{Tday\_first\_valid}$ is the value of the recording period total revenue parameter closest to the beginning of the day
- $R_{Tfirst\_valid}$ is the value of the recording period total revenue parameter closest to the beginning of the reporting period

Due to the cumulative nature of the total revenue parameter, the completeness criterion of section 4.7 has to be modified. More specifically, $R_{T,\tau}$ is produced only if the timestamps of the first and last valid recording period values participating in the calculation are apart at least more than 75% of the length of the reporting period.

### *6.2.8.3 Total CO2 reduction*

The total CO2 reduction is calculated upon user demand for a specific PV plant or a group of PV plants using the following formula:

$$R_{CO_2} = \left(E_{TU,tot,T_{last\_valid}} - E_{FU,tot,T_{last\_valid}}\right) \times R$$

where:

- $E_{TU,tot,Tlast\_valid}$ is the cumulative energy towards the utility network (power grid) measured by the PCC multimeter as per section 6.2.4.3. Only the last valid value is used, since this parameter is recorded in a cumulative manner.
- $E_{FU,tot,Tlast\_valid}$ is cumulative energy from the utility network (power grid) measured by the PCC multimeter as per section 6.2.4.3. Only the last valid value is used, since this parameter is recorded in a cumulative manner.
- $R$ is a setting equal to the kilos of CO2 saved per kWh of energy produced and consumed compared to the same amount of energy produced using a specific fossil fuel, based on data available in the literature

No historical data are retained by the inSolar solution.

### 6.2.9 State-sets related to the PV plant production

The current state of these state-sets is updated directly using data from equipment related to the production facilities of the PV plant. These state-sets encode the status of the inverters/string monitors and the status of environmental parameters related to the production facilities.

#### *6.2.9.1 Primary state-sets related to the inverter and string monitors*

It is understandable that the state-sets used by the inverters to encode their status are manufacturer dependent, since each inverter model is different. Based on prior knowledge of inverters built by SMA, KACO and FRONIUS, which are the 3 leaders in the European market, a state-set which can be considered as the superset has been compiled and is presented below:

- Inverter in normal operation: The inverter is operating normally and produces electrical power which is then fed into the utility grid
- Inverter in derating mode: The inverter is reducing its power output in order to protect itself or the grid against failure
- Inverter in disturbance mode: The inverter is not feeding electrical power into the utility grid, since it has detected conditions that led it to disconnect from the network
- Inverter in error mode: The inverter is in error (not damaged) and needs service
- Inverter in stopped mode: The inverter is stopped and needs service in order to resume operation
- Inverter in night shutdown mode: The inverter has shutdown for the night. It has to be noted that not all inverters notify external entities (such as the inSolar solution) that they are shutting down, so there are cases that the inSolar solution has to determine the occurrence of this state using other information. This is further dealt with within section 6.8.1.

This state-set is supported for the inverter entity presented within Figure 3. One more state-set that is supported for the same entity contains the following 2 states:

- Inverter warning present: The inverter has detected a condition that calls for the attention of technical personnel
- No inverter warning present

The reason for the existence of this additional state-set is that warnings may be present regardless of the operational state the inverter is into.

Additionally, all examined inverters provide extensive status information for each state via dedicated status and error codes. If only the abovementioned state-sets are treated by the inSolar solution, this information goes to waste, so the inSolar solution records and presents to the user the time-series of these codes in the form of 2 raw, real-time data parameters (text containing the inverter status and error messages and time-stamps) as per section 4.8 for each inverter separately.

Finally, one state-set is supported for each string monitor, encoding the following states:

- String monitor in normal operation (measuring): The string monitor is operating normally and measures the string current
- String monitor in stopped mode: The string monitor is stopped and needs service in order to resume operation

String monitors also provide status and error codes and, in analogy to the inverters, the inSolar solution records and presents to the user the time-series of these codes in the form of 2 raw, real-time data parameters (text containing the string monitor status and error messages and time-stamps) as per section 4.8 for each string monitor separately.
The reader has to be advised that the state-set list provided in the present section is indicative and may be extended or narrowed down according to the inverter and string monitors being employed at the monitored PV plant. For this reason, the user has to consult the documentation of the respective components. Moreover, in order for the user to interpret the textual representation of error and status codes provided by the inverter and string monitors, he/she has to consult the respective manual.

#### *6.2.9.2 Primary state-sets related to environmental parameters affecting production*

The only environmental parameter which is related to the production of the PV plant is the module temperature presented in section 6.2.1. The real-time data used for the calculation of the respective primary parameter are thresholded against high/low limits treated as settings by the inSolar solution and the PV panel module temperature status is updated. 3 states are supported, namely:

- Module temperature normal
- Module temperature low
- Module temperature high

One state-set is supported for each PV panel group which is monitored by a temperature sensor.

#### *6.2.9.3 Derived state-sets*

The only derived parameter that updates a state-set is the performance ratio, since it is the only indicator that normalizes all free parameters related to the PV plant performance. Performance ratios are calculated per PV array, PV array group and PV plant as per section 6.2.6.4. When the recording period (and not the daily/monthly/yearly) performance ratios drop below pre-defined values, treated as settings by the inSolar solution (different for PV arrays, PV array groups and the PV plant as a whole), the respective states (also encoding using different state-sets for PV arrays, PV array groups and the PV plant as a whole) are updated. The states are 2, namely performance ratio low and normal.

 

## 6.3 Monitoring of the interconnection of the PV plant with the EPS

The PV plant is essentially a distributed energy production resource connected to the Electrical Power System via the Point of Common Coupling (PCC). As already mentioned and depending on the PV plant output power, such an interconnection may be implemented at low-voltage (220 Volts) or at high-voltage (which in Greece is 20 kVolts). In the former case, the inverters are directly connected to the EPS, while in the second, step-up transformers are employed, which are considered part of the interconnection.

Other equipment installed at the PV plant, interconnected with the EPS are the loads. Loads can also be connected either at low or high-voltage via a transformer, which is also considered a load. 3 different points of the electrical installation of the PV plant, which implement its interconnection with the EPS are separately monitored by the inSolar solution, namely:

- The Point of Common Coupling (PCC)
- The Transformer Connection Point (TCP)
- The Load Connection Point (LCP)

### 6.3.1 Monitoring of the PCC

The PCC is the point at which all requirements imposed by the EPS apply. These requirements, standardized within Reference 5, mandate:

- The response of the PV plant to abnormal utility grid conditions in order to enhance EPS stability, since the PV plant production equipment (inverters) uses the EPS as a voltage and frequency reference
- The quality of the power the PV plant produces

As far as the first requirement set is concerned, the PV plant is required to automatically detect when specific limits on specific parameters related to the operational characteristics of the utility grid are exceeded or the utility grid experiences certain faults and react accordingly by disconnecting from the EPS (a “trip” in related bibliography). The “trip” has to adhere to very strict timing requirements, so the sampling interval and the averaging used for the production monitoring (section 4.3) cannot be employed, but large sampling rates, coupled with sophisticated event detection mechanisms will have to be implemented, something that is outside the scope of the inSolar solution. For these reasons, external equipment is employed (the PCC protection device depicted in Figure 1), so that the inSolar solution gets informed only on the occurrence of such events.

In the case of a “trip”, this protection device has to monitor the PCC after the “trip” and automatically re-connect the PV plant when the parameters return to their normal values and remain there for a period of time (this action is denoted as “reclosure” in the bibliography). According to Reference 5, PCC “trips” must be triggered upon occurrence of:

- <u>Over/under frequency conditions:</u> The utility controls system frequency and the PV system must operate in synchronism with the utility. Any deviation must cause a disconnection of the PV system from the utility grid. The frequency limits that lead to “tripping” are stated within Reference 5 and are programmed within the protection device at installation time.
- <u>Rate of change of frequency (ROCOF) violation:</u> This protective function targets at tripping the PV plant if an islanding condition is detected. Islanding is a situation

potentially threatening to human lives during which the PV plant supplies with power local loads even when the utility grid is offline. During the transition to islanding, the ROCOF is increased.

- Over/under voltage conditions (per phase): Utility-interconnected PV systems do not regulate voltage; they inject current into the utility. Therefore, the voltage operating range for PV inverters is selected as a protection function that responds to abnormal utility conditions, not as a voltage regulation function. Voltage values outside the pre-defined range must cause a disconnection of the PV system from the utility grid. The voltage limits that lead to "tripping" are stated within Reference 5, apply per phase and are programmed within the protection device at installation time.
- Overcurrent conditions (per phase): 2 thresholds are employed to detect overcurrent conditions. If the current exceeds the first, an overcurrent condition is indicated (no trip) while if the second (set higher than the first) is exceeded, a "trip" must occur and a short-circuit condition is indicated.
- Grid faults: There exist 2 types of faults (neutral voltage displacement and earth fault) that may occur within the grid which must lead to a "trip" of the PV plant.

All tripping events and their causes must be reported to the user, since they represent critical errors. Trip codes, depending on the cause, are encoded as per Reference 8. It has to be noted that the inverters also monitor the voltage and frequency and go "off-grid" (i.e. they "trip") regardless of the state of the protection device. This is the strategy to be followed for low-power plants, since the protection device is mandated by the EPS operator only in the case of PV plants being connected to the high-voltage. The inverters are expected, in any case, to report "off-grid" events through their state-sets as per section 6.2.9.1.

As far as the second requirement set is concerned, Reference 5 dictates limits on several power quality parameters, measured at the PCC, the deviation of which from specified limits must be monitored for compliance, even though it does not lead to a "trip". These are the following and apply only when the irradiance is above a certain threshold, i.e. only when the PV plant is producing power:

- Reduced power factor (true power factor, lagging or leading): Most modern inverters are designed to operate at a unity power factor. Deviations may indicate malfunctions, though only during the day, i.e. during the time the PV plant produces power.
- Increased DC injection: Excessive DC current on the grid can damage attached utility equipment and customer loads. Monitoring of this parameter is unnecessary for systems with dedicated output transformers.
- Increased waveform (current harmonic) distortion (per phase and the neutral): The PV system output should have low current distortion levels to ensure that no adverse effects are caused to other equipment connected to the utility system. Harmonic distortion is contributed by modern inverters due to their switching mode of operation. The current harmonic distortion is measured as the ratio of all harmonics over the current fundamental.
- Increased phase current unbalance (per phase): The phase currents of the inverter or system output must be reasonably balanced and pose no phase balance problem at the point of utility interconnect, since unbalanced phase currents may cause

overheating of the utility transformer. Unbalance within an inverter generally indicates a problem.

- Voltage flicker: This is the fluctuation of the PV plant output voltage. Since the monitoring of voltage flicker calls for sophisticated power quality meters which, most probably cannot be afforded by PV plant owners, it is not further dealt with within the present document.

For the already presented reasons, the detection of the occurrence of events leading to trips has to be monitored through the protection device depicted in Figure 1. Events that do not lead to trips are monitored through both the PCC multimeter and protection device of Figure 1. All thresholds for event detection are configured within the PCC protection device and multimeter by the PV plant designer at installation time so that the PV plant adheres to national codes of conduct. Consequently, the inSolar solution is not required to set them.

As mandated within relevant standards, both the PCC multimeter and protection device must be connected to the AC output of the inverters, ideally between the inverter and the PCC. If there is a dedicated transformer at the output of the inverter, the connection has to be made on the utility side of this transformer, as depicted in Figure 1.

#### *6.3.1.1 State-sets*

The occurrence of the abovementioned events is encoded by the inSolar solution using the following state-sets, supported for the PCC entity presented in Figure 3:

- State-sets related to “trips”:
    - Connectivity status:
        - Relay tripped (off-grid)
        - Relay closed (on-grid)
    - Frequency status:
        - Overfrequency
        - Underfrequency
        - Frequency normal
    - ROCOF status:
        - ROCOF high
        - ROCOF normal
    - Voltage status:
        - Overvoltage on one or more phases
        - Undervoltage on one or more phases
        - Voltage normal on all phases
    - Current status:
        - Short-circuit on one or more phases
        - Overcurrent on one or more phases
        - Current normal on all phases
    - Earth fault status:
        - Earth fault present
        - Earth fault not present
    - Neutral voltage displacement status:
        - Neutral voltage displacement present

        - Neutral voltage displacement not present
- State-sets not related to “trips”:
    - Phase current imbalance status:
        - Phase current imbalance high on one or more phases
        - Phase current imbalance normal on all phases
    - Reduced power factor status (during the day):
        - Power factor low
        - Power factor normal
    - Increased DC injection status:
        - DC injection high
        - DC injection normal
    - Increased current harmonic distortion status:
        - Current harmonic distortion high on one or more phases or the neutral
        - Current harmonic distortion normal on all phases

Since the selection of the PCC multimeter and protection device is made by the PV plant designer, some variations are anticipated. It is, however, desirable for them to adhere to the list of state-sets presented above. If, in any case, a state-set is not monitored by either the multimeter or the protection device, it is not presented to the user by the inSolar solution.

#### *6.3.1.2 Parameters*

Apart from the abovementioned states, it is desirable that the inSolar solution monitors a number of parameters for the PCC entity, some of which have already been presented within section 6.1 since they are also necessary in the monitoring of the PV plant production facilities. These parameters, which have been discussed with potential customers and can be provided either by the PCC multimeter or the protection device or both, are presented in the table below.

Since the selection of the PCC multimeter and protection device is made by the PV plant designer, some of the parameters presented below may not be available. It is, however, desirable that the set adheres to the list below.

Since the 2 devices monitor different points of the electrical installation of the PV plant, all available parameters from both are monitored with a proper designation. Readings from the PCC multimeter are indicated with a “PCC” prefix while for readings from the protection device, the “Production” prefix is used.

| | Measured parameter | Parameter type | Units |
|---|---|---|---|
| *Power* | Reactive power of phase L1 | Real-time monitored | kVAr |
| | Reactive power of phase L2 | Real-time monitored | kVAr |
| | Reactive power of phase L3 | Real-time monitored | kVAr |
| | Active power of phase L1 | Real-time monitored | kW |
| | Active power of phase L2 | Real-time monitored | kW |
| | Active power of phase L3 | Real-time monitored | kW |
| | Apparent power of phase L1 | Real-time monitored | kVA |
| | Apparent power of phase L2 | Real-time monitored | kVA |
| | Apparent power of phase L3 | Real-time monitored | kVA |
| | Total reactive power across phases | Primary/Real-time monitored | kVAr |
| | Total active power across phases | Primary/Real-time monitored | kW |
| | Total apparent power across phases | Primary/Real-time monitored | kVA |
| *Energy* | Apparent energy | Maximum | kVAh |
| | Active Energy Import | Maximum | kWh |
| | Active Energy Export | Maximum | kWh |
| | Reactive Energy Import | Maximum | kVArh |
| | Reactive Energy Export | Maximum | kVArh |
| *Current* | Current of phase L1 | Primary/Real-time monitored | Amps |
| | Current of phase L2 | Primary/Real-time monitored | Amps |
| | Current of phase L3 | Primary/Real-time monitored | Amps |
| | Maximum current of phase L1 | Maximum | Amps |
| | Minimum current of phase L1 | Minimum | Amps |
| | Maximum current of phase L2 | Maximum | Amps |
| | Minimum current of phase L2 | Minimum | Amps |
| | Maximum current of phase L3 | Maximum | Amps |
| | Minimum current of phase L3 | Minimum | Amps |
| | Average Current (across L1, L2, L3) | Real-time monitored | Amps |
| *Voltage* | Average Voltage (across L1, L2, L3) | Real-time monitored | Volts |
| | Average Voltage (across L1-L2, L2-L3, L3-L1) | Real-time monitored | Volts |
| | Voltage of phase L1 | Real-time monitored | Volts |
| | Voltage of phase L2 | Real-time monitored | Volts |
| | Voltage of phase L3 | Real-time monitored | Volts |
| | Voltage L1-L2 | Primary/Real-time monitored | Volts |
| | Voltage L2-L3 | Primary/Real-time monitored | Volts |
| | Voltage L3-L1 | Primary/Real-time | Volts |

| | | | |
|---|---|---|---|
| | | monitored | |
| | Maximum voltage L1-L2 | Maximum | Volts |
| | Minimum voltage L1-L2 | Minimum | Volts |
| | Maximum voltage L2-L3 | Maximum | Volts |
| | Minimum voltage L2-L3 | Minimum | Volts |
| | Maximum voltage L3-L1 | Maximum | Volts |
| | Minimum voltage L3-L1 | Minimum | Volts |
| | Frequency | Primary/Real-time monitored | Hz |
| | Maximum frequency | Maximum | Hz |
| | Minimum frequency | Minimum | Hz |
| *Power factor* | Power factor of phase L1 | Primary | Dimensionless |
| | Power factor of phase L2 | Primary | Dimensionless |
| | Power factor of phase L3 | Primary | Dimensionless |
| | Total power factor across phases | Primary | Dimensionless |
| *THD* | THD-R of the current of phase L1 | Primary | Percentage |
| | THD-R of the current of phase L2 | Primary | Percentage |
| | THD-R of the current of phase L3 | Primary | Percentage |
| | THD-R of the current of the neutral | Primary | Percentage |
| *Current Unbalance* | Current unbalance for phase L1 | Primary | Percentage |
| | Current unbalance for phase L2 | Primary | Percentage |
| | Current unbalance for phase L3 | Primary | Percentage |

**Table 1. Parameters related to the monitoring of the PCC**

Parameters of the "maximum/minimum" types adhere to the requirements of sections 4.4 and 4.7. Parameters of the "real-time monitored" type adhere to the requirements of section 4.9. Daily, monthly and yearly values of the parameters indicated as primary in the table above are calculated using the formula below, taking into consideration the completeness threshold of section 4.7:

$$Value_{T,\tau} = \frac{\sum_{t=T-\tau}^{T} Parameter_t}{\sum_{t=T-\tau}^{T} V_t}$$

where:

- $\tau$ is the length of the reporting period in hours (days, months, years)
- $T$ denotes the recording period during which the calculation is carried out
- $Parameter_t$ is a vector of the values of the proper recording period primary parameter spanning the reporting period

- $V_t$ is a data validity vector, containing ones for recording periods where all formula inputs contain valid parameter values and zeros for recording periods where at least one formula input has an invalid value

Cumulative parameter values (i.e. energies) may be either provided by the PCC multimeter/protection device as primary parameters (section 6.2.4.1) or calculated from power readings as per section 6.2.4.2 (derived parameters). The energies are then corrected as per section 6.2.4.3 through the use of related offsets, implemented as settings within the inSolar solution. Energy values for the current day as well as for past days/months/years are calculated as per section 6.2.4.4.

### 6.3.2 Monitoring of a TCP

The main component of a Transformer Connection Point is, of course, the transformer, whose duty is to raise the LV output of the inverters to medium voltage, so as to connect the PV plant to the distribution network. Depending on the nominal power of the plant, transformers may or may not be employed.

Transformers can generally be categorized as follows:

- Dry-type transformers, whose dielectric is a solid substance
- Oil-type transformers, whose dielectric is mineral oil. This transformer class is further sub-divided in 2 categories:
    - The air-breathing conservator-tank type, operating at atmospheric pressure
    - The hermetically-sealed totally-filled tank type

Transformers are usually monitored through multimeters placed either along the HV side or the LV side or both. Protection against internal faults is provided by a variety of devices, depending on the transformer type and a protective relay is used to disconnect the transformer from the grid for safety reasons.

#### *6.3.2.1 Parameters*

In order to unify the way the inSolar solution tackles electrical parameter monitoring, it has been decided to treat TCPs (see Figure 3) in the same way as the PCC presented in section 6.3, so the electrical parameters are the same as those presented in Table 1 for all multimeters and protection devices related to transformers, regardless of whether they are placed at low or high voltage (see Figure 1). Parameters related to the transformer input and output are different and distinguished using a suitable prefix (High Voltage – HV or Low Voltage – LV). If a transformer protection device is installed, the parameters it measures are the same as those monitored through the HV multimeter, since it is placed at the secondary transformer winding.

#### *6.3.2.2 State-sets*

The state of a transformer can be monitored through a variety of status monitoring devices, depending on their type (for a complete justification, please consult Reference 10):

- For <u>both dry and oil-type transformers</u>, the temperature is monitored using a thermostat with 2 contacts, one indicating a mild increase of the temperature above the nominal level (temperature alarm) and the second indicating a serious deviation, probably caused by an internal fault. When this situation is encountered, the protective relay is tripped and the transformer is automatically disconnected from the grid

- Oil-type transformers of the air-breathing conservator-type are protected against internal failures by an ingenious electromechanical device called the Buchholz relay. This device issues an alarm and disconnects the transformer from the grid under certain conditions.
- Oil-type transformers of the hermetically-sealed totally-filled tank type are protected against internal failures by a device called DGPT (Detection of Gas, Pressure and Temperature), since the principles of operation of the Buchholz relay do not apply. The DGPT detects the existence of gas as well as changes in the internal pressure of the transformer (both indicating electrical arcs within the transformer). Protection from overtemperature is implemented as presented above.

Additionally, a protection device may be installed across the HV winding of the transformer (regardless of its type), detecting overcurrent conditions which indicate possible short-circuits as well as earth faults of the cabling between the transformer output and the PCC. This protection device can also trip the protective relay.

The following state-sets are thus supported by the inSolar solution to properly record the state of the transformer of each TCP:

- State-sets updated by the protection device:
  - Current status:
    - Short-circuit on one or more phases
    - Overcurrent on one or more phases
    - Current normal on all phases
  - Earth fault status:
    - Earth fault present
    - Earth fault not present
- State-sets updated by the Buchholz relay:
  - Buchholz relay status:
    - Buchholz relay alarm
    - Buchholz relay trip
    - Buchholz relay normal
- State-sets updated by the DGPT device:
  - Gas accumulation status:
    - Gas accumulation present
    - Gas accumulation not present
  - Gas pressure status:
    - Overpressure
    - Pressure normal
- State-sets updated by the thermostat or DGPT device:
  - Temperature status:
    - Temperature alarm
    - Temperature trip
    - Temperature normal

Additionally, the state of the protection relay is monitored by the inSolar solution:

- Connectivity status:

  - Relay tripped (off-grid)
  - Relay closed (on-grid)

Apart from the state-sets presented above, the inSolar solution additionally supports the state-sets of section 6.3.1.1 that encode events which do not lead to trips, though different for the input and output.

### 6.3.3 Monitoring of the LCP

For the same reasons presented in the sections above, it has been decided to treat the load connection point (see Figure 1) in the same way as the PCC of section 6.3. The electrical parameters are the same as those presented in Table 1.

The state-sets for the LCP entity are used to monitor the power supply to the loads as follows (whenever the support provided by multimeters allows it):

- Current status:
  - Overcurrent on one or more phases
  - Current normal on all phases
- Voltage status:
  - Overvoltage on one or more phases
  - Undervoltage on one or more phases
  - Voltage normal on all phases

If a transformer exists and its status is monitored, the state-sets supported by the inSolar solution are the same as those presented in section 6.3.2.2. Apart from the multimeter and the transformer, the LCP entity also supports monitoring of a transfer switch, which connects the loads to various electrical power sources, depending on its position. These sources include:

- A mechanical generator
- A wind generator
- An auxiliary electrical power source
- The EPS/PV plant output

The abovementioned sources encode a state-set, which is augmented by an OFF state to be used when the loads are not connected to any power supply. The transfer switch notifies the inSolar solution on its position using contacts. Up to 8 contacts are supported to cater for more states that may appear in the future.

 

## 6.4 Monitoring of electrical panels

The electrical installation of the PV plant includes a number of electrical panels. These panels may include:

1. Circuit breakers, switches and fuses which isolate parts of the electrical installation
2. Surge arresters, protecting circuits which enter the PV plant shelters from outdoors against lighting strikes
3. Multimeters and protection devices, which are not related to the PCC, TCPs or LCP, as presented in Figure 1

Circuit breakers provide auxiliary contacts which encode their state as “closed” or “tripped”. The same applies for switches (the respective states are “closed” and “open”), fuses (the respective states are “Normal” and “Blown”) and surge arresters (the respective states are “Normal” and “Needs replacement”, if the surge arrester has been destroyed by lightning). It has to be noted that the auxiliary contacts which encode the state of circuit breakers, switches, fuses and surge arresters are usually bused to the inSolar solution in groups, i.e. the state of each one is not encoded separately, but instead an OR operation is applied. It is understandable that each such bus has to contain only components of the same type.

The parameters/state-sets related to multimeters and protection devices are those presented in section 6.3.1. The parameters provided by multimeters and protection devices follow the sign conventions presented in section 4.13, depending on whether they are monitoring loads or production facilities.

## 6.5 Monitoring of the shelter room status

The temperature of each shelter room (in ºC) is measured separately through the temperature sensors/transducers presented in Figure 3 as per section 4.3 (primary parameter). Daily/monthly/yearly derived parameters are calculated by the inSolar solution as per section 6.1.1. This parameter is also real-time monitored as per 4.9.

State-sets that encode the status of a shelter room include:

- Temperature status: The real-time data provided by the temperature sensor located in the room is thresholded against high and low limits treated as settings by the inSolar solution and the state is updated correspondingly. 3 states related to temperature are supported, namely low, high and normal.
- Flood status: Determined via related sensors, located inside the shelter rooms, which detect the presence of water. The encoded states are:
    - Room not flooded
    - Room flooded
- Fire status: Determined via related sensors, located inside the shelter rooms, which detect the presence of fire or smoke or both. The encoded states are:
    - Room not on fire
    - Room on fire
- Occupancy status: Determined via related sensors, located inside the shelter rooms, which detect the presence of movement. The encoded states are:
    - Room not occupied
    - Room occupied

## *6.6 Monitoring of the shelter room access-controlled door status*

The inSolar solution supports 2 state-sets for each access-controlled door:

- One state-set encoding the door status (“door left open/closed”), which is updated by a magnetic contact (“door open/closed” contact) and the electric door bolt (“door locked/unlocked” contact), located on each shelter room access-controlled door. More specifically, the door status is considered as “closed”, whenever it is closed and locked and as “open” in any other case. If the door is left open beyond a configurable user-defined time setting, a “door left open” alarm is activated. The alarm is de-activated when the door is closed again.
- One state-set encoding the door lock override status (door lock overridden/under local control or not).

The access-controlled door entity also supports one setting, namely the door left-open hysteresis.

## *6.7 Monitoring of the site perimeter status*

State-sets that encode the status of the PV plant site perimeter include:

- The alarm zone activity:
    - Alarm zone active (at least one alarm belonging to the specific alarm zone is active)
    - Alarm zone inactive
- The actuator zone activity:
    - Actuator zone active (all actuators of the specific actuator zone have been activated)
    - Alarm zone inactive

## 6.8 Monitoring of the operational and communication status of equipment monitored/controlled by the inSolar solution

Apart from the monitoring of the vast set of parameters and state-sets presented in previous sections, the inSolar solution also supervises all equipment being monitored and controlled (sensors, multimeters, protection devices, inverters, etc.) for failures and loss of communication with them, since such events indicate that either the operation of the PV plant is affected or the capability of the inSolar solution to monitor/control the PV plant has been impaired.

2 related state-sets are independently supported for each piece of equipment:

- Malfunction status:
  - Equipment is malfunctioning
  - Equipment is operating normally
- Communication status:
  - The inSolar solution is communicating with the equipment
  - The inSolar solution is not communicating with the equipment

The reason for the existence of 2 different state-sets is that since, depending on the equipment type, it is not always possible to discern between malfunctions and failures in the communication. A multimeter, for example, may notify the inSolar solution on its failure or may be damaged in such a way that the communication with it is interrupted before such a notification is issued.

### 6.8.1 Treatment of malfunctions/communication errors related to the inverter and string monitors

Modeling the inverter behavior as far as malfunctions are concerned is a complicated task, since many inverters models cease to communicate every time the sun goes down. Since string monitors almost always communicate using the inverter as a bridge, the same problem applies.

The reason is that the communication modules of the inverters are powered by solar energy. Unfortunately, this loss of communication also indicates damaged inverters. In order for the inSolar solution to discern between these 2 conditions, one more piece of information is necessary. This clue is the irradiance.

More specifically, the inSolar solution determines whether it is day or night based on the total irradiance parameter of the PV plant entity (Figure 3) and a pre-configured threshold. This approach enables some redundancy, i.e. for plants employing multiple pyranometers, the day/night indication continues to function, based on the readings of the remaining pyranometers, even when a single pyranometer malfunctions.

During the day, any interruption of the communication with the inverter is interpreted as an inverter failure. Whenever communication is possible, the inSolar records the actual states of the inverter and string monitors related with it as per section 6.2.9.1.

During the night, the interruption of the communication with the inverter is inevitable for certain inverter models, so the state that is recorded is “Night shutdown”, again as per section 6.2.9.1.

It is evident that if no pyranometer is installed, the inSolar solution cannot implement the algorithm specified above. In that case, the inSolar solution operates as if the sun is always up.

 

During a communication error situation, the inSolar solution does not record any of the parameters provided by the inverter or the string monitors, except for:

- The PV panel group and array output powers. The gaps for these 2 parameters are filled with zeros as per section 6.2.2.1.
- The cumulative measured PV panel group and array output energies (see 6.2.4.1). The gaps for these 2 parameters are filled with the last valid value for the duration of the communication error.

No state-change events, except for those related to the communication error/malfunction are recorded during this period.

### 6.8.2 Treatment of malfunctions/communication errors related to equipment directly providing parameters

Certain pieces of the equipment presented in Figure 3 directly provide the real-time data necessary for the calculation of one or more primary parameters. These are:

- Certain sensors/transducers (temperature, irradiance, barometric pressure, rain gauge, humidity, wind speed, wind direction)
- Multimeters
- Protection devices

The inSolar solution does not record the values for any of the primary parameters provided by such equipment throughout a communication error condition with them. No state-change events, except for those related to the communication error/malfunction are recorded during this period.

### 6.8.3 Treatment of malfunctions/communication errors related to equipment directly encoding state-sets

Certain pieces of the equipment presented in Figure 3 directly encode one or more state-sets. These are:

- Certain sensors (flood, occupancy, fire)
- Contacts/actuators
- Protective relays
- The electric door bolt

The inSolar solution retains the last known state throughout a communication error condition with the abovementioned equipment.

## 6.9 Monitoring of the status of IT equipment

The IT segment (only one for each PV plant) is monitored through:

- The average, maximum and minimum round trip time primary parameters (recording period only, measured in milliseconds), whose values are measured by pinging the modem/router of the PV plant for a number of times during the recording period and performing an average (as per section 4.3) or maximum/minimum operation (as per section 4.4) on the ping results
- The site link status state-set, consisting of 2 states:
  - Site online, when the ping to the router/modem presented in Figure 11 is successful
  - Site offline, otherwise

Each controller is monitored through 2 state-sets:

- The controller connection link status state-set, consisting of 2 states:
  - Controller online, when the ping to the controller is successful
  - Controller offline, otherwise
- The controller status state-set, consisting of 2 states:
  - Controller available, when the interface to the controller is operational (i.e. data transfer is possible). The condition for entering this state depends on the controller being employed.
  - Controller unavailable, otherwise

Additional state-sets/parameters may be used for monitoring purposes, depending on the controller being employed.

Each video server is monitored through the video server connection link status state-set, consisting of 2 states:

- Video server online, when the ping to the video server is successful
- Video server offline, otherwise

# 7 The control service

Apart from the extensive monitoring functionality presented in the sections above, the inSolar solution also implements some control functionalities specified in the sections below.

## *7.1 PV plant security control*

Upon occurrence and persistence of an event from an alarm zone for a preset period of time, the inSolar solution activates the actuators of any related actuator zones (for a definition of alarm and actuator zones please consult section 6.7). The inSolar solution provides all means necessary to configure the relationship among alarm and actuator zones. This functionality does not depend on user interaction. The user is provided with feedback on the status of each alarm and actuator zone independently.

## *7.2 Access control*

The inSolar solution controls the access to the shelter rooms. The visiting person holds an RFID tag. He/she approaches it to a proximity reader. If the identity is valid, entry to the room is granted through unlocking an electric door bolt. In any other case, the access is denied. Notification of the access status (granted or not) is provided to the local user through an optical and/or sound indication. The access control scheme also includes a “door open” button, placed inside the room, to allow the visitor to exit the shelter and a magnetic contact on the door together with a contact within the electric bolt to indicate whether the door is open, closed, locked or unlocked. The inSolar solution activates a buzzer if the door is left open, so as to notify the person there to close it. The access control system is preset as to which RFID tag opens which door and it operates without the intervention of the remote user(s). However, the latter are informed on the status of the access controlled room door as per section 6.6.

## *7.3 Handling of settings*

The inSolar solution provides the user with the capability to view and configure the settings that are supported for each of the entities presented in Figure 3. The user is notified on the status of the setting configuration (pending, failed or successful).

## *7.4 State-change event acknowledgements*

The inSolar solution provides the user with the capability to acknowledge and view the acknowledgement status of each state-change event independently for each of the entities presented in Figure 3 according to section 4.10.

## *7.5 Handling of real-time monitors*

The inSolar solution provides the user with the capability to start and stop the real-time monitors that are supported for each of the entities presented in Figure 3. The user is notified on the status of the monitor setup status (pending, failed or successful).

 

# 8 The notification service

Except for the monitoring and control services described in the previous sections, the inSolar solution is also capable of notifying the user on the occurrence of critical events via SMS or e-mail.

The generation of such notifications is not triggered by individual state-change events but rather by the status changes of each of the entities presented in Figure 3 affected by them, according to section 4.10. More specifically, if the severity of the status goes above a certain user configurable threshold and remains there for a preset period of time, a notification is generated to alert the user. Additionally, if the status falls below the same threshold, another notification is generated in order to notify the user on the expiration of the criticality.

The user is capable of subscribing to the status changes of and configuring the severity thresholds for each entity separately. It is understandable that if he/she, depending on the subscription he/she has configured, may receive multiple notifications upon occurrence of the same event or set of events affecting the statuses of a certain set of entities. For example, if the user subscribes for status changes of an inverter and the PV plant which contains it, he/she shall be notified twice when the inverter breaks down. Once because of the subscription to the status of the inverter and once more because of the subscription to the PV plant status.

Moreover, the user is capable of subscribing separately to the positive edge (status going above threshold) or the negative edge (status going below threshold) for each entity separately.

The notification text includes the full path to the entity within the PV plant hierarchy as per Figure 3 and the unacknowledged state-change event which triggered the notification, as well as the time/date of the event and the severity after the status change.

Finally, the inSolar solution logs all notifications sent with the following information:

- Notified user
- Notification method used
- Which subscription triggered the notification
- Time/date
- Delivery status

# 9 The graphing service

In order for the user of the solution to reach useful conclusions regarding the state of the monitored PV plant and possible faults as well as their probable causes, there is a definite need for concise presentation of the numerous monitored parameters and events using a multitude of graphs. This functionality is offered by the graphing service, which is built around 3 graph types:

- The x vs. y plot, presenting, in a scatter form, the values of 1 parameter vs. the values of the other
- The time plot of up to 2 parameters on a time axis, as a bar, stacked bar or line plot.
- The timeline plot of state-change events vs. their time of occurrence

For the time plot and if 2 parameters are presented on the same plot, 2 y-axis shall be used to cater for parameters which do not share the same range.

The inSolar solution provides the user with the capability to configure:

- The time period for which a plot shall be produced ("from/to" selection, aligned on day boundaries as well as pre-sets including current day, yesterday, last month, last year, from the beginning of operation)
- The data values to be used for the plot (recording period, daily, monthly, yearly) as per sections 4.6 & 4.7

The necessary graphing functionality is specified in detail within the sections below, which also provide guidelines on the interpretation of the graphs. This specification follows the hierarchical structure presented in Figure 3.

## 9.1 Plots related to PV arrays

The class of plots described within the present section deals with the presentation of data from PV arrays alone and is treated separately due to the level of detail that is necessary for failure identification.

### 9.1.1 Relative array efficiency vs. temperature

The mean array efficiency ($\eta_{A,mean}$) as per section 6.2.6.1.1 represents the mean energy conversion efficiency of the PV array, which is useful for the comparison with the nominal array efficiency $\eta_{A0}$ at its rated power $P_0$. The difference between these 2 efficiency values represents diode, wiring and mismatch losses as well as energy wasted during plant operation.

In general, when the module temperature raises, the ratio of the mean array efficiency over the nominal array efficiency (the relative array efficiency) drops in a linear fashion. The angle of the best fit straight line indicates the temperature coefficient of the PV module, which should match the manufacturer datasheets. An exemplary scatter plot, taken from Reference 6, is presented below and depicts data from 43 PV plants in Switzerland.

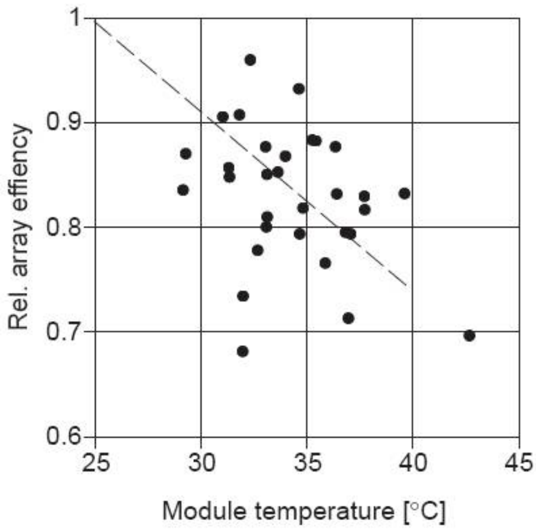


**Figure 14. Exemplary plot of relative array efficiency vs. module temperature**

### 9.1.2 Relative array efficiency vs. irradiance

At low irradiance levels, the array does not have the same ability to convert sunlight into electric energy, so its relative efficiency drops. The effect is depicted in the exemplary scatter plot below, taken from Reference 7 and presenting data from a Canadian PV plant.

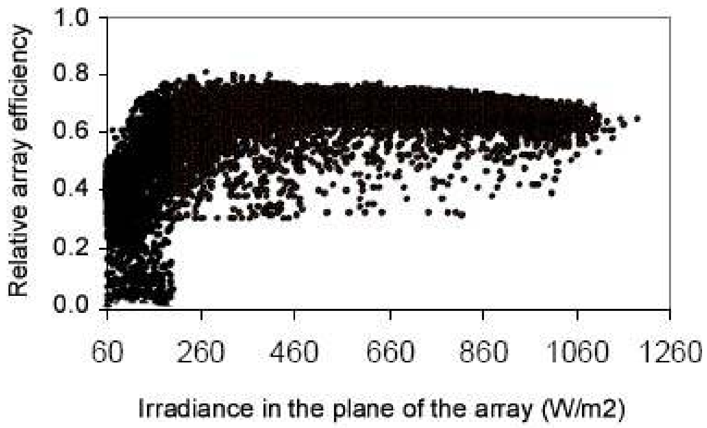


**Figure 15. Exemplary plot of relative array efficiency vs. irradiance**

## 9.1.3 Relative array efficiency vs. performance ratio

The relative array efficiency (ratio of mean and nominal array efficiencies) of a grid-connected PV system is a good indicator for the evaluation of the system operational performance. Good operation of systems (Performance Ratio > 0.7, the performance ratio being defined according to section 6.2.6.4) can be expected from systems with a ratio $\eta_{A,mean}/\eta_{A0}$ > 80 %.

The following scatter plot (taken from Reference 6 and presenting the results from 34 German PV plants interconnected with the grid), together with the least-squares linear fit proves this rationale. Deviations from the expected behavior indicate probable inverter-related problems. It has to be noted that the performance ratio used in the graph is defined after the inverter.

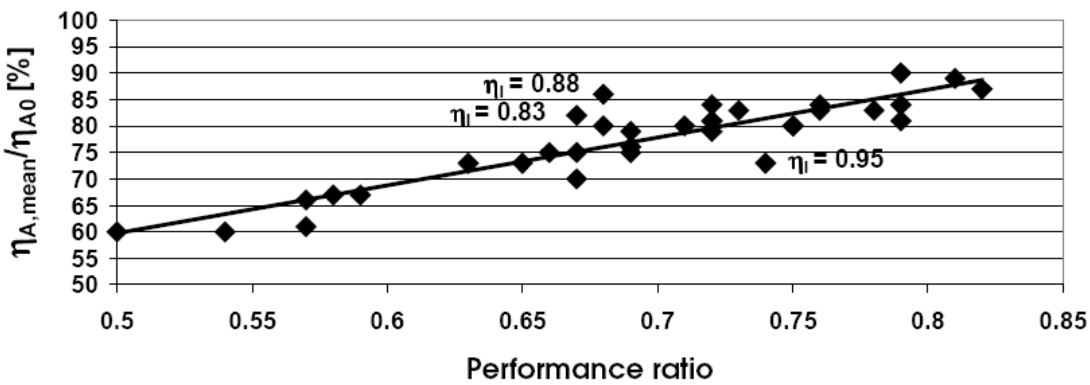


**Figure 16. Exemplary plot of relative array efficiency vs. performance ratio (post-inverter)**

## 9.1.4 Inverter efficiency vs. DC power

The efficiency of the inverter (as per section 6.2.6.1.2) depends on the panel group output power (as per section 6.2.2.1.1) and at low DC power levels, which correspond to low irradiation, it drops significantly. The exemplary scatter plot taken from Reference 7 and presented below, depicts data from a grid-connected PV system in Canada. Large deviations from this performance curve indicate possible problems with the inverter.

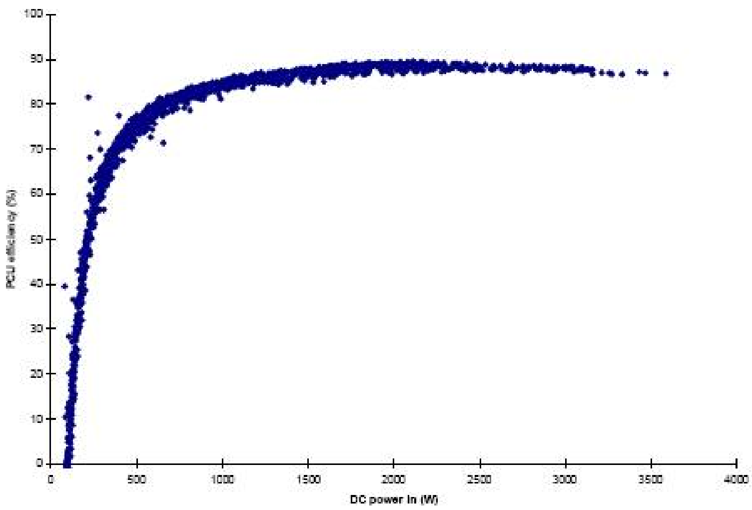


**Figure 17. Exemplary plot of inverter efficiency vs. output power**

### 9.1.5 Final yield/array capture losses/system losses vs. time

The performance ratio (see section 6.2.6.4), being the ratio of the final yield to the reference yield, is the most useful normalized performance indicator. It indicates the overall effect of losses on the array's nominal energy output due to array temperature, incomplete utilization of irradiation, and system component inefficiencies or failures. For the comparison of different PV plants, particularly grid-connected PV systems, the performance ratio is widely used because it is independent of the system size and location of the plant.

If, however, the performance ratio is low, which points to potential problems, it is not feasible to identify the cause. In that case, a stacked bar graph of the final yield (as per section 6.2.6.2.3), the array capture losses (as per section 6.2.6.3.1) and the system losses (as per section 6.2.6.3.2) can provide valuable information. In the following 2 figures, taken from Reference 6, the stacked bar graphs of 2 systems are presented. Both demonstrate a performance ratio of 0.3, but they have substantial differences, since for the one related to the graph on the left hand side, most of the energy goes wasted while converting solar energy to electricity (low array efficiency), while for the one on the right, the inverter is the inefficient component.

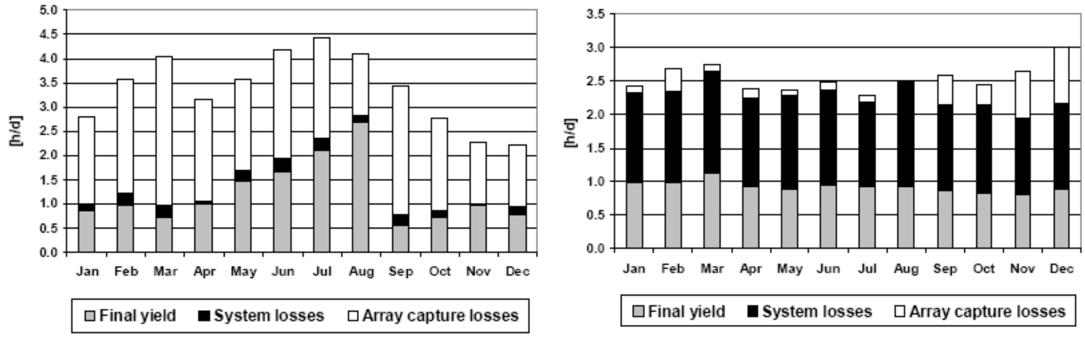


**Figure 18. Exemplary bar graph of final yield/array capture losses/system losses vs. time**

In the figure above, the array capture and system losses are quite constant across time, so they represent a steady-state behavior. A malfunction or inverter failure, on the other hand, shall result in a remarkable rise of array capture losses across time. If the grid-connected system fails completely, the values of array yield, final yield and performance ratio will drop to zero, while capture losses will rise towards the reference yield and system losses become negligible. To sum this section up, the temporal evolution of the final yield/array capture losses/system losses vs. time bar graph can provide valuable information on panel and inverter operation.

### 9.1.6 State-change events vs. time

This group includes a plot of the state-change events related with PV arrays (see section 13.4 for a complete list) versus time as a timeline.

### 9.1.7 Parameter(s) vs. time

This group includes plots of any of the parameters related with PV arrays (see section 13.4 for a complete list) versus time, independently of whether they are presented in other plots (see sections above) or not. The user is capable of selecting up to 2 of the available parameters for the same plot (4 if they have the same engineering unit). 2 different y axis are employed to aid in the presentation of 2 parameters which do not have the same range and engineering unit.

## 9.2 Plots related to PV array groups

The class of plots described within the present section deals with the presentation of data from PV array groups.

### 9.2.1 Final yield/array capture losses/system losses vs. time

This plot is the same as the one presented in section 9.1.5, but using the parameters for the PV array group as per section 13.3.

### 9.2.2 State-change events vs. time

This plot is the same as the one presented in section 9.1.6, but using the state-change events for the PV array group as per section 13.3.

### 9.2.3 Parameter(s) vs. time

This plot is the same as the one presented in section 9.1.7, but using the parameters for the PV array group as per section 13.3.

## 9.3 *Plots related to the PV plant*

The class of plots described within the present section deals with the presentation of data from the PV plant as a whole.

### 9.3.1 Final yield/array capture losses/system losses vs. time

This plot is the same as the one presented in section 9.1.5, but using the parameters for the PV plant as per section 13.2.

### 9.3.2 State-change events vs. time

This plot is the same as the one presented in section 9.1.6, but using the state-change events for the PV plant as per section 13.2.

### 9.3.3 Parameter(s) vs. time

This plot is the same as the one presented in section 9.1.7, but using the parameters for the PV plant as per section 13.2.

## 9.4 *Plots related to a shelter room*

The class of plots described within the present section deals with the presentation of data from shelter rooms.

### 9.4.1 Shelter room temperature vs. time

This plot is the same as the one presented in section 9.1.7, but presenting this one parameter with 1 y-axis.

### 9.4.2 State-change events vs. time

This plot is the same as the one presented in section 9.1.6, but using the events for the specific shelter room as per section 13.9.

## 9.5 *Plots related to a PV panel group*

The class of plots described within the present section deals with the presentation of data from PV panel groups.

### 9.5.1 Tracker tilt angle/tracker azimuth angle vs. time

This plot is the same as the one presented in section 9.1.7, but presenting these 2 parameters for the specific PV panel group with 1 y-axis.

### 9.5.2 Module temperature/total irradiance in the plane of the array vs. time

This plot is the same as the one presented in section 9.1.7, but presenting these 2 parameters for the specific PV panel group with 2 y-axis.

### 9.5.3 State-change events vs. time

This plot is the same as the one presented in section 9.1.6, but using the state-change events for the specific PV panel group as per section 13.5.

## *9.6 Plots related to other nodes of the PV plant hierarchy*

The first graph type to be supported by the PV plant monitoring solution for hierarchy other than those explicitly covered in the sections above is a plot of the state-change events related with them as per section 13 vs. time.

The second graph type (parameters vs. time) is the same as the one presented in section 9.1.7, but using the parameters for the specific node.

# 10 The report generation and delivery service

The inSolar solution user is capable of subscribing to multiple reporting periods (days, weeks, months or years), in order for the inSolar solution to deliver to him/her, via e-mail, 2 different types of reports for the PV plant of interest:

- A production report
- A financial report

Production reports include:

- General plant information such as name and type of the plant, latitude and longitude, nominal power, panel area and others
- A summary table, summing up the most important production metrics such as produced power, consumed power, performance ratio, irradiance and others with their minimum/maximum values and their mean value
- A section containing value tables and plots of the most important normalized performance indicators (efficiencies, performance ratio, yields and losses) with mimimum, maximum and mean values
- A section containing value tables and plots of the most important energy measurements (panel group output energy, array output energy, transformer input/output energies) with mimimum, maximum and mean values
- A section containing value tables and plots of the most important environmental measurements (irradiance, temperature and others) with mimimum, maximum and mean values

Financial reports include:

- General plant information such as name and type of the plant, latitude and longitude, nominal power, panel area and others
- A summary table, summing up the most important financial metrics such as produced power, consumed power and revenue with their minimum/maximum values and their mean value. Plots of these parameters are also provided
- A summary table, summing up the prices for the energy sold to and bought from the utility

 

# 11 The export service

The users of the inSolar solution are capable of exporting data from any entity presented in Figure 3 in a Comma Separated Value (CSV) format. The source of data to be exported may be:

- Primary parameters
- Derived parameters

The users are capable of selecting the time-span for the export as well as the reporting period for the exported data for primary and derived parameters. The format of the CSV file is as presented in the table below.

| Date/Time; | Parameter 1 name; | Parameter 2 name; | Parameter 3 name; |
|---|---|---|---|
| 1/20/10 8:30:00 AM EET; | 7.41906281E+01; | 8.30038910E+01; | 6.94309387E+01; |
| 1/20/10 8:45:00 AM EET; | 8.86931152E+01; | 8.83470993E+01; | 8.76818314E+01; |
| 1/20/10 9:00:00 AM EET; | 1.30789200E+02; | 1.34262375E+02; | 1.32416534E+02; |
| 1/20/10 9:15:00 AM EET; | 7.41906281E+01; | 8.30038910E+01; | 6.94309387E+01; |

Parameter values are exported in scientific format with 8 decimal digits and 2 exponent digits. The timestamps refer to the timezone of the plant and the parameter names are indicative of the entity they belong to. If no value exists for a reporting period, the value shall also be missing from the export file.

The values of real-time monitors are also exportable using the same format but this applies only for the period a real-time monitor was activated by the user.

 

# 12 General and system properties

Apart from the parameters and events presented in the previous sections, the inSolar solution is capable of storing, using and presenting various properties known a priori. The following sections record the properties of this type that are generally needed for each of the entities presented in Figure 3. The IEA-PVPS Task 2 Database (please consult Reference 6) has been used as a guideline.

The approach is different, depending on the entity type:

- For virtual segments, the general properties are stored by the inSolar solution for each entity separately
- For asset segments, the system properties can generally be categorized as follows:
    - System properties pertinent to the specific asset installed at the PV plant, such as the name, serial number, installation date, which are stored for each entity separately
    - System properties pertinent to the model of the asset, such as linearity, accuracy and others, depending on the type of the asset, which are stored for each asset model. Multiple instances of the specific asset model may be present at each PV plant.

## *12.1 General properties for virtual segments*

### 12.1.1 General properties for a PV plant

The following list includes the general properties supported for each PV plant:

- Mandatory fields:
    - PV plant ID
    - PV plant descriptive name
    - Country
    - City
    - Installation date
    - Plant owner
    - Plant designer
    - Plant installer
    - Type of project:
        - R & D
        - Production
    - Type of plant:
        - Grid-connected
        - Stand-alone
        - Grid-connected hybrid
        - Stand-alone hybrid
    - Typical use:
        - Telecommunication
        - Cath. Protection
        - Warning/monitor

        - Other Professional
        - Water pumping
        - Fencing
        - Mountain farm
        - Other Rural
        - Domestic
        - Apartments
        - Office
        - Factory
        - Vacation house
        - Hotel
        - Refuges
        - Other Housing
        - Power station
        - Other
    - Latitude
    - Longitude
    - Altitude
    - AC load:
        - Yes
        - No
    - Grid connected:
        - Yes
        - No
    - Number of transformers
    - Timezone
    - Recording period
- Optional fields:
    - Plant operator
    - Plant photo
    - Contact person
    - Zip
    - Street
    - PO Box
    - Phone
    - Fax
    - E-mail
    - Number of PV panels
    - Number of PV panel groups
    - Number of PV arrays
    - Number of PV array groups

### 12.1.2 General properties for a PV array group

The following list includes the general properties supported for each PV array group:

- Mandatory fields:

  - PV array group ID
  - PV array group descriptive name
  - Number of PV arrays
- Optional fields:
  - Number of PV panels
  - Number of PV panel groups

### 12.1.3 General properties for a PV array

The following list includes the general properties supported for each PV array:

- Mandatory fields:
  - PV array ID
  - PV array descriptive name
  - Multi-string PV array:
    - Yes
    - No
  - Number of PV panel groups
  - Number of strings in the PV array
  - Number of PV panels per string
  - By-pass diodes:
    - Yes
    - No
  - Blocking diodes:
    - Yes
    - No
  - String fuse:
    - Yes
    - No
  - Concentrator:
    - Yes
    - No
- Optional fields:
  - Number of PV panels

### 12.1.4 General properties for a PV panel group

The following list includes the general properties supported for each PV panel group:

- Mandatory fields:
  - PV panel group ID
  - PV panel group descriptive name
  - Number of PV panels
  - PV panel group mounting:
    - Free-standing
    - Sloped roof
    - Flat roof
    - Facade
    - Sound barrier

        - Other
    - PV panel group integrated mounting:
        - Yes
        - No
- Optional fields:
    - PV panel group tracking type:
        - Seasonal
        - One-axis
        - Two-axis
    - PV panel group tilt angle
    - PV panel group azimuth angle
    - PV panel group tracking axis:
        - None
        - Polar
        - Vertical
        - North-south
        - East-west
    - Panel type
    - Per PV panel Vnom @ STC
    - Per PV panel short circuit current ISC
    - Per PV panel open circuit voltage VOC
    - Nominal efficiency

### 12.1.5 General properties for the PCC

The following list includes the general properties supported for the PCC:

- Mandatory fields:
    - Connection voltage
    - Protection present:
        - Yes
        - No
    - Production metering present:
        - Yes
        - No
    - PCC metering present:
        - Yes
        - No
    - EPS operator name

### 12.1.6 General properties for the LCP

The following list includes the general properties supported for the LCP:

- Mandatory fields:
    - Connection voltage
    - Metering present:
        - Yes
        - No

- EPS operator name
- Transformer present:
  - Yes
  - No
- DGPT relay present:
  - Yes
  - No
- Buchholz relay present:
  - Yes
  - No
- Thermostat present:
  - Yes
  - No
- Transfer switch present:
  - Yes
  - No

### 12.1.7 General properties for the TCPs

The following list includes the general properties supported for each TCP:

- Mandatory fields:
  - Transformer connection point ID
  - Transformer connection point descriptive name
  - Protection present:
    - Yes
    - No
  - DGPT relay present:
    - Yes
    - No
  - Buchholz relay present:
    - Yes
    - No
  - Thermostat present:
    - Yes
    - No
  - LV metering present:
    - Yes
    - No
  - HV metering present:
    - Yes
    - No
  - Number of PV array groups

### 12.1.8 General properties for a room

The following list includes the general properties supported for each room:

- Mandatory fields:

- Room ID
- Room descriptive name
- Air conditioned room:
    - Yes
    - No
- Temperature sensor:
    - Yes
    - No
- Fire sensor:
    - Yes
    - No
- Flood sensor:
    - Yes
    - No
- Occupancy sensor:
    - Yes
    - No
- Room type:
    - Control room
    - Storage room
    - Metering room
    - Pump room
    - Transformer room
    - HV electrical panel room
    - LV electrical panel room
    - Toilet
    - Generator room
    - Office
    - Warehouse
    - Kitchen
- Number of access-controlled doors

- Optional fields:
    - Width
    - Length
    - Height

### 12.1.9 General properties for a shelter

The following list includes the general properties supported for each shelter:

- Mandatory fields:
    - Shelter ID
    - Shelter descriptive name
    - Latitude
    - Longitude
    - Number of rooms
    - Shelter type:

        - Substation
        - Other
- Optional fields:
    - Width
    - Length
    - Height

### 12.1.10 General properties for an alarm zone

The following list includes the general properties supported for each alarm zone:

- Mandatory fields:
    - Alarm zone ID
    - Alarm zone descriptive name
- Optional fields:
    - Number of alarms
    - Zone location:
        - North
        - Northeast
        - East
        - Southeast
        - South
        - Southwest
        - West
        - Northwest

### 12.1.11 General properties for an actuator zone

The following list includes the general properties supported for each actuator zone:

- Mandatory fields:
    - Actuator zone ID
    - Actuator zone descriptive name
- Optional fields:
    - Number of actuators
    - Zone location:
        - North
        - Northeast
        - East
        - Southeast
        - South
        - Southwest
        - West
        - Northwest

### 12.1.12 General properties for an access controlled door

The following list includes the general properties supported for each access controlled door:

- Mandatory fields:
    - Door ID
    - Door descriptive name

### 12.1.13 General properties for a weather station

The following list includes the general properties supported for each weather station:

- Mandatory fields:
    - Weather station ID
    - Weather station descriptive name
- Optional fields:
    - Horizontal pyranometer present:
        - Yes
        - No
    - Wind direction sensor present:
        - Yes
        - No
    - Wind speed sensor present:
        - Yes
        - No
    - Barometric pressure sensor present:
        - Yes
        - No
    - Relative humidity sensor present:
        - Yes
        - No
    - Ambient temperature sensor present:
        - Yes
        - No
    - Rain gauge present:
        - Yes
        - No

### 12.1.14 General properties for a reference cell

The following list includes the general properties supported for each reference cell:

- Mandatory fields:
    - Reference cell ID
    - Reference cell descriptive name
- Optional fields:
    - Reference cell irradiance sensor present:
        - Yes
        - No
    - Reference cell temperature sensor present:
        - Yes
        - No

 

### 12.1.15 General properties for an electrical panel

The following list includes the general properties supported for each electrical panel:

- Mandatory fields:
  - Electrical panel ID
  - Electrical panel descriptive name

### 12.1.16 General properties for a circuit breaker group

The following list includes the general properties supported for each circuit breaker group:

- Mandatory fields:
  - Circuit breaker group ID
  - Circuit breaker group descriptive name

### 12.1.17 General properties for a surge arrester group

The following list includes the general properties supported for each surge arrester group:

- Mandatory fields:
  - Surge arrester group ID
  - Surge arrester group descriptive name

### 12.1.18 General properties for a switch group

The following list includes the general properties supported for each switch group:

- Mandatory fields:
  - Switch group ID
  - Switch group descriptive name

### 12.1.19 General properties for a fuse group

The following list includes the general properties supported for each fuse group:

- Mandatory fields:
  - Fuse group ID
  - Fuse group descriptive name

### 12.1.20 General properties for the IT

The following list includes the general properties supported for the IT virtual segment:

- Internet connection type:
  - ADSL
  - 3G
  - GPRS
- Connection download speed
- Connection upload speed
- Site network address
- IP address
- Netmask
- Video application port
- Video application username

- Video application password
- Video application type

## 12.2 System properties for asset segments

Each asset segment mandatorily includes the following set of system properties, pertinent to the specific instance of the asset:

- Asset ID
- Asset descriptive name
- Asset serial number
- Asset installation date

The following paragraphs present the system properties that can be stored by the inSolar solution for each asset model.

### 12.2.1 Transformer system properties

The following list includes the system properties supported for each transformer model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Nominal transformer power
- Number of Phases
- Primary voltage
- Secondary voltage
- Frequency
- Vector group
- Transformer type:
  - Dry
  - Oil-immersed, hermetically sealed
  - Oil-immersed, conservator type
- Cooling type:
  - ONAN
  - ONAF
- Voltage regulation:
  - Yes
  - No
- Maximum temperature rise
- Maximum ambient temp
- Impedance voltage
- Core losses
- Winding losses

### 12.2.2 Inverter system properties

The following list includes the system properties supported for each inverter model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Mode of operation:
    - Parallel
    - One per array
    - Master/Slave
    - Other
- Number of MPP trackers
- Inverter control:
    - Line-commutated
    - Self-commutated
- Operational frequency
- Grid connection type:
    - Phase
    - 3 Phase
    - 3 P+N
- DC overvoltage protection:
    - Yes
    - No
- Maximum input voltage
- Maximum input power
- Maximum input current
- Maximum number of strings
- Nominal output voltage
- Nominal output power
- Maximum output power
- Maximum output current
- Nominal efficiency
- Standby losses
- Losses at no load
- Galvanic insulation:
    - Yes
    - No
- Output THD
- Output power factor

### 12.2.3 String monitor system properties

The following list includes the system properties supported for each string monitor model:

- Manufacturer

- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Maximum DC voltage
- Maximum DC current per DC input
- Number of DC inputs
- DC switch present:
    - Yes
    - No
- DC fuse present:
    - Yes
    - No
- Surge arrester present:
    - Yes
    - No
- Communication interface:
    - Modbus
    - CAN
    - Custom

#### 12.2.4 Temperature sensor system properties

The following list includes the system properties supported for each temperature sensor model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Sensor type:
    - PT50
    - PT100
    - PT500
    - PT1000
    - Cu50
    - Cu100
    - Type “J”
    - Type “K”
    - Type “E”

    - Type “L”
    - Type “T”
    - Type “U”
- Accuracy class:
    - “A”
    - “B”
    - “C”
- Measurement range high limit
- Measurement range low limit
- Mounting:
    - Silicon body, self-adhesive
    - Probe
- Protection class:
    - IP20
    - IP30
    - IP32
    - IP34
    - IP41
    - IP44
    - IP54
    - IP65
    - IP66
    - IP67

### 12.2.5 Temperature transducer system properties

The following list includes the system properties supported for each temperature transducer model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Transducer type:
    - PT50
    - PT100
    - PT500
    - PT1000
    - Cu50
    - Cu100
    - Type “J”
    - Type “K”
    - Type “E”

    - Type “L”
    - Type “T”
    - Type “U”
- Accuracy
- Measurement range high limit
- Measurement range low limit
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Transducer output:
    - 0..20mA
    - 4..20mA
    - 0..1V
    - 0..10V
    - 2-10V
- Mounting:
    - Head type
    - On DIN rail
    - In box
- Protection class:
    - IP20
    - IP30
    - IP32
    - IP34
    - IP41
    - IP44
    - IP54
    - IP65
    - IP66
    - IP67

### 12.2.6 Pyranometer system properties

The following list includes the system properties supported for each pyranometer model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- WMO standard:
    - Good quality
    - High quality
- ISO standard:

 

  - First class
  - Secondary class
- Spectral range
- Field of view
- Maximum irradiance
- Sensitivity
- Impedance
- Response time
- Sensitivity temperature dependency
- Non-stability
- Uncertainty in daily total
- Zero offset
- Protection class:
  - IP20
  - IP30
  - IP32
  - IP34
  - IP41
  - IP44
  - IP54
  - IP65
  - IP66
  - IP67

### 12.2.7 Pyranometer transducer system properties

The following list includes the system properties supported for each pyranometer transducer model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Input impedance
- Accuracy
- Gain/Conversion factor
- Transducer output:
  - 0..20mA
  - 4..20mA
  - 0..1V

    - 0..10V
    - 2-10V
- Protection class:
    - IP20
    - IP30
    - IP32
    - IP34
    - IP41
    - IP44
    - IP54
    - IP65
    - IP66
    - IP67

### 12.2.8 Wind-speed indicator system properties

The following list includes the system properties supported for each wind-speed indicator model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Max supply current
- Output type
- Wind-speed sensor measurement range low limit
- Wind-speed sensor measurement range hi limit
- Accuracy
- Threshold
- Distance constant
- Output impedance

### 12.2.9 Wind direction sensor system properties

The following list includes the system properties supported for each wind direction sensor model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length

- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Max supply current
- Output type
- Electrical azimuth
- Mechanical azimuth
- Linearity
- Accuracy
- Threshold
- Damping ratio
- Delay distance
- Output impedance

### 12.2.10 Rain gauge system properties

The following list includes the system properties supported for each rain gauge model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Max supply current
- Output type
- Measuring principle
- Orifice
- Accuracy
- Resolution

### 12.2.11 Barometric pressure sensor system properties

The following list includes the system properties supported for each barometric pressure sensor model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight

- Supply voltage range high limit
- Supply voltage range low limit
- Max supply current
- Output type
- Measurement range low limit
- Measurement range hi limit
- Resolution
- Accuracy
- Linearity
- Hysteresis

### 12.2.12 Relative humidity sensor system properties

The following list includes the system properties supported for each relative humidity sensor model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Max supply current
- Output type
- Measurement range low limit
- Measurement range hi limit
- Resolution
- Accuracy

### 12.2.13 Reference cell irradiance sensor system properties

The following list includes the system properties supported for each reference cell irradiance sensor model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Solar cell type
- Supply voltage range high limit
- Supply voltage range low limit

- Max supply current
- Output type
- Measurement range low limit
- Measurement range hi limit
- Non-linearity
- Accuracy

### 12.2.14 Multimeter system properties

The following list includes the system properties supported for each multimeter model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Connection voltage:
  - LV
  - HV
- Current transformer ratio
- Voltage transformer ratio
- Communication module:
  - Yes
  - No
- Communication interface:
  - RS-485 half-duplex
  - RS-485 full-duplex
  - RS-422
  - RS-232
  - Ethernet
- Communication protocol:
  - Modbus RTU
  - Modbus TCP

### 12.2.15 Protection device system properties

The following list includes the system properties supported for each protection device model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height

- Weight
- ANSI C37.2 protections (text entry for informative purposes)
- Communication module:
    - Yes
    - No
- Communication interface:
    - RS-485 half-duplex
    - RS-485 full-duplex
    - RS-422
    - RS-232
    - Ethernet
    - Contacts
- Communication protocol:
    - Modbus RTU
    - Modbus TCP
- Contact interface:
    - Yes
    - No
- Contact type:
    - Relay
    - Voltage
- Contact voltage
- Contact current
- Contact polarity:
    - NO (normally open)
    - NC (normally closed)

### 12.2.16 Flood sensor system properties

The following list includes the system properties supported for each flood sensor model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Output type:
    - Relay
    - Open collector
    - Voltage
    - Current increase

- Polarity:
    - NO (normally open)/Active high
    - NC (normally closed)/Active low
- Voltage
- Current
- Protection class:
    - IP20
    - IP30
    - IP32
    - IP34
    - IP41
    - IP44
    - IP54
    - IP65
    - IP66
    - IP67

### 12.2.17 Fire sensor system properties

The following list includes the system properties supported for each fire sensor model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Fire sensor type:
    - Smoke sensor
    - Thermal sensor
    - Smoke + thermal sensor
- Monitoring area
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Output type:
    - Relay
    - Open collector
    - Voltage
    - Current increase
- Polarity:
    - NO (normally open)/Active high
    - NC (normally closed)/Active low
- Voltage
- Current

- Protection class:
    - IP20
    - IP30
    - IP32
    - IP34
    - IP41
    - IP44
    - IP54
    - IP65
    - IP66
    - IP67

### 12.2.18 Occupancy sensor system properties

The following list includes the system properties supported for each occupancy sensor model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Monitoring angle
- Monitoring distance
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Output type:
    - Relay
    - Open collector
    - Voltage
    - Current increase
- Polarity:
    - NO (normally open)/Active high
    - NC (normally closed)/Active low
- Voltage
- Current
- Protection class:
    - IP20
    - IP30
    - IP32
    - IP34
    - IP41
    - IP44

  - IP54
  - IP65
  - IP66
  - IP67

### 12.2.19 Proximity reader system properties

The following list includes the system properties supported for each proximity reader model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- RF frequency:
  - 125 kHz
  - 13.56 MHz
- Transponder Chip:
  - Mifare – MF1 standard cards
  - Mifare Ultralight cards
  - Mifare DESfire
  - EM4001
  - TEMPIC e5550
- Output format:
  - RS-485
  - RS-232
  - Wiegand
  - Mag Stripe emulation
  - Clock/data
- Number of output bits
- Protection class:
  - IP20
  - IP30
  - IP32
  - IP34
  - IP41
  - IP44
  - IP54
  - IP65
  - IP66

    - IP67

### 12.2.20 Electric bolt system properties

The following list includes the system properties supported for each electric bolt model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Actuator input type:
    - Voltage
    - 1 terminal – connect to ground
    - 2 terminal – connect together
- Voltage
- Current
- Polarity:
    - NO (normally open)/Active high
    - NC (normally closed)/Active low
- Striking plate supported:
    - Yes
    - No
- Striking plate part number
- Mechanical override function:
    - Yes
    - No
- Override contact:
    - Yes
    - No
- Door closed contact:
    - Yes
    - No
- Door locked contact:
    - Yes
    - No
- Protection class:
    - IP20
    - IP30
    - IP32
    - IP34

  - IP41
  - IP44
  - IP54
  - IP65
  - IP66
  - IP67

### 12.2.21 Buchholz relay system properties

The following list includes the system properties supported for each Buchholz relay model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Alarm contact present:
  - Yes
  - No
- Trip contact present:
  - Yes
  - No
- Output type:
  - Relay
  - Open collector
  - Voltage
  - Current increase
- Voltage
- Current
- Polarity:
  - NO (normally open)/Active high
  - NC (normally closed)/Active low
- Protection class:
  - IP20
  - IP30
  - IP32
  - IP34
  - IP41
  - IP44
  - IP54
  - IP65

  - IP66
  - IP67

### 12.2.22 DGPT relay system properties

The following list includes the system properties supported for each DGPT relay model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Gas presence alarm contact present:
  - Yes
  - No
- Pressure high alarm contact present:
  - Yes
  - No
- Temperature alarm contact present:
  - Yes
  - No
- Trip contact present:
  - Yes
  - No
- Output type:
  - Relay
  - Open collector
  - Voltage
  - Current increase
- Voltage
- Current
- Polarity:
  - NO (normally open)/Active high
  - NC (normally closed)/Active low
- Protection class:
  - IP20
  - IP30
  - IP32
  - IP34
  - IP41
  - IP44

  - IP54
  - IP65
  - IP66
  - IP67

### 12.2.23 Thermostat system properties

The following list includes the system properties supported for each thermostat model:

- Manufacturer
- Part Number
- Operational temperature range
- Operational humidity range
- Width
- Length
- Height
- Weight
- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Temperature alarm setting
- Temperature trip setting
- Temperature alarm contact present:
  - Yes
  - No
- Trip contact present:
  - Yes
  - No
- Output type:
  - Relay
  - Open collector
  - Voltage
  - Current increase
- Voltage
- Current
- Polarity:
  - NO (normally open)/Active high
  - NC (normally closed)/Active low
- Protection class:
  - IP20
  - IP30
  - IP32
  - IP34
  - IP41
  - IP44
  - IP54
  - IP65

  - IP66
  - IP67

### 12.2.24 Generic alarm contact system properties

The following list includes the system properties supported for each generic alarm contact model:

- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Output type:
  - Relay
  - Open collector
  - Voltage
  - Current increase
- Polarity:
  - NO (normally open)/Active high
  - NC (normally closed)/Active low
- Voltage
- Current

### 12.2.25 Generic actuator contact system properties

The following list includes the system properties supported for each generic actuator contact model:

- Supply voltage range high limit
- Supply voltage range low limit
- Supply current (max)
- Actuator input type:
  - Voltage
  - 1 terminal – connect to ground
  - 2 terminal – connect together
- Polarity:
  - NO (normally open)/Active high
  - NC (normally closed)/Active low
- Voltage
- Current

### 12.2.26 Controller system properties

The following list includes the system properties supported for each controller:

- IP address
- Gateway
- Netmask
- Update period for events
- Update period for parameters

Additional properties may be required, according to the type of controller being employed.

### 12.2.27 Video server system properties

The following list includes the system properties supported for each video server:

- IP address
- Netmask

Additional properties may be required, according to the type of video server being employed.

# 13 Summary of the information management supported by the inSolar solution

At this point, it is deemed necessary to summarize the parameters, state-sets and states, status information, settings, plots as well as the general and system properties presented within the previous sections. This categorization is based on the entities of the hierarchical model presented in Figure 3.
It is understandable that the relevant list is provisional and can be extended or narrowed down according to user requirements; however, it is expected to fulfill all baseline and further user requirements.

## *13.1 For a group of PV plants*

**Parameters:**

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| Total produced active energy | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Total consumed active energy | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Total revenue | Derived | Section 6.2.8.1 | inSolar solution | Recording period | N/A | N/A | Euro |
| Total CO2 reduction | Derived | Section 6.2.8.3 | inSolar solution | Recording period | N/A | N/A | Kilo |
| Nominal output power | Derived | Section 6.2.5.1 | inSolar solution | Recording period | N/A | N/A | kW |
| Performance ratio | Derived | Section 6.2.6.4 | inSolar solution | Recording period | N/A | N/A | % |

**State-sets & states:**
None

**Settings:**
None

**Status:**
None

**General properties:**
None

**Plots:**
None

## 13.2 For a PV plant

**Parameters:**

| **Parameter** | **Parameter type** | **Reference** | **Source** | **Reporting period(s)** | **Required measurement accuracy** | **Required minimum sampling rate** | **Units** |
|---|---|---|---|---|---|---|---|
| (Average) total irradiance, at the horizontal plane | Primary/derived/ real-time monitored | Sections 6.1.1 & 6.1.8 | Weather station irradiance sensor/transducer or inSolar solution | Recording period, daily, monthly, yearly | Better than 5% when primary | 1 sec when primary | $W \bullet m^{-2}$ |
| (Average) ambient air temperature | Primary/derived/ real-time monitored | Sections 6.1.2 & 6.1.8 | Weather station ambient air temperature sensor/transducer or inSolar solution | Recording period, daily, monthly, yearly | Better than 1 K when primary | 1 min when primary | ºC |
| (Average) wind speed | Primary/derived/ real-time monitored | Sections 6.1.3 & 6.1.8 | Weather station wind speed sensor or inSolar solution | Recording period, daily, monthly, yearly | Better than 0.5 $m \bullet s^{-1}$ for wind speeds <5 $m \bullet s^{-1}$, better than 10 % for wind speeds > 5 $m \bullet s^{-1}$ when primary | 1 min when primary | $M \bullet s^{-1}$ |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| (Average) wind direction | Primary/derived/ real-time monitored | Sections 6.1.4 & 6.1.8 | Weather station wind direction sensor or inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | Degrees from the north |
| (Average) rain precipitation | Primary/derived/ real-time monitored | Sections 6.1.5 & 6.1.8 | Weather station rain gauge or inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | mm/h |
| (Average) barometric pressure | Primary/derived/ real-time monitored | Sections 6.1.6 & 6.1.8 | Weather station barometric pressure sensor or inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kPa |
| (Average) relative humidity | Primary/derived/ real-time monitored | Sections 6.1.7 & 6.1.8 | Weather station relative humidity sensor or inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| (Average) total irradiance, in the plane of the array | Primary/derived/ real-time monitored | Section 6.2.1.1 | Irradiance sensor/transducer/inSolar solution | Recording period, daily, monthly, yearly | Better than 5% when primary | 1 sec when primary | $W \bullet m^{-2}$ |
| Average module temperature | Derived | Section 6.2.1.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | °C |
| (Average) reference cell irradiance, in the plane of the array | Primary/derived/ real-time monitored | Sections 6.2.1.3.1 & 6.2.1.3.3 | Reference cell/inSolar solution | Recording period, daily, monthly, yearly | Better than 5% when primary | 1 sec when primary | $W \bullet m^{-2}$ |
| (Average) reference cell temperature | Primary/derived/ real-time monitored | Sections 6.2.1.3.2 & 6.2.1.3.3 | Reference cell/inSolar solution | Recording period, daily, monthly, yearly | Better then 1K when primary | 1 min when primary | °C |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Solar energy | Derived | Section 6.2.1.5 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $kWh \bullet m^{-2} \bullet d^{-1}$, $kWh \bullet m^{-2} \bullet mo^{-1}$, $kWh \bullet m^{-2} \bullet y^{-1}$ |
| PV panel group output power | Derived | Section 6.2.2.1.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kW |
| PV array output power | Derived | Section 6.2.2.1.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kW |
| Transformer input active power | Derived | Section 6.2.2.2.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kW |
| Transformer output active power | Derived | Section 6.2.2.2.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kW |
| Active power towards the loads | Derived | Section 6.2.2.2.3 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kW |
| Active power towards the utility grid | Derived | Section 6.2.2.2.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kW |
| Active power from to the utility grid | Derived | Section 6.2.2.2.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kW |
| Cumulative active energy towards the loads | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Cumulative active energy towards the utility grid | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Cumulative active energy from the utility grid | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Cumulative active energy input to the transformers | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Cumulative active energy output from the transformers | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Cumulative PV panel group output energy | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Cumulative PV array output energy | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Active energy towards the loads | Derived | Sections 6.2.4.3 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Active energy towards the utility grid | Derived | Sections 6.2.4.3 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Active energy from the utility grid | Derived | Sections 6.2.4.3 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Transformer input active energy | Derived | Sections 6.2.4.3 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Transformer output active energy | Derived | Sections 6.2.4.3 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| PV panel group output energy | Derived | Sections 6.2.4.3 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| PV array output energy | Derived | Sections 6.2.4.3 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Net energy towards the utility network | Derived | Section 6.2.4.5.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Net energy from the utility network | Derived | Section 6.2.4.5.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Total system output energy | Derived | Section 6.2.4.5.3 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Total system input energy | Derived | Section 6.2.4.5.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Nominal output power | Derived | Section 6.2.5.1 | inSolar solution | Recording period | N/A | N/A | kWp |
| PV panel area | Derived | Section 6.2.5.2 | inSolar solution | Recording period | N/A | N/A | $m^2$ |
| Mean array efficiency | Derived | Section 6.2.6.1.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Inverter efficiency | Derived | Section 6.2.6.1.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Load efficiency | Derived | Section 6.2.6.1.3 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| System efficiency | Derived | Section 6.2.6.1.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Total efficiency | Derived | Section 6.2.6.1.5 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Solar fraction | Derived | Section 6.2.6.1.6 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Transformer efficiency | Derived | Section 6.2.6.1.7 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Energy yield | Derived | Section 6.2.6.2.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| Reference yield | Derived | Section 6.2.6.2.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| Final yield | Derived | Section 6.2.6.2.3 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| Array capture losses | Derived | Section 6.2.6.3.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| System losses | Derived | Section 6.2.6.3.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| Maintenance performance ratio | Derived | Section 6.2.6.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Total revenue | Derived | Section 6.2.8.1 | inSolar solution | Recording period | N/A | N/A | Euro |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Revenue | Derived | Section 6.2.8.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | Euro |
| Total CO2 reduction | Derived | Section 6.2.8.3 | inSolar solution | N/A | N/A | N/A | Kilo |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Performance ratio status | Performance ratio low | Derived | Section 6.2.9.3 | inSolar solution | 50 |
| | Performance ratio normal | Derived | Section 6.2.9.3 | inSolar solution | 0 |
| Ambient temperature status | Ambient temperature high | Primary | Section 6.1.9 | Weather station ambient temperature sensor/transducer | 50 |
| | Ambient temperature low | Primary | Section 6.1.9 | Weather station ambient temperature sensor/transducer | 30 |
| | Ambient temperature normal | Primary | Section 6.1.9 | Weather station ambient temperature sensor/transducer | 0 |
| Wind speed status | Wind speed high | Primary | Section 6.1.9 | Weather station wind speed sensor | 100 |
| | Wind speed normal | Primary | Section 6.1.9 | Weather station wind speed sensor | 0 |

**Settings:**

| Setting | Reference | Units |
|---|---|---|
| Irradiance threshold | Sections 6.2.6.1.1, 6.2.6.1.2, 6.2.6.1.4, 6.2.6.1.5, 6.2.6.1.7 & 6.2.6.4 | $W \bullet m^{-2}$ |
| Ambient temperature high threshold | Section 6.1.9 | ºC |
| Ambient temperature low threshold | Section 6.1.9 | ºC |
| Wind speed high threshold | Section 6.1.9 | $m*s^{-1}$ |
| Performance ratio low threshold | Section 6.2.9.3 | Dimensionless |
| Price of the energy sold to the distribution company | Section 6.2.8.1 | Euro |
| Price of the energy bought from the distribution company | Section 6.2.8.1 | Euro |
| Kilos of CO2 per kWh of energy produced/consumed | Section 6.2.8.3 | Kilos/kWh |

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.1

**Plots:**

As per section 9.3

 

## *13.3 For a PV array group*

**Parameters:**

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| Total irradiance, in the plane of the array | Derived | Section 6.2.1.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $W \bullet m^{-2}$ |
| Module temperature | Derived | Section 6.2.1.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | °C |
| Solar energy | Derived | Section 6.2.1.5 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $kWh \bullet m^{-2} \bullet d^{-1}$, $kWh \bullet m^{-2} \bullet mo^{-1}$, $kWh \bullet m^{-2} \bullet y^{-1}$ |
| PV panel group output power | Derived | Section 6.2.2.1.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kW |
| PV array output power | Derived | Section 6.2.2.1.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kW |
| Cumulative PV panel group output energy | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Cumulative PV array output energy | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PV panel group output energy | Derived | Sections 6.2.4.3 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| PV array output energy | Derived | Sections 6.2.4.3 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Nominal output power | Derived | Section 6.2.5.1 | inSolar solution | Recording period | N/A | N/A | kW |
| PV panel area | Derived | Section 6.2.5.2 | inSolar solution | Recording period | N/A | N/A | $m^2$ |
| Mean array efficiency | Derived | Section 6.2.6.1.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Inverter efficiency | Derived | Section 6.2.6.1.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Energy yield | Derived | Section 6.2.6.2.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h{\bullet}d^{-1}$, $h{\bullet}mo^{-1}$, $h{\bullet}y^{-1}$ |
| Reference yield | Derived | Section 6.2.6.2.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h{\bullet}d^{-1}$, $h{\bullet}mo^{-1}$, $h{\bullet}y^{-1}$ |
| Final yield | Derived | Section 6.2.6.2.3 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h{\bullet}d^{-1}$, $h{\bullet}mo^{-1}$, $h{\bullet}y^{-1}$ |
| Array capture losses | Derived | Section 6.2.6.3.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h{\bullet}d^{-1}$, $h{\bullet}mo^{-1}$, $h{\bullet}y^{-1}$ |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| System losses | Derived | Section 6.2.6.3.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| Maintenance performance ratio | Derived | Section 6.2.6.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |

**State-sets & states:**

| **State-set** | **State** | **State type** | **Reference** | **Update source** | **Default Severity** |
|---|---|---|---|---|---|
| Performance ratio status | Performance ratio low | Derived | Section 6.2.9.3 | inSolar solution | 50 |
| | Performance ratio normal | Derived | Section 6.2.9.3 | inSolar solution | 0 |

**Settings:**

| **Setting** | **Reference** | **Units** |
|---|---|---|
| Performance ratio low threshold | Section 6.2.9.3 | Dimensionless |

**Status:**
As per section 4.10

**General properties:**
As per section 12.1.2

**Plots:**
As per section 9.2

## *13.4 For a PV array*

**Parameters:**

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| Total irradiance, in the plane of the array | Derived | Section 6.2.1.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $W \bullet m^{-2}$ |
| Module temperature | Derived | Section 6.2.1.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | °C |
| Solar energy | Derived | Section 6.2.1.5 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $kWh \bullet m^{-2} \bullet d^{-1}$, $kWh \bullet m^{-2} \bullet mo^{-1}$, $kWh \bullet m^{-2} \bullet y^{-1}$ |
| PV panel group output power | Primary/real-time monitored or derived | Section 6.2.2.1.1 | Inverter/string monitor | Recording period, daily, monthly, yearly | Better than 2% when primary | 1 sec when primary | kW |
| PV array output power | Primary/real-time monitored | Section 6.2.2.1.2 | Inverter | Recording period, daily, monthly, yearly | Better than 2% | 1 sec | kW |
| PV panel group output voltage | Primary/real-time monitored | Sections 6.2.3.1.1 | Inverter | Recording period | Better than 1% | 1 sec | Volts DC |
| PV panel group output current | Primary/real-time monitored | Sections 6.2.3.1.1 | Inverter | Recording period | Better than 1% | 1 sec | Amps |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PV array output voltage (single or 3-phase) | Primary/real-time monitored | Sections 6.2.3.1.2 | Inverter | Recording period | Better than 1% | 1 sec | Volts AC |
| PV array output current | Primary/real-time monitored | Sections 6.2.3.1.2 | Inverter | Recording period | Better than 1% | 1 sec | Amps |
| Cumulative measured PV panel group output energy | Maximum/Derived | Sections 6.2.4.1 & 6.2.4.2 | Inverter/inSolar solution | Recording period | N/A | N/A | kWh |
| Cumulative measured PV array output energy | Maximum/Derived | Sections 6.2.4.1 & 6.2.4.2 | Inverter/inSolar solution | Recording period | N/A | N/A | kWh |
| Cumulative PV panel group output energy | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| Cumulative PV array output energy | Derived | Section 6.2.4.3 | inSolar solution | Recording period | N/A | N/A | kWh |
| PV panel group output energy | Derived | Section 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| PV array output energy | Derived | Section 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | kWh |
| Nominal output power | Derived | Section 6.2.5.1 | inSolar solution | Recording period | N/A | N/A | kW |
| PV panel area | Derived | Section 6.2.5.2 | inSolar solution | Recording period | N/A | N/A | $m^2$ |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Mean array efficiency | Derived | Section 6.2.6.1.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Inverter efficiency | Derived | Section 6.2.6.1.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Energy yield | Derived | Section 6.2.6.2.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| Reference yield | Derived | Section 6.2.6.2.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| Final yield | Derived | Section 6.2.6.2.3 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| Array capture losses | Derived | Section 6.2.6.3.1 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| System losses | Derived | Section 6.2.6.3.2 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | $h \bullet d^{-1}$, $h \bullet mo^{-1}$, $h \bullet y^{-1}$ |
| Maintenance performance ratio | Derived | Section 6.2.6.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |
| Array reactive output power | Primary | Section 6.2.7.6 | Inverter | Recording period | N/A | N/A | kVAr |
| Array apparent output power | Primary | Section 6.2.7.7 | Inverter | Recording period | N/A | N/A | kVA |
| Total power factor | Primary | Section 6.2.7.8 | Inverter | Recording period | N/A | N/A | Dimensionless |
| Panel group open-circuit voltage | Primary | Section 6.2.7.9 | Inverter | Recording period | N/A | N/A | Volts DC |
| Panel group setpoint voltage | Primary | Section 6.2.7.10 | Inverter | Recording period | N/A | N/A | Volts DC |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Array output frequency | Primary/real-time monitored | Section 6.2.7.11 | Inverter | Recording period | N/A | N/A | Hz |
| Inverter leakage current | Real-time monitored | Section 6.2.7.12 | Inverter | N/A | N/A | N/A | Amperes |
| Inverter insulation resistance | Real-time monitored | Section 6.2.7.13 | Inverter | N/A | N/A | N/A | Ohms |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Performance ratio status | Performance ratio low | Derived | Section 6.2.9.3 | inSolar solution | 50 |
| | Performance ratio normal | Derived | Section 6.2.9.3 | inSolar solution | 0 |

**Settings:**

| Setting | Reference | Units |
|---|---|---|
| Performance ratio low threshold | Section 6.2.9.3 | Dimensionless |
| Cumulative panel group output energy offset | Section 6.2.4.3 | kWh |
| Cumulative array output energy offset | Section 6.2.4.3 | kWh |

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.3

**Plots:**

As per section 9.1

## 13.5 For a PV panel group

**Parameters:**

| **Parameter** | **Parameter type** | **Reference** | **Source** | **Reporting period(s)** | **Required measurement accuracy** | **Required minimum sampling rate** | **Units** |
|---|---|---|---|---|---|---|---|
| Total irradiance, in the plane of the array | Primary/derived/ real-time monitored | Section 6.2.1.1 | Irradiance sensor/transducer/inSolar solution | Recording period, daily, monthly, yearly | Better than 5% when primary | 1 sec when primary | W•m$^{-2}$ |
| Module temperature | Primary/derived/ real-time monitored | Section 6.2.1.2 | Temperature sensor/transducer/inSolar solution | Recording period, daily, monthly, yearly | Better then 1K when primary | 1 min when primary | ºC |
| Panel group tracker tilt angle | Primary/Real-time monitored | Section 6.2.1.4 | PV panel tracker | Recording period, daily, monthly, yearly | N/A | 1 min | Degrees |
| Panel group tracker azimuth angle | Primary/Real-time monitored | Section 6.2.1.4 | PV panel tracker | Recording period, daily, monthly, yearly | N/A | 1 min | Degrees |
| PV panel group output power | Primary/real-time monitored | Section 6.2.2.1.1 | Inverter/String monitor | Recording period, daily, monthly, yearly | Better than 2% | 1 sec | kW |
| PV panel group output voltage | Primary/real-time monitored | Sections 6.2.3.1.1 | Inverter/String monitor | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts DC |
| PV panel group output current | Primary/real-time monitored | Sections 6.2.3.1.1 | Inverter/String monitor | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |

 

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Module temperature status | Module temperature high | Primary | Section 6.2.9.2 | Temperature sensor & transducer/inSolar solution | 80 |
| | Module temperature low | Primary | Section 6.2.9.2 | Temperature sensor & transducer/inSolar solution | 60 |
| | Module temperature normal | Primary | Section 6.2.9.2 | Temperature sensor & transducer/inSolar solution | 0 |

**Settings:**

| Setting | Reference | Units |
|---|---|---|
| Module temperature high threshold | Section 6.2.9.2 | ºC |
| Module temperature low threshold | Section 6.2.9.2 | ºC |
| Nominal output power | Section 6.2.5.1 | kWp |
| PV panel area | Section 6.2.5.2 | $M^2$ |

**Status:**
As per section 4.10

**General properties:**
As per section 12.1.4

**Plots:**
As per section 9.5

## *13.6 For the PCC*

**Parameters:**

The list of parameters presented in the table below applies twice, one set provided by the PCC multimeter and the other provided by the PCC protection device multimeter. Each parameter is distinguished through the use of a "PCC/Production" prefix.

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| Reactive power of phase L1 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 2% | 1 sec | kVAr |
| Reactive power of phase L2 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 2% | 1 sec | kVAr |
| Reactive power of phase L3 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 2% | 1 sec | kVAr |
| Active power of phase L1 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 2% | 1 sec | kW |
| Active power of phase L2 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 2% | 1 sec | kW |
| Active power of phase L3 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 2% | 1 sec | kW |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Apparent power of phase L1 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 2% | 1 sec | kVA |
| Apparent power of phase L2 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 2% | 1 sec | kVA |
| Apparent power of phase L3 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 2% | 1 sec | kVA |
| Total reactive power across phases | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 2% | 1 sec | kVAr |
| Total active power across phases | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 2% | 1 sec | kW |
| Total apparent power across phases | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 2% | 1 sec | kVA |
| Cumulative measured apparent energy | Maximum/Derived | Section 6.3.1.2 | PCC multimeter/Protection device or inSolar solution | Recording period | N/A | 1 min | kVAh |
| Cumulative measured active energy import | Maximum/Derived | Sections 6.2.4.1 & 6.3.1.2 | PCC multimeter/Protection device or inSolar solution | Recording period | N/A | 1 min | kWh |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Cumulative measured active energy export | Maximum/Derived | Sections 6.2.4.1 & 6.3.1.2 | PCC multimeter/Protection device or inSolar solution | Recording period | N/A | 1 min | kWh |
| Cumulative measured reactive energy import | Maximum/Derived | Section 6.3.1.2 | PCC multimeter/Protection device or inSolar solution | Recording period | N/A | 1 min | kVArh |
| Cumulative measured reactive energy export | Maximum/Derived | Section 6.3.1.2 | PCC multimeter/Protection device or inSolar solution | Recording period | N/A | 1 min | kVArh |
| Cumulative apparent energy | Derived | Sections 6.3.1.2 & 6.2.4.3 | inSolar solution | Recording period | N/A | 1 min | kVAh |
| Cumulative active energy import | Derived | Sections 6.3.1.2 & 6.2.4.3 | inSolar solution | Recording period | N/A | 1 min | kWh |
| Cumulative active energy export | Derived | Sections 6.3.1.2 & 6.2.4.3 | inSolar solution | Recording period | N/A | 1 min | kWh |
| Cumulative reactive energy import | Derived | Sections 6.3.1.2 & 6.2.4.3 | inSolar solution | Recording period | N/A | 1 min | kVArh |
| Cumulative reactive energy export | Derived | Sections 6.3.1.2 & 6.2.4.3 | inSolar solution | Recording period | N/A | 1 min | kVArh |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Apparent energy | Derived | Sections 6.3.1.2 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | 1 min | kVAh |
| Active Energy Import | Derived | Sections 6.3.1.2 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | 1 min | kWh |
| Active Energy Export | Derived | Sections 6.3.1.2 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | 1 min | kWh |
| Reactive Energy Import | Derived | Sections 6.3.1.2 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | 1 min | kVArh |
| Reactive Energy Export | Derived | Sections 6.3.1.2 & 6.2.4.4 | inSolar solution | Recording period, daily, monthly, yearly | N/A | 1 min | kVArh |
| Current of phase L1 | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |
| Current of phase L2 | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |
| Current of phase L3 | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |
| Maximum current of phase L1 | Maximum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Minimum current of phase L1 | Minimum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |
| Maximum current of phase L2 | Maximum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |
| Minimum current of phase L1 | Minimum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |
| Maximum current of phase L2 | Maximum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |
| Minimum current of phase L3 | Minimum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Amps |
| Average Current (across L1, L2, L3) | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 1% | 1 sec | Amps |
| Average Voltage (across L1, L2, L3) | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 1% | 1 sec | Volts AC |
| Average Voltage (across L1-L2, L2-L3, L3-L1) | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 1% | 1 sec | Volts AC |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Voltage of phase L1 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 1% | 1 sec | Volts AC |
| Voltage of phase L2 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 1% | 1 sec | Volts AC |
| Voltage of phase L3 | Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | N/A | Better than 1% | 1 sec | Volts AC |
| Voltage L1-L2 | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts AC |
| Voltage L2-L3 | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts AC |
| Voltage L3-L1 | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts AC |
| Maximum voltage L1-L2 | Maximum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts AC |
| Minimum voltage L1-L2 | Minimum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts AC |
| Maximum voltage L2-L3 | Maximum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts AC |
| Minimum voltage L2-L3 | Minimum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts AC |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Maximum voltage L3-L1 | Maximum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts AC |
| Minimum voltage L3-L1 | Minimum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | Better than 1% | 1 sec | Volts AC |
| Frequency | Primary/Real-time monitored | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Hz |
| Maximum frequency | Maximum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Hz |
| Minimum frequency | Minimum | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Hz |
| Power factor of phase L1 | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Dimensionless |
| Power factor of phase L2 | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Dimensionless |
| Power factor of phase L3 | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Dimensionless |
| Total power factor across phases | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Dimensionless |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| THD-R of the current of phase L1 | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Percentage |
| THD-R of the current of phase L2 | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Percentage |
| THD-R of the current of phase L3 | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Percentage |
| THD-R of the current of the neutral | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Percentage |
| Current unbalance for phase L1 | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Percentage |
| Current unbalance for phase L2 | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Percentage |
| Current unbalance for phase L3 | Primary | Section 6.3.1.2 | PCC multimeter/Protection device | Recording period, daily, monthly, yearly | N/A | 1 sec | Percentage |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Connectivity status | Off-grid/relay tripped | Primary | Section 6.3.1.1 | PCC protection device | 100 |
| | On-grid | Primary | Section 6.3.1.1 | PCC protection device | 0 |
| Frequency status | Overfrequency | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 80 |
| | Underfrequency | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 80 |
| | Frequency normal | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 0 |
| Rate of change of frequency status | ROCOF high | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 80 |
| | ROCOF normal | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 0 |
| Voltage status | Overvoltage on one or more phases | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 80 |
| | Undervoltage on one or more phases | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 80 |
| | Voltage normal on all phases | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 0 |
| Current status | Short-circuit on one or more phases | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 70 |
| | Overcurrent on one or more phases | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 10 |
| | Current normal on all phases | Primary | Section 6.3.1.1 | PCC multimeter/Protection device | 0 |
| Earth fault status | Earth fault present | Primary | Section 6.3.1.1 | PCC protection device | 100 |
| | No earth fault present | Primary | Section 6.3.1.1 | PCC protection device | 0 |
| Neutral voltage displacement status | Neutral voltage displacement present | Primary | Section 6.3.1.1 | PCC protection device | 80 |
| | No neutral voltage displacement present | Primary | Section 6.3.1.1 | PCC protection device | 0 |
| PCC current imbalance status | Current imbalance high on one or more phases | Primary | Section 6.3.1.1 | PCC multimeter | 30 |
| | Current imbalance normal on all phases | Primary | Section 6.3.1.1 | PCC multimeter | 0 |
| PCC power factor status | Power factor low | Primary | Section 6.3.1.1 | PCC multimeter | 30 |
| | Power factor normal | Primary | Section 6.3.1.1 | PCC multimeter | 0 |

| | | | | | |
|---|---|---|---|---|---|
| PCC DC injection status | DC injection high | Primary | Section 6.3.1.1 | PCC multimeter | 30 |
| | DC injection normal | Primary | Section 6.3.1.1 | PCC multimeter | 0 |
| PCC harmonic distortion status | Harmonic distortion high on one or more phases or the neutral | Primary | Section 6.3.1.1 | PCC multimeter | 30 |
| | Harmonic distortion normal on all phases | Primary | Section 6.3.1.1 | PCC multimeter | 0 |
| Production current imbalance status | Current imbalance high on one or more phases | Primary | Section 6.3.1.1 | Production multimeter | 30 |
| | Current imbalance normal on all phases | Primary | Section 6.3.1.1 | Production multimeter | 0 |
| Production power factor status | Power factor low | Primary | Section 6.3.1.1 | Production multimeter | 30 |
| | Power factor normal | Primary | Section 6.3.1.1 | Production multimeter | 0 |
| Production DC injection status | DC injection high | Primary | Section 6.3.1.1 | Production multimeter | 30 |
| | DC injection normal | Primary | Section 6.3.1.1 | Production multimeter | 0 |
| Production harmonic distortion status | Harmonic distortion high on one or more phases or the neutral | Primary | Section 6.3.1.1 | Production multimeter | 30 |
| | Harmonic distortion normal on all phases | Primary | Section 6.3.1.1 | Production multimeter | 0 |

**Settings:**

The list of settings presented in the table below applies twice, one set provided for the PCC multimeter and the other provided for the PCC protection device multimeter. Each setting is distinguished through the use of a “PCC/Production” prefix.

| Setting | Reference | Units |
|---|---|---|
| Cumulative active energy export offset | Section 6.2.4.3 | kWh |
| Cumulative active energy import offset | Section 6.2.4.3 | kWh |
| Cumulative reactive energy positive offset | Section 6.2.4.3 | kVArh |
| Cumulative reactive energy negative offset | Section 6.2.4.3 | kVArh |
| Cumulative apparent energy offset | Section 6.2.4.3 | kVAh |

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.5

**Plots:**

As per section 9.6

## *13.7 For the LCP*

**<u>Parameters:</u>**

Same as in section 13.6, but measured through the load connection point multimeter

**<u>State-sets & states:</u>**

| State-sets | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Connectivity status | Off-grid/relay tripped | Primary | Section 6.3.3 | Trip relay | 100 |
| | On-grid | Primary | Section 6.3.3 | Trip relay | 0 |
| Current status | Overcurrent on one or more phases | Primary | Section 6.3.3 | Load connection point multimeter | 70 |
| | Current normal on all phases | Primary | Section 6.3.3 | Load connection point multimeter | 0 |
| Voltage status | Overvoltage on one or more phases | Primary | Section 6.3.3 | Load connection point multimeter | 80 |
| | Undervoltage on one or more phases | Primary | Section 6.3.3 | Load connection point multimeter | 80 |
| | Voltage normal on all phases | Primary | Section 6.3.3 | Load connection point multimeter | 0 |
| Transfer switch status | Loads connected to a mechanical generator | Primary | Section 6.3.3 | Transfer switch contacts | 0 |
| | Loads connected to a wind generator | Primary | Section 6.3.3 | Transfer switch contacts | 0 |
| | Loads connected to the auxiliary power source | Primary | Section 6.3.3 | Transfer switch contacts | 0 |
| | Loads connected to the EPS/PV plant output | Primary | Section 6.3.3 | Transfer switch contacts | 0 |
| | Loads set to OFF | Primary | Section 6.3.3 | Transfer switch contacts | 0 |

**Settings:**

| Setting | Reference | Units |
|---|---|---|
| Cumulative active energy export offset | Section 6.2.4.3 | kWh |
| Cumulative active energy import offset | Section 6.2.4.3 | kWh |
| Cumulative reactive energy positive offset | Section 6.2.4.3 | kVArh |
| Cumulative reactive energy negative offset | Section 6.2.4.3 | kVArh |
| Cumulative apparent energy offset | Section 6.2.4.3 | kVAh |

**Status:**
As per section 4.10

**General properties:**
As per section 12.1.6

**Plots:**
As per section 9.6

## 13.8 For a TCP

**Parameters:**

Same as in section 13.6, but measured through the transformer connection point multimeters (LV/HV) or protection device. Each parameter is distinguished through the use of an LV/HV prefix. One additional parameter is supported and presented in the table below:

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| Transformer efficiency | Derived | Section 6.2.6.1.7 | inSolar solution | Recording period, daily, monthly, yearly | N/A | N/A | % |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Connectivity status | Off-grid/relay tripped | Primary | Section 6.3.2.2 | Protection device | 100 |
| | On-grid | Primary | Section 6.3.2.2 | Protection device | 0 |
| Current status | Short-circuit on one or more phases | Primary | Section 6.3.2.2 | HV multimeter/Protection device | 70 |
| | Overcurrent on one or more phases | Primary | Section 6.3.2.2 | HV multimeter/Protection device | 10 |
| | Current normal on all phases | Primary | Section 6.3.2.2 | HV multimeter/Protection device | 0 |
| Earth fault status | Earth fault present | Primary | Section 6.3.2.2 | Protection device | 100 |
| | No earth fault present | Primary | Section 6.3.2.2 | Protection device | 0 |
| LV current imbalance status | Current imbalance high on one or more phases | Primary | Section 6.3.2.2 | LV transformer multimeter | 30 |

| | | | | | |
|---|---|---|---|---|---|
| | Current imbalance normal on all phases | Primary | Section 6.3.2.2 | LV transformer multimeter | 0 |
| LV power factor status | Power factor low | Primary | Section 6.3.2.2 | LV transformer multimeter | 30 |
| | Power factor normal | Primary | Section 6.3.2.2 | LV transformer multimeter | 0 |
| LV DC injection status | DC injection high | Primary | Section 6.3.2.2 | LV transformer multimeter | 30 |
| | DC injection normal | Primary | Section 6.3.2.2 | LV transformer multimeter | 0 |
| LV current THD-R status | Current THD-R high on one or more phases or the neutral | Primary | Section 6.3.2.2 | LV transformer multimeter | 30 |
| | Current THD-R normal on all phases | Primary | Section 6.3.2.2 | LV transformer multimeter | 0 |
| HV current imbalance status | Current imbalance high on one or more phases | Primary | Section 6.3.2.2 | HV transformer multimeter/Protection device | 30 |
| | Current imbalance normal on all phases | Primary | Section 6.3.2.2 | HV transformer multimeter/Protection device | 0 |
| HV power factor status | Power factor low | Primary | Section 6.3.2.2 | HV transformer multimeter/Protection device | 30 |
| | Power factor normal | Primary | Section 6.3.2.2 | HV transformer multimeter/Protection device | 0 |
| HV DC injection status | DC injection high | Primary | Section 6.3.2.2 | HV transformer multimeter | 30 |
| | DC injection normal | Primary | Section 6.3.2.2 | HV transformer multimeter | 0 |
| HV current THD-R status | Current THD-R high on one or more phases or the neutral | Primary | Section 6.3.2.2 | HV transformer multimeter/Protection device | 30 |
| | Current THD-R normal on all phases | Primary | Section 6.3.2.2 | HV transformer multimeter/Protection device | 0 |

**Settings:**

2 sets of settings as per the table below exist, referring to each multimeter/protection device (LV/HV) separately. Each setting is distinguished through the use of an LV/HV prefix.

| Setting | Reference | Units |
|---|---|---|
| Cumulative active energy export offset | Section 6.2.4.3 | kWh |
| Cumulative active energy import offset | Section 6.2.4.3 | kWh |
| Cumulative reactive energy positive offset | Section 6.2.4.3 | kVArh |
| Cumulative reactive energy negative offset | Section 6.2.4.3 | kVArh |
| Cumulative apparent energy offset | Section 6.2.4.3 | kVAh |

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.7

**Plots:**

As per section 9.6

## *13.9For a shelter room*

**Parameters:**

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| Room temperature | Primary/Real-time monitored | Section 6.5 | Temperature sensor/transducer | Recording period, daily, monthly, yearly | Better than 1K | 1 min | ºC |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Room flood status | Room flooded | Primary | Section 6.5 | Room flood sensor | 100 |
| | Room not flooded | Primary | Section 6.5 | Room flood sensor | 0 |
| Room fire status | Room on fire | Primary | Section 6.5 | Room fire sensor | 100 |
| | Room not on fire | Primary | Section 6.5 | Room fire sensor | 0 |
| Room occupancy status | Room occupied | Primary | Section 6.5 | Room occupancy sensor | 30 |
| | Room not occupied | Primary | Section 6.5 | Room occupancy sensor | 0 |
| Room temperature status | Room temperature high | Primary | Section 6.5 | Temperature sensor/transducer | 50 |
| | Room temperature low | Primary | Section 6.5 | Temperature sensor/transducer | 30 |
| | Room temperature normal | Primary | Section 6.5 | Temperature sensor/transducer | 0 |

**Settings:**

| Setting | Reference | Units |
|---|---|---|
| Room temperature high threshold | As per section 6.5 | ℃ |
| Room temperature low threshold | As per section 6.5 | ℃ |

**Status:**
As per section 4.10

**General properties:**
As per section 12.1.8

**Plots:**
As per section 9.4

## 13.10 For a shelter

**Parameters:**

None

**State-sets & states:**

None

**Settings:**

None

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.9

**Plots:**

None

 

## *13.11 For an alarm zone*

**Parameters:**

None

**State-sets & states:**

None

**Settings:**

None

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.10

**Plots:**

As per section 9.6

## *13.12 For an actuator zone*

**Parameters:**

None

**State-sets & states:**

None

**Settings:**

None

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.11

**Plots:**

As per section 9.6

 

## 13.13 For an access-controlled door

**Parameters:**

None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Door status | Door left open | Primary | Section 6.6 | inSolar solution | 100 |
| | Door closed | Primary | Section 6.6 | inSolar solution | 0 |
| Door lock override status | Door lock overridden | Primary | Section 6.6 | Electric door bolt | 30 |
| | Door lock not overridden | Primary | Section 6.6 | Electric door bolt | 0 |

**Settings:**

| Setting | Reference | Units |
|---|---|---|
| Door left open alarm hysteresis time | Section 6.6 | Sec |

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.12

**Plots:**

As per section 9.6

## *13.14For a weather station*

**Parameters:**

| **Parameter** | **Parameter type** | **Reference** | **Source** | **Reporting period(s)** | **Required measurement accuracy** | **Required minimum sampling rate** | **Units** |
|---|---|---|---|---|---|---|---|
| Total irradiance, at the horizontal plane | Primary/Real-time monitored | Section 6.1.1 | Weather station irradiance sensor/transducer | Recording period, daily, monthly, yearly | Better than 5% | 1 sec | $W \bullet m^{-2}$ |
| Ambient air temperature | Primary/Real-time monitored | Section 6.1.2 | Weather station ambient air temperature sensor/transducer | Recording period, daily, monthly, yearly | Better than 1 K | 1 min | ºC |
| Wind speed | Primary/Real-time monitored | Section 6.1.3 | Weather station wind speed sensor | Recording period, daily, monthly, yearly | Better than 0.5 $m \bullet s^{-1}$ for wind speeds <5 $m \bullet s^{-1}$, better than 10 % for wind speeds > 5 $m \bullet s^{-1}$ | 1 min | $M \bullet s^{-1}$ |
| Wind direction | Primary/Real-time monitored | Section 6.1.4 | Weather station wind direction sensor | Recording period, daily, monthly, yearly | N/A | N/A | Degrees from the north |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Rain precipitation | Primary/Real-time monitored | Section 6.1.5 | Weather station rain gauge | Recording period, daily, monthly, yearly | N/A | N/A | mm/h |
| Barometric pressure | Primary/Real-time monitored | Section 6.1.6 | Weather station barometric pressure sensor | Recording period, daily, monthly, yearly | N/A | N/A | kPa |
| Relative humidity | Primary/Real-time monitored | Section 6.1.7 | Weather station relative humidity sensor | Recording period, daily, monthly, yearly | N/A | N/A | % |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Ambient temperature status | Ambient temperature high | Primary | Section 6.1.9 | Weather station ambient temperature sensor/transducer | 50 |
| | Ambient temperature low | Primary | Section 6.1.9 | Weather station ambient temperature sensor/transducer | 30 |
| | Ambient temperature normal | Primary | Section 6.1.9 | Weather station ambient temperature sensor/transducer | 0 |
| Wind speed status | Wind speed high | Primary | Section 6.1.9 | Weather station wind speed sensor | 100 |
| | Wind speed normal | Primary | Section 6.1.9 | Weather station wind speed sensor | 0 |

**Settings:**

| Setting | Reference | Units |
|---|---|---|
| Ambient temperature high threshold | Section 6.1.9 | ºC |
| Ambient temperature low threshold | Section 6.1.9 | ºC |
| Wind speed high threshold | Section 6.1.9 | $m*s^{-1}$ |

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.13

**Plots:**

As per section 9.6

## 13.15 For a reference cell

**Parameters:**

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| Reference cell irradiance, in the plane of the array | Primary/Real-time monitored | Section 6.2.1.3.1 | Reference cell | Recording period, daily, monthly, yearly | Better than 5% | 1 sec | $W \bullet m^{-2}$ |
| Reference cell temperature | Primary/Real-time monitored | Section 6.2.1.3.2 | Reference cell | Recording period, daily, monthly, yearly | Better then 1K | 1 min | ℃ |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Malfunction status | Reference cell temperature sensor/irradiance sensor malfunction | Primary | Section 6.8.2 | Reference cell temperature sensor/irradiance sensor | 100 |
| | Reference cell temperature sensor/irradiance sensor in normal operation | Primary | Section 6.8.2 | Reference cell temperature sensor/irradiance sensor | 0 |
| Communication status | Error in the communication with the reference cell temperature sensor/irradiance sensor | Primary | Section 6.8.2 | inSolar solution | 50 |
| | Communication with the reference cell temperature sensor/irradiance sensor operational | Primary | Section 6.8.2 | inSolar solution | 0 |

**Settings:**

None

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.14

**Plots:**

As per section 9.6

 

## *13.16 For an electrical panel*

**Parameters:**

None

**State-sets & states:**

None

**Settings:**

None

**Status:**

As per section 4.10

**General properties:**

As per section 12.1.15

**Plots:**

None

 

## 13.17 For circuit breaker, surge arrester, fuse and switch groups

**Parameters:**

None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Group status | “Tripped” for circuit breaker groups, “Open” for switch groups, “Blown” for fuse groups and “Needs replacement” for surge arrester groups | Primary | Section 6.4 | Contact | 100 for fuses, circuit breakers and surge arresters, 0 for switches |
| | “Closed” for circuit breaker/switch groups, “Normal” for fuse/ surge arrester groups | Primary | Section 6.4 | Contact | 0 for all |
| Malfunction status | Group contact malfunction | Primary | Section 6.8.3 | Contact | 100 |
| | Group contact in normal operation | Primary | Section 6.8.3 | Contact | 0 |
| Communication status | Error in the communication with the group contact | Primary | Section 6.8.3 | inSolar solution | 50 |
| | Communication with the group contact operational | Primary | Section 6.8.3 | inSolar solution | 0 |

**Settings:**

None

**Status:**

As per section 4.10

**General properties:**

As per sections 12.1.16, 12.1.17, 12.1.18 & 12.1.19

**Plots:**

As per section 9.6

 

## *13.18 For IT*

**Parameters:**

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| Average round-trip time (RTT) | Primary | Section 6.9 | Router/modem | Recording period | N/A | N/A | msec |
| Maximum round-trip time (RTT) | Maximum | Section 6.9 | Router/modem | Recording period | N/A | N/A | msec |
| Minimum round-trip time (RTT) | Minimum | Section 6.9 | Router/modem | Recording period | N/A | N/A | msec |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Site link status | Site offline | Primary | Section 6.9 | Router/modem | 30 |
| | Site online | Primary | Section 6.9 | Router/modem | 0 |

**Settings:**
None

**Status:**
As per section 4.10

**General properties:**
As per section 12.1.20

**Plots:**
As per section 9.6

 

## 13.19 For a transformer

**Parameters:**

None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Buchholz relay status | Buchholz relay alarm | Primary | Section 6.3.2.2 | Buchholz relay/Protection device | 50 |
| | Buchholz relay trip | Primary | Section 6.3.2.2 | Buchholz relay/Protection device | 100 |
| | Buchholz relay normal | Primary | Section 6.3.2.2 | Buchholz relay/Protection device | 0 |
| Gas accumulation status | Gas accumulation present | Primary | Section 6.3.2.2 | DGPT relay/Protection device | 80 |
| | Gas accumulation not present | Primary | Section 6.3.2.2 | DGPT relay/Protection device | 0 |
| Temperature status | Temperature alarm | Primary | Section 6.3.2.2 | Thermostat relay/DGPT relay/Protection device | 50 |
| | Temperature trip | Primary | Section 6.3.2.2 | Thermostat relay/DGPT relay/Protection device | 100 |
| | Temperature normal | Primary | Section 6.3.2.2 | Thermostat relay/DGPT relay/Protection device | 0 |
| Gas pressure status | Overpressure | Primary | Section 6.3.2.2 | DGPT relay/Protection device | 80 |
| | Pressure normal | Primary | Section 6.3.2.2 | DGPT relay/Protection device | 0 |

**Settings:**

None

**Status:**

As per section 4.10

**System properties:**

As per section 12.2.1

**Plots:**

As per section 9.6

## 13.20 For an inverter

**Parameters:**

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| Inverter internal temperature | Primary/real-time monitored | Section 6.2.7.1 | Inverter | Recording period | N/A | N/A | °C |
| Inverter cooler temperature | Primary/real-time monitored | Section 6.2.7.2 | Inverter | Recording period | N/A | N/A | °C |
| Inverter availability | Primary | Section 6.2.7.3 | Inverter | Recording period | N/A | N/A | % |
| Inverter total hours of operation | Maximum | Section 6.2.7.4 | Inverter | Recording period | N/A | N/A | Hours |
| Inverter total hours of grid feed-in operation | Maximum | Section 6.2.7.5 | Inverter | Recording period | N/A | N/A | Hours |
| Inverter status code log | Raw | Section 6.2.9.1 | Inverter | Raw data | N/A | N/A | N/A |
| Inverter error code log | Raw | Section 6.2.9.1 | Inverter | Raw data | N/A | N/A | N/A |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Inverter status | Inverter in normal operation | Primary | Section 6.2.9.1 | Inverter | 0 |
| | Inverter in derating mode | Primary | Section 6.2.9.1 | Inverter | 30 |
| | Inverter in disturbance mode | Primary | Section 6.2.9.1 | Inverter | 80 |
| | Inverter in error mode | Primary | Section 6.2.9.1 | Inverter | 80 |
| | Inverter in stopped mode | Primary | Section 6.2.9.1 | Inverter | 100 |
| | Inverter in night shutdown mode | Primary | Sections 6.2.9.1 & 6.8.1 | Inverter | 0 |
| | Error in the communication with the inverter | Primary | Section 6.8.1 | inSolar solution | 80 |
| Inverter warning status | Inverter warning present | Primary | Section 6.2.9.1 | Inverter | 50 |
| | Inverter warning not present | Primary | Section 6.2.9.1 | Inverter | 0 |

**Settings:**
None

**Status:**
As per section 4.10

**System properties:**
As per section 12.2.2

**Plots:**
As per section 9.6. The status and error codes cannot be plotted.

## 13.21 For a string monitor

**Parameters:**

| Parameter | Parameter type | Reference | Source | Reporting period(s) | Required measurement accuracy | Required minimum sampling rate | Units |
|---|---|---|---|---|---|---|---|
| String monitor status code log | Raw | Section 6.2.9.1 | String monitor | Raw data | N/A | N/A | N/A |
| String monitor error code log | Raw | Section 6.2.9.1 | String monitor | Raw data | N/A | N/A | N/A |

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| String monitor status | String monitor in normal operation | Primary | Section 6.2.9.1 | String monitor | 0 |
| | String monitor in stopped mode | Primary | Section 6.2.9.1 | String monitor | 100 |
| | Error in the communication with the string monitor | Primary | Section 6.8.1 | inSolar solution | 100 |

**Settings:**
None

**Status:**
As per section 4.10

**System properties:**
As per section 12.2.3

**Plots:**
As per section 9.6. The status and error codes cannot be plotted.

## *13.22 For a temperature sensor/pyranometer*

**Parameters:**

None

**State-sets & states:**

None

**Settings:**

None

**Status:**

None

**System properties:**

As per section 12.2.4 for a temperature sensor and section 12.2.6 for a pyranometer

**Plots:**

None

 

## *13.23 For a temperature transducer, pyranometer transducer, wind-speed indicator, wind direction sensor, rain gauge, barometric pressure sensor, relative humidity sensor & reference cell irradiance sensor*

**Parameters:**
None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Malfunction status | Transducer/sensor/indicator malfunction | Primary | Section 6.8.2 | Transducer/ sensor/indicator | 100 |
| | Transducer/sensor/indicator in normal operation | Primary | Section 6.8.2 | Transducer/ sensor/indicator | 0 |
| Communication status | Error in the communication with the transducer/sensor/indicator | Primary | Section 6.8.2 | inSolar solution | 50 |
| | Communication with the transducer/sensor/indicator operational | Primary | Section 6.8.2 | inSolar solution | 0 |

**Settings:**
None

**Status:**
As per section 4.10

**System properties:**
As per section 12.2.5 for a temperature transducer, section 12.2.7 for a pyranometer transducer, section 12.2.8 for a wind-speed indicator, section 12.2.9 for a wind direction sensor, section 12.2.10 for a rain gauge, section 12.2.11 for a barometric pressure sensor, section 12.2.12 for a relative humidity sensor and section 12.2.13 for a reference cell irradiance sensor

**Plots:**
As per section 9.6

 

## *13.24 For a multimeter*

**Parameters:**

None for multimeters belonging to the PCC, LCP and TCPs. For multimeters under electrical panels the parameters are the same as those presented within section 13.6.

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Malfunction status | Multimeter malfunction | Primary | Section 6.8.2 | Multimeter | 100 |
| | Multimeter in normal operation | Primary | Section 6.8.2 | Multimeter | 0 |
| Communication status | Error in the communication with the multimeter | Primary | Section 6.8.2 | inSolar solution | 50 |
| | Communication with the multimeter operational | Primary | Section 6.8.2 | inSolar solution | 0 |

Multimeters under electrical panels additionally support the state-sets presented within the table below:

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Frequency status | Overfrequency | Primary | Section 6.3.1.1 | Electrical panel multimeter | 80 |
| | Underfrequency | Primary | Section 6.3.1.1 | Electrical panel multimeter | 80 |
| | Frequency normal | Primary | Section 6.3.1.1 | Electrical panel multimeter | 0 |
| Rate of change of frequency status | ROCOF high | Primary | Section 6.3.1.1 | Electrical panel multimeter | 80 |
| | ROCOF normal | Primary | Section 6.3.1.1 | Electrical panel multimeter | 0 |
| Voltage status | Overvoltage on one or more phases | Primary | Section 6.3.1.1 | Electrical panel multimeter | 80 |
| | Undervoltage on one or more phases | Primary | Section 6.3.1.1 | Electrical panel multimeter | 80 |
| | Voltage normal on all phases | Primary | Section 6.3.1.1 | Electrical panel multimeter | 0 |

| | | | | | |
|---|---|---|---|---|---|
| Current status | Short-circuit on one or more phases | Primary | Section 6.3.1.1 | Electrical panel multimeter | 70 |
| | Overcurrent on one or more phases | Primary | Section 6.3.1.1 | Electrical panel multimeter | 10 |
| | Current normal on all phases | Primary | Section 6.3.1.1 | Electrical panel multimeter | 0 |
| Current imbalance status | Current imbalance high on one or more phases | Primary | Section 6.3.1.1 | Electrical panel multimeter | 30 |
| | Current imbalance normal on all phases | Primary | Section 6.3.1.1 | Electrical panel multimeter | 0 |
| Power factor status | Power factor low | Primary | Section 6.3.1.1 | Electrical panel multimeter | 30 |
| | Power factor normal | Primary | Section 6.3.1.1 | Electrical panel multimeter | 0 |
| DC injection status | DC injection high | Primary | Section 6.3.1.1 | Electrical panel multimeter | 30 |
| | DC injection normal | Primary | Section 6.3.1.1 | Electrical panel multimeter | 0 |
| Harmonic distortion status | Harmonic distortion high on one or more phases or the neutral | Primary | Section 6.3.1.1 | Electrical panel multimeter | 30 |
| | Harmonic distortion normal on all phases | Primary | Section 6.3.1.1 | Electrical panel multimeter | 0 |

**Settings:**
Multimeters under electrical panels support the settings presented within section 13.6

**Status:**
As per section 4.10

**System properties:**
As per section 12.2.14

**Plots:**
As per section 9.6

## 13.25 For a protection device

**Parameters:**

None for protection devices belonging to the PCC, LCP and TCPs. For protection devices under electrical panels the parameters are the same as those presented within section 13.6.

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Malfunction status | Protection device malfunction | Primary | Section 6.8.2 | Protection device | 100 |
| | Protection device in normal operation | Primary | Section 6.8.2 | Protection device | 0 |
| Communication status | Error in the communication with the protection device | Primary | Section 6.8.2 | inSolar solution | 50 |
| | Communication with the protection device operational | Primary | Section 6.8.2 | inSolar solution | 0 |

Protection devices under electrical panels additionally support the state-sets presented within the table below:

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Connectivity status | Off-grid/relay tripped | Primary | Section 6.3.1.1 | Electrical panel protection device | 100 |
| | On-grid | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |
| Frequency status | Overfrequency | Primary | Section 6.3.1.1 | Electrical panel protection device | 80 |
| | Underfrequency | Primary | Section 6.3.1.1 | Electrical panel protection device | 80 |
| | Frequency normal | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |

| | | | | | |
|---|---|---|---|---|---|
| Rate of change of frequency status | ROCOF high | Primary | Section 6.3.1.1 | Electrical panel protection device | 80 |
| | ROCOF normal | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |
| Voltage status | Overvoltage on one or more phases | Primary | Section 6.3.1.1 | Electrical panel protection device | 80 |
| | Undervoltage on one or more phases | Primary | Section 6.3.1.1 | Electrical panel protection device | 80 |
| | Voltage normal on all phases | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |
| Current status | Short-circuit on one or more phases | Primary | Section 6.3.1.1 | Electrical panel protection device | 70 |
| | Overcurrent on one or more phases | Primary | Section 6.3.1.1 | Electrical panel protection device | 10 |
| | Current normal on all phases | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |
| Earth fault status | Earth fault present | Primary | Section 6.3.1.1 | Electrical panel protection device | 100 |
| | No earth fault present | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |
| Neutral voltage displacement status | Neutral voltage displacement present | Primary | Section 6.3.1.1 | Electrical panel protection device | 80 |
| | No neutral voltage displacement present | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |
| Current imbalance status | Current imbalance high on one or more phases | Primary | Section 6.3.1.1 | Electrical panel protection device | 30 |
| | Current imbalance normal on all phases | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |

| | | | | | |
|---|---|---|---|---|---|
| Power factor status | Power factor low | Primary | Section 6.3.1.1 | Electrical panel protection device | 30 |
| | Power factor normal | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |
| DC injection status | DC injection high | Primary | Section 6.3.1.1 | Electrical panel protection device | 30 |
| | DC injection normal | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |
| Harmonic distortion status | Harmonic distortion high on one or more phases or the neutral | Primary | Section 6.3.1.1 | Electrical panel protection device | 30 |
| | Harmonic distortion normal on all phases | Primary | Section 6.3.1.1 | Electrical panel protection device | 0 |

**Settings:**
Protection devices under electrical panels support the settings presented within section 13.6.

**Status:**
As per section 4.10

**System properties:**
As per section 12.2.15

**Plots:**
As per section 9.6

 

## 13.26 For a flood sensor, fire sensor, occupancy sensor, Buchholz relay, DGPT relay and thermostat

**Parameters:**

None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Malfunction status | Sensor/ relay/thermostat malfunction | Primary | Section 6.8.3 | Sensor/ relay/thermostat | 100 |
| | Sensor/ relay/thermostat in normal operation | Primary | Section 6.8.3 | Sensor/ relay/thermostat | 0 |
| Communication status | Error in the communication with the sensor/ relay/thermostat | Primary | Section 6.8.3 | inSolar solution | 50 |
| | Communication with the sensor/ relay/thermostat operational | Primary | Section 6.8.3 | inSolar solution | 0 |

**Settings:**

None

**Status:**

As per section 4.10

**System properties:**

As per section 12.2.16 for a flood sensor, section 12.2.17 for a fire sensor, section 12.2.18 for an occupancy sensor, section 12.2.21 for a Buchholz relay, section 12.2.22 for a DGPT relay and section 12.2.23 for a thermostat

**Plots:**

As per section 9.6

## *13.27 For a proximity reader*

**Parameters:**

None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Malfunction status | Proximity reader malfunction | Primary | Section 6.8.2 | Proximity reader | 100 |
| | Proximity reader in normal operation | Primary | Section 6.8.2 | Proximity reader | 0 |
| Communication status | Error in the communication with the proximity reader | Primary | Section 6.8.2 | inSolar solution | 50 |
| | Communication with the proximity reader operational | Primary | Section 6.8.2 | inSolar solution | 0 |

**Settings:**

None

**Status:**

As per section 4.10

**System properties:**

As per section 12.2.19

**Plots:**

As per section 9.6

## *13.28 For an electric bolt*

**Parameters:**

None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Malfunction status | Electric bolt malfunction | Primary | Section 6.8.3 | Electric bolt | 100 |
| | Electric bolt in normal operation | Primary | Section 6.8.3 | Electric bolt | 0 |
| Communication status | Error in the communication with the electric bolt | Primary | Section 6.8.3 | inSolar solution | 50 |
| | Communication with the electric bolt operational | Primary | Section 6.8.3 | inSolar solution | 0 |

**Settings:**

None

**Status:**

As per section 4.10

**System properties:**

As per section 12.2.20

**Plots:**

As per section 9.6

## 13.29 For a generic alarm

**Parameters:**

None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Alarm status | Alarm active | Primary | Section 6.7 | Alarm contact | 100 |
| | Alarm inactive | Primary | Section 6.7 | Alarm contact | 0 |
| Malfunction status | Alarm contact malfunction | Primary | Section 6.8.3 | Alarm contact | 100 |
| | Alarm contact in normal operation | Primary | Section 6.8.3 | Alarm contact | 0 |
| Communication status | Error in the communication with the alarm contact | Primary | Section 6.8.3 | inSolar solution | 50 |
| | Communication with the alarm contact | Primary | Section 6.8.3 | inSolar solution | 0 |

**Settings:**

None

**Status:**

As per section 4.10

**System properties:**

As per section 12.2.24

**Plots:**

As per section 9.6

## 13.30 For a generic actuator

**Parameters:**

None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Actuator status | Actuator active | Primary | Section 6.7 | Actuator contact | 100 |
| | Actuator inactive | Primary | Section 6.7 | Actuator contact | 0 |
| Malfunction status | Actuator contact malfunction | Primary | Section 6.8.3 | Actuator contact | 100 |
| | Actuator contact in normal operation | Primary | Section 6.8.3 | Actuator contact | 0 |
| Communication status | Error in the communication with the actuator contact | Primary | Section 6.8.3 | inSolar solution | 50 |
| | Communication with the actuator contact | Primary | Section 6.8.3 | inSolar solution | 0 |

**Settings:**

None

**Status:**

As per section 4.10

**System properties:**

As per section 12.2.25

**Plots:**

As per section 9.6

## *13.31 For a controller*

**Parameters:**

None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Controller connection link status | Controller online | Primary | Section 6.9 | Controller | 0 |
| | Controller offline | Primary | Section 6.9 | Controller | 30 |
| Controller status | Controller available | Primary | Section 6.9 | Controller | 0 |
| | Controller unavailable | Primary | Section 6.9 | Controller | 100 |

**Settings:**

None

**Status:**

As per section 4.10

**System properties:**

As per section 12.2.26

**Plots:**

As per section 9.6

 

## *13.32 For a video server*

**Parameters:**
None

**State-sets & states:**

| State-set | State | State type | Reference | Update source | Default Severity |
|---|---|---|---|---|---|
| Video server connection link status | Video server online | Primary | Section 6.9 | Video server | 0 |
| | Video server offline | Primary | Section 6.9 | Video server | 30 |

**Settings:**
None

**Status:**
As per section 4.10

**System properties:**
As per section 12.2.27

**Plots:**
As per section 9.6

# 14 References

**Reference 1: IEC 61724 standard, “Photovoltaic system performance monitoring - Guidelines for measurement, data exchange and analysis”, 04/1998**

**Reference 2: Environmental Protection Agency 454/B-08-002, “Quality Assurance Handbook for Air Pollution Measurement Systems Volume IV: Meteorological Measurements Version 2.0 (Final)”, March 2008**

**Reference 3: IEEE Std 762-2006, “IEEE Standard Definitions for Use in Reporting Electric Generating Unit Reliability, Availability, and Productivity”, 15 March 2007**

**Reference 4: “Utility use of PV system ratings”, Firor, K.; Whitaker, C.M.; Jennings, C., Photovoltaic Specialists Conference, 1990, Conference Record of the Twenty First IEEE Volume , Issue , 21-25 May 1990 Page(s):932 - 936 vol.2**

**Reference 5: IEEE Std 929-2000 (Revision of IEEE Std 929-1988), “IEEE Recommended Practice for Utility Interface of Photovoltaic (PV) Systems”, 30 January 2000**

**Reference 6: International Energy Association, “Analysis Of Photovoltaic Systems”, Task 2 Report IEA-PVPS T2-01: 2000, 2000**

**Reference 7: Thevenard, Didier, Lisa Dignard, Sylvain Martel, Michael Ross, Dave Turcotte and Sharon Troke, "Evaluation of the Monitored Data from a Grid-Connected Photovoltaic System at Nunavut Arctic College, Iqaluit, NWT (Canada)", Renewable Energy Technologies in Cold Climates: Proceedings of the 24th Annual Conference of the Solar Energy Society of Canada, Montréal, Québec, Canada: May 4-6, 1998, pp. 263-268.**

**Reference 8: IEEE Std C37.2-1996(R2001, Revision of C37.2-1991), “IEEE Standard Electrical Power System Device Function Numbers and Contact Designations”, 2/05/2002**

**Reference 9: IEEE 754-1985, “IEEE standard for binary floating-point arithmetic”, Standards Committee of the IEEE Computer Society, USA, 12 Aug 1985**

**Reference 10: “The Electrical Installation Guide”, Schneider Electric, 2009 edition**

 